

Riemannian and Complex Geometry

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— Lecture Notes —

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Preface

We're excited to share these notes as a companion to the course on Riemannian and Complex Geometry. The course offers a hands-on introduction to two fascinating and closely connected areas of differential geometry, both of which provide essential tools for understanding the shape and structure of spaces known as manifolds.

The main goal is to introduce the core ideas and methods used in both Riemannian and complex geometry, while also exploring how they relate to and differ from each other. The course is designed to build a clear and connected understanding of these topics, and to show how they appear in both pure mathematics and theoretical physics.

This edition of the course ran from September 2025 to January 2026, guided by Daniel Galviz and Bowen Liu, as part of a broader effort to support advanced education and collaboration in mathematics across Latin America and beyond.

We want to make it clear that these notes are not meant to be an original work, but rather a well-organized and thoughtful collection of materials and references that support and expand on what was covered in the lectures. Throughout these pages, we've brought together ideas and insights from a range of respected sources that have helped shape our current understanding of geometry.

We are very grateful to the institutions and individuals whose support made this course and these notes possible. Special thanks to the Abdus Salam International Centre for Theoretical Physics (ICTP) – Physics Without Frontiers, for their commitment to building strong academic connections and educational opportunities around the world.

Daniel Galviz would like to sincerely thank Nicolai Reshetikhin for his generous support throughout this project. He also wishes to thank Akito Futaki for his meaningful teaching on Complex Geometry and engaging conversations, which played a key role in shaping the course's content and structure. A very personal note of gratitude goes to Daniel Huybrechts and Ursula Hamenstädt, who introduced him to differential geometry and complex geometry when he "secretly" attended their lectures at the University of Bonn.

Bowen Liu wishes to express his thanks to Chenglong Yu and Mao Sheng for their support and guidance, which played an important role in shaping this material and helping bring the course to completion.

Both authors are especially thankful to the Yau Mathematical Sciences Center at Tsinghua University, whose support was essential in making this effort a success. A heartfelt thanks to Shing-Tung Yau, who allowed both of us to enter the beautiful world of mathematics and geometry.

Finally, we thank all the students who joined the course. Their curiosity and dedication greatly enriched the experience. We hope these notes serve as a useful guide for those deepening their understanding of differential geometry, and as a friendly starting point for newcomers to Riemannian and complex geometry. Above all, we hope this material inspires future students and researchers to keep exploring and expanding the frontiers of modern geometry.

Physics Latam, 2025

Daniel Galviz
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Part I

Quick Review on Differential Geometry

1 Smooth manifolds

1.1 Topological spaces

The notion of manifolds is built upon topological spaces. The concept of topological spaces provides open and closed sets that allow defining concepts such as continuity, connectedness, convergence and so on.

Definition 1.1. Let X be a set and $\mathcal{T}$ be a collection of subsets of X satisfying:

- (1) $\emptyset$ and X are in $\mathcal{T}$;
- (2) For any subcollection $\{U_i \in \mathcal{T} \mid i \in I\}$, the union $\bigcup_i U_i \in \mathcal{T}$;
- (3) If $U_1, U_2 \in \mathcal{T}$, then $U_1 \cap U_2 \in \mathcal{T}$.

Then, $\mathcal{T}$ is called a *topology* and $(X, \mathcal{T})$ is called a *topological space*.

Notation 1.2. We usually denote X as a topological space for brevity, if there is no ambiguous.

Definition 1.3. Let $(X, \mathcal{T})$ be a topological space.

1. An element of $\mathcal{T}$ is called an *open set*;
2. A subset V of X is *closed* if its complement in X is an open set, i.e., $X - V \in \mathcal{T}$.

Lemma 1.4.

- (1) For a collection $\{V_i \mid i \in I\}$ of closed sets, the intersection $\bigcap_i V_i$ is also closed.
- (2) For closed sets V_1, V_2 , $V_1 \cup V_2$ is closed.

Definition 1.5. A family of (open) sets $\{U_i\}_{i \in I}$ of X is called an (*open*) *cover* of X if $\bigcup_{i \in I} U_i = X$.

Definition 1.6. A topological space is *compact* if every open cover of X has a finite subcover, i.e., if $X = \bigcup_{i \in I} U_i$ for a collection of open sets $\{U_i \mid i \in I\}$, then we can find finitely many U_{i_k} ($k = 1, \dots, n$) such that $X = \bigcup_{k=1}^n U_{i_k}$.

Definition 1.7. Let $(X, \mathcal{T})$ and $(X', \mathcal{T}')$ be two topological spaces.

- (1) Given a map $f: X \rightarrow X'$, it is called a *continuous map* if $f^{-1}(V) \in \mathcal{T}$ for all $V \in \mathcal{T}'$.
- (2) If f is bijective and both f and f^{-1} are continuous, then f is called a *homeomorphism*.

Definition 1.8. A topological space X is *connected* if it cannot be written as $X = X_1 \cup X_2$, where X_1 and X_2 are non-empty open sets and $X_1 \cap X_2 = \emptyset$. Otherwise, X is called *disconnected*.

Definition 1.9. A topological space is *path-connected* if for every pair of points x and x' in X , there is a path in X from x to x' , i.e., a continuous map $\gamma: [0, 1] \rightarrow X$ such that:

$$\gamma(0) = x \quad \text{and} \quad \gamma(1) = x'.$$

Definition 1.10. Let $(X, \mathcal{T})$ be a topological space and $S \subseteq X$ be a subset. The *subspace topology* on S is defined as

$$\mathcal{T}_S = \{U \cap S \mid U \in \mathcal{T}\}.$$

Example 1.11. Let $\mathbb{S}^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ with the subspace topology from $\mathbb{R}^2$. Take any pair of points $p, q \in \mathbb{S}^1$. We can construct a function:

$$\gamma(t) = (\cos \theta(t), \sin \theta(t)), \quad \theta(t) = (1-t)\theta_0 + t\theta_1,$$

where θ_0 and θ_1 are the angles corresponding to p and q . This gives a continuous path on $\mathbb{S}^1$, so $\mathbb{S}^1$ is path-connected.

Definition 1.12. Let $(X, \mathcal{T})$ be a topological space and $\sim$ an equivalence relation on X . We define a topology on the quotient set $X/\sim$ by:

$$\mathcal{T}' = \{U \subseteq X/\sim \mid \pi^{-1}(U) \in \mathcal{T}\},$$

where $\pi: X \rightarrow X/\sim$ is the projection map. Then $(X/\sim, \mathcal{T}')$ is the *quotient space* of $(X, \mathcal{T})$.

Example 1.13. Let $X = \mathbb{R}^2$ and $W = \{(x, 0) \mid x \in \mathbb{R}\}$. Then X/W is the set of horizontal lines in the plane. Two points (x_1, y_1) and (x_2, y_2) are equivalent if they have the same y -value, regardless of x . Thus $(x_1, y_1) \sim (x_2, y_2)$ if and only if $y_1 = y_2$. The new space X/W is homeomorphic to $\mathbb{R}$.

Definition 1.14. Let $X_1, \dots, X_k$ be topological spaces. Their Cartesian product is $X_1 \times \dots \times X_k = \{(x_1, \dots, x_k) \mid x_i \in X_i\}$. The *ith projection map* is

$$\pi_i: X_1 \times \dots \times X_k \rightarrow X_i, \quad \pi_i(x_1, \dots, x_k) = x_i.$$

The *product topology* on $X_1 \times \dots \times X_k$ is the topology having as a basis all subsets of the form

$$U_1 \times \dots \times U_k, \quad U_i \subseteq X_i \text{ open.}$$

The space $X_1 \times \dots \times X_k$ with this topology is called a *product space*, and its basis elements are called *product open subsets*.

Definition 1.15. A topological space is called a *Hausdorff space* if for every pair of distinct points x_1, x_2 in X , there exist neighborhoods U_1 and U_2 of x_1 and x_2 respectively that are disjoint.

Definition 1.16. An open cover $\{U_\alpha\}$ of M is *locally finite* if every point $p \in M$ has a neighborhood intersecting only finitely many U_α .

Definition 1.17. A (open) cover $\{V_\beta\}$ is an *(open) refinement* of $\{U_\alpha\}$ if each V_β is contained in some U_α .

Definition 1.18. A topological space is called *paracompact* if every open cover has an open refinement that is locally finite.

Example 1.19. Let X be an infinite set, and τ the topology where open sets are:

- $\emptyset$;
- Any subset $U \subseteq X$ such that $X \setminus U$ is finite.

This topology is called the *finite complement topology*. Take two distinct points $x, y \in X$. Suppose we want to find disjoint open sets $U \ni x, V \ni y$. Since $X \setminus U$ and $X \setminus V$ are finite, their union is finite. Thus:

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

is finite. Since $U \cap V \neq \emptyset$, it is impossible to separate x, y with disjoint open sets. Therefore this topology is not Hausdorff.

Remark 1.20. For the case $X = \mathbb{A}^n$ is an affine space, the finite complement topology is exactly the Zariski topology on $\mathbb{A}^n$, which plays an important role in algebraic geometry.

Example 1.21. Take two copies of $\mathbb{R}$, denoted $\mathbb{R}_1$ and $\mathbb{R}_2$. Identify each point $x \in \mathbb{R} \setminus \{0\}$ in $\mathbb{R}_1$ and $\mathbb{R}_2$. Thus $x_1 \sim x_2$ for all $x \neq 0$, but the points $p = 0_1 \in \mathbb{R}_1$ and $q = 0_2 \in \mathbb{R}_2$ are distinct. The points p and q are distinct in the topological space X , but every neighborhood of p intersects every neighborhood of q because they share all points except the origin. Therefore there are no disjoint neighborhoods around these points, so this topology is not Hausdorff.

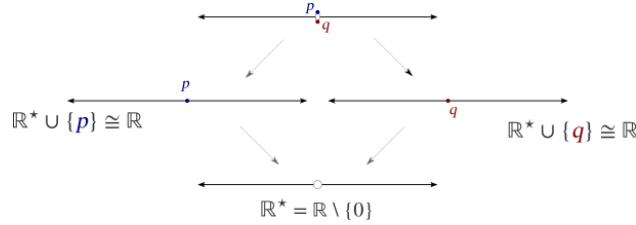


Figure 1: The line with two origins is a non-Hausdorff manifold

Example 1.22. The space $\mathbb{R}^n$ is paracompact. Let $X = \mathbb{R}^n$, and let $\mathcal{A}$ be an open covering of X . Set $\mathbb{B}_0 = \emptyset$, and for each positive integer m , let $\mathbb{B}_m$ denote the open ball of radius m centered at the origin. Given m , choose finitely many elements of $\mathcal{A}$ that cover $\mathbb{B}_m$, and intersect each one with the open set $X - \mathbb{B}_{m-1}$. Let this finite collection of open sets be denoted $\mathcal{C}_m$. Define $\mathcal{C} = \bigcup_{m=1}^{\infty} \mathcal{C}_m$. Then $\mathcal{C}$ is a refinement of $\mathcal{A}$.

It is locally finite: indeed, the open set $\mathbb{B}_k$ intersects only finitely many elements of $\mathcal{C}$, namely those belonging to the finite union $\mathcal{C}_1 \cup \dots \cup \mathcal{C}_k$. Finally, $\mathcal{C}$ covers X . For given $x \in X$, let m be the smallest integer such that $x \in \mathbb{B}_m$. Then by construction, x belongs to some element of $\mathcal{C}_m$. Thus $\mathcal{C}$ is a locally finite open refinement of $\mathcal{A}$, and $\mathbb{R}^n$ is paracompact.

1.2 Smooth manifolds

Definition 1.23. A Hausdorff paracompact topological space M is called a *topological n -dimensional manifold* if it has the following structure:

1. $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$ is an open cover of M .
2. For each $\alpha \in I$, there is a homeomorphism $\varphi_\alpha: U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.

The collection of $\{\varphi: U_\alpha \rightarrow \mathbb{R}^n\}$ is called an *atlas*, and each element in an atlas is called a *chart*.

Remark 1.24. Roughly speaking, a topological manifold is a topological space which locally "looks like" a Euclidean space. However, we need to put extra conditions, such as Hausdorffness and paracompactness, to avoid weird cases.

- (1) There exists a topological space which admits an open cover such that each open subset in that open cover is homeomorphic to a Euclidean space, but not Hausdorff (Example 1.21).
- (2) The paracompactness guarantees the existence of the partition of the unity, which is a useful tool.

Definition 1.25. A topological manifold is called a *smooth manifold*, if there exists an atlas $\{\varphi: U_\alpha \rightarrow \mathbb{R}^n\}$ such that for all U_α, U_β , the transition functions:

$$\varphi_\alpha \circ \varphi_\beta^{-1}: \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$$

are smooth maps between $\mathbb{R}^n$, and such an atlas is called a *smooth structure*.

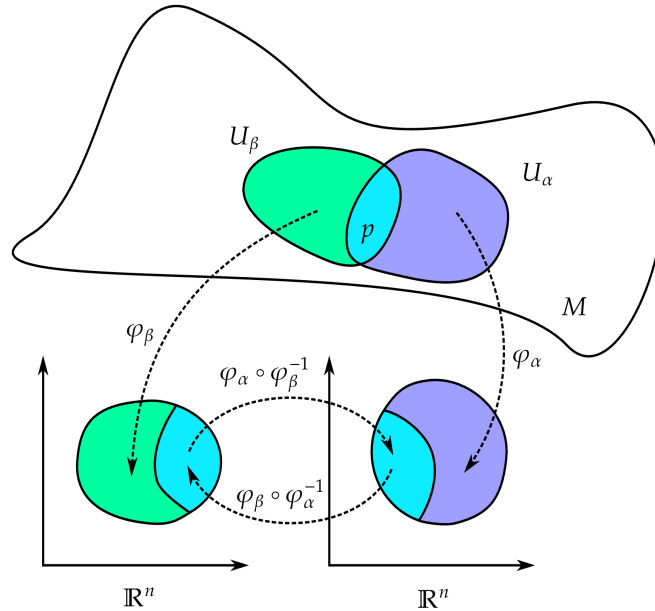


Figure 2: Charts and transition functions

Example 1.26. Define the n -sphere:

$$\mathbb{S}^n = \{x = (x^0, \dots, x^n) \in \mathbb{R}^{n+1} : \|x\|^2 = 1\}.$$

Let $U^+ = \mathbb{S}^n \setminus \{\text{north pole}\}$ and $U^- = \mathbb{S}^n \setminus \{\text{south pole}\}$. Define an atlas:

$$\begin{aligned}\varphi^+ : U^+ &\rightarrow \mathbb{R}^n, & (x^0, \dots, x^n) &\mapsto \frac{1}{1-x^0}(x^1, \dots, x^n), \\ \varphi^- : U^- &\rightarrow \mathbb{R}^n, & (x^0, \dots, x^n) &\mapsto \frac{1}{1+x^0}(x^1, \dots, x^n).\end{aligned}$$

These maps are called *stereographic projections*. The inverse maps are:

$$\begin{aligned}(\varphi^+)^{-1}(x^1, \dots, x^n) &= \frac{1}{1+\|x\|^2} (\|x\|^2 - 1, 2x^1, \dots, 2x^n). \\ (\varphi^-)^{-1}(x^1, \dots, x^n) &= \frac{1}{1+\|x\|^2} (1 - \|x\|^2, 2x^1, \dots, 2x^n).\end{aligned}$$

The transition functions are given by:

$$\varphi^+ \circ (\varphi^-)^{-1}(x) = \frac{x}{\|x\|^2}, \quad \varphi^- \circ (\varphi^+)^{-1}(x) = \frac{x}{\|x\|^2}.$$

The n -sphere is a compact manifold.

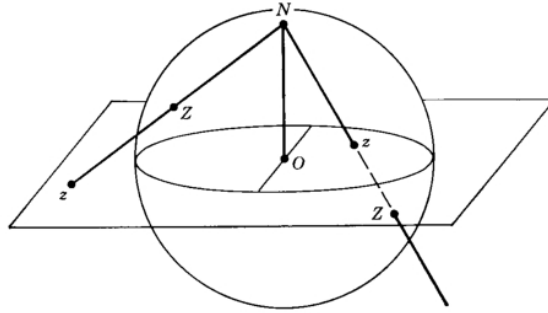


Figure 3: Stereographic projection from north pole

Example 1.27. Let $\mathbb{T}^2 = \{(x, y, z, w) \in \mathbb{R}^4 \mid x^2 + y^2 = 1, z^2 + w^2 = 1\}$. $\mathbb{T}^2$ is a subspace of $\mathbb{R}^4$ with subspace topology. The product of two compact Hausdorff spaces is compact and Hausdorff. Each $\mathbb{S}^1$ has an atlas with two smooth charts. Construct an atlas for $\mathbb{T}^2$ as the product atlas:

$$U_{ij} = U_i \times V_j, \quad \psi_{ij} = \psi_i \times \phi_j : U_{ij} \rightarrow \mathbb{R}^2 \quad (i, j = 1, 2),$$

where ψ_i, ϕ_j are charts of $\mathbb{S}^1$. The transition functions are products of smooth functions, hence smooth.

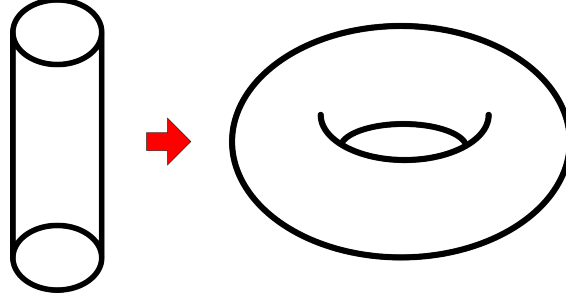


Figure 4: 2-dimensional torus $\mathbb{T}^2$.

Definition 1.28. Let M be an n -dimensional manifold. A subset $M' \subset M$ is called a *submanifold* if it satisfies: For each $q \in M'$, there exists a local coordinate system $(U; x^1, \dots, x^n)$ such that:

$$M' \cap U = \{q \in U \mid x^{k+1}(q) = \dots = x^n(q) = 0\}.$$

Then M' is k -dimensional. If $\dim(M') = n - k$, then M' is called a submanifold of codimension k in M .

Example 1.29.

$$\mathbb{S}^k \subset \mathbb{S}^n = \{x = (x^0, \dots, x^k, 0, \dots, 0) \in \mathbb{R}^{n+1} : \|x\|^2 = 1\}$$

is a submanifold of $\mathbb{S}^n$.

Definition 1.30. Let M and N be m - and n -dimensional manifolds respectively.

- (1) A continuous map $f: M \rightarrow N$ is called *smooth* at $p \in M$, if for every chart $(U_\alpha, \varphi_\alpha)$ containing p and every chart (V_β, ψ_β) containing $f(p) \in N$, the map:

$$\psi_\beta \circ f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap f^{-1}(V_\beta)) \rightarrow \psi_\beta(V_\beta)$$

is smooth at $\varphi_\alpha(p)$.

- (2) A continuous map $f: M \rightarrow N$ is called *smooth* if it is smooth at every point $p \in M$.
- (3) A continuous map $f: M \rightarrow N$ is called *diffeomorphism*, if f is smooth, bijective and it has a smooth inverse.

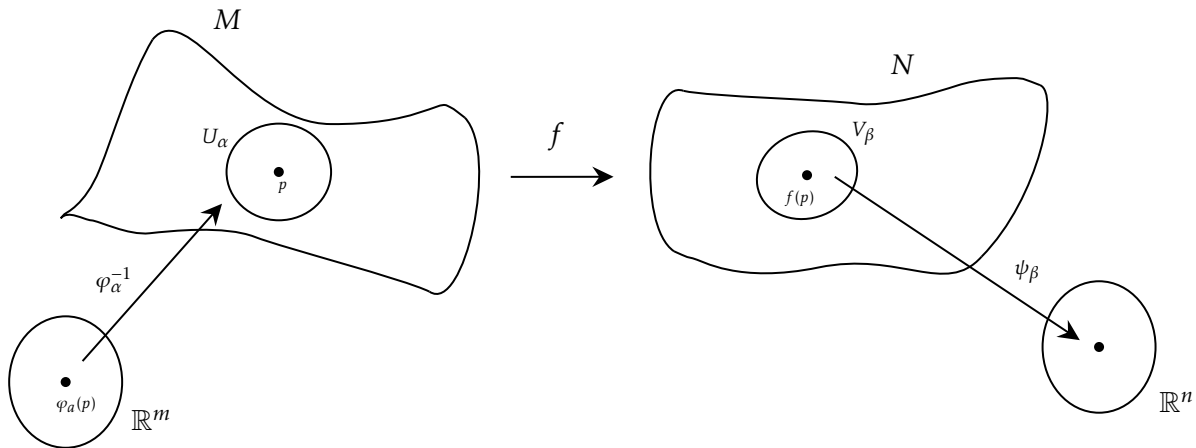


Figure 5: The map $f : M \rightarrow N$ is C^∞ at p .

Example 1.31. Let M and N be smooth manifolds. Then the projection $M \times N \rightarrow N$ is a smooth map.

Example 1.32. A rotation of $\mathbb{S}^1$ is a diffeomorphism.

1.3 Tangent bundle and cotangent bundle

1.3.1 Tangent spaces

Definition 1.33. A *tangent vector* X_p at $p \in M$ is a map $X_p : C^\infty(M) \rightarrow \mathbb{R}$ satisfying:

1. *Linearity*: $X_p(af + bg) = aX_p(f) + bX_p(g)$ for $a, b \in \mathbb{R}$;
2. *Leibniz Rule*: $X_p(fg) = f(p)X_p(g) + X_p(f)g(p)$.

A tangent vector X_p behaves like a derivative. The set of tangent vectors at p forms a vector space:

$$(X_p + Y_p)(f) = X_p(f) + Y_p(f), \quad (aX_p)(f) = a(X_p(f)).$$

This vector space is called the *tangent space* $T_p M$ at $p \in M$.

Definition 1.34. An *integral curve* of a vector field $X \in \mathfrak{X}(M)$ is a smooth map $\gamma : I \rightarrow M$ (where $I \subset \mathbb{R}$ is an open interval) such that $\dot{\gamma}(t) = X_{\gamma(t)}$.

Example 1.35. For $X = \alpha \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y}$ in $\mathbb{R}^2$, the integral curve is a translation:

$$\gamma_t(x, y) = (x + t\alpha, y + t\beta).$$

Given local coordinates $(x^1, \dots, x^n)$, the integral curve satisfies:

$$\frac{d}{dt} x^i = \alpha^i(x^1(t), \dots, x^n(t)).$$

These are ODEs. By the existence theorem, there is a unique solution $\gamma : (-\epsilon, \epsilon) \rightarrow M$ with $\gamma(0) = p$ for sufficiently small $\epsilon > 0$.

Theorem 1.36. For $X \in \mathfrak{X}(M)$ and $p \in M$, there exists an open interval $I \subset \mathbb{R}$ containing 0 and an integral curve $\gamma: I \rightarrow M$ for X with $\gamma(0) = p$. If $\tilde{\gamma}: \tilde{I} \rightarrow M$ is another integral curve for X with $\tilde{\gamma}(0) = p$, then $\gamma = \tilde{\gamma}$ on $I \cap \tilde{I}$.

Definition 1.37. If $\gamma(t)$ is defined for all $t \in \mathbb{R}$, then X is *complete*. If M is compact, every vector field is complete. In this case, the flow is a smooth map:

$$\Phi: \mathbb{R} \times M \rightarrow M, \quad (t, p) \mapsto \Phi_t(p)$$

satisfying:

$$\Phi_{t+s}(p) = \Phi_t(\Phi_s(p)), \quad \frac{d}{dt}\Phi_t(p) = X_{\Phi_t(p)}.$$

There is another way to think about tangent vectors. A curve starting from p is a smooth map $\gamma: I \rightarrow M$ with $\gamma(0) = p$ where $I \subset \mathbb{R}$ is a non-empty open interval. Two curves γ_1, γ_2 are *tangent* at p if:

$$\gamma_1(0) = p = \gamma_2(0), \quad \left. \frac{d}{dt} \right|_{t=0} \varphi(\gamma_1(t)) = \left. \frac{d}{dt} \right|_{t=0} \varphi(\gamma_2(t))$$

for any chart (U, φ) about p . This gives an equivalent relation, and the set of all curves starting from p up to this equivalence classes forms $T_p M$. For $f \in C^\infty(M)$, we can take the derivative of f along γ :

$$X_p(f) = \left. \frac{d}{dt} \right|_{t=0} f(\gamma(t)),$$

where $X_p = \left. \frac{d}{dt} \right|_{t=0} \psi(\gamma(t))$ and φ is any chart about p .

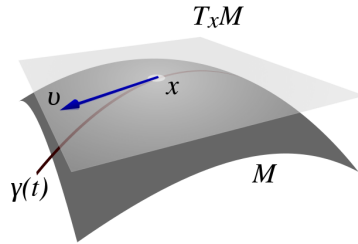


Figure 6: Tangent vector at $x \in M$

Given local coordinates $\varphi = (x^1, \dots, x^n)$, the tangent vector along a curve γ can be written as:

$$X_p = \sum_{i=1}^n X_p^i \frac{\partial}{\partial x^i} \Big|_p \quad \text{where } dx_p^i = \left. \frac{d}{dt} x^i(\gamma(t)) \right|_{t=0}$$

Thus $(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$ form a basis for $T_p M$. For another coordinate system $(y^1, \dots, y^n)$:

$$\frac{\partial}{\partial y^j} \Big|_p = \sum_{i=1}^n \frac{\partial x^i}{\partial y^j}(p) \frac{\partial}{\partial x^i} \Big|_p,$$

where $\frac{\partial x^i}{\partial y^j}(p)$ is the Jacobian at p .

Definition 1.38. Let $f: M \rightarrow N$ be a smooth map between manifolds. For $p \in M$ and each $v \in T_p M$, choose a curve $\gamma: (-\epsilon, \epsilon) \rightarrow M$ with $\gamma(0) = p$ and $\gamma'(0) = v$. The *tangent map* of f at p is:

$$(df)_p: T_p M \rightarrow T_{f(p)} N$$

defined by:

$$((df)_p)(v) = \beta'(0),$$

where $\beta = f \circ \gamma: (-\epsilon, \epsilon) \rightarrow N$.

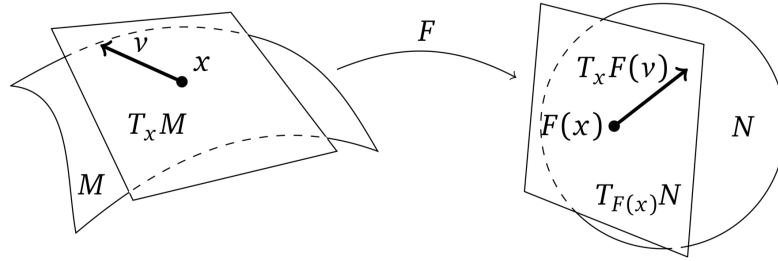


Figure 7: Tangent map at $x \in M$ such that $(dF)_x: T_x M \rightarrow T_{F(x)} N$.

Proposition 1.39. Let M, N, P be manifolds, $f: M \rightarrow N$ and $g: N \rightarrow P$ smooth maps. For $q \in M$:

$$(dg \circ f)_q = dg_{f(q)} \circ (df)_q: T_q M \rightarrow T_{g(f(q))} P.$$

Definition 1.40. Let M, N be smooth manifolds.

1. A map $f: M \rightarrow N$ is a *diffeomorphism* if it is smooth, bijective, and f^{-1} is smooth.
2. $f: M \rightarrow N$ is a *local diffeomorphism* at $p \in M$ if there exist neighborhoods $U \ni p$ and $V \ni f(p)$ such that $f: U \rightarrow V$ is a diffeomorphism.

Example 1.41. Consider $f: \mathbb{R} \rightarrow \mathbb{S}^1$, $f(\theta) = e^{\sqrt{-1}\theta}$. This is a local diffeomorphism but not a diffeomorphism.

Lemma 1.42. Every bijective local diffeomorphism is a diffeomorphism.

Definition 1.43. Let $f: M \rightarrow N$ be a smooth map. The *rank* of f at p is defined as the rank of the tangent map:

$$\text{rank}((df)_p: T_p M \rightarrow T_{f(p)} N).$$

Definition 1.44. Let $f: M \rightarrow N$ be a smooth map and $(df)_p: T_p M \rightarrow T_{f(p)} N$ the tangent map.

1. f is an *immersion* if $(df)_p$ is injective for all $p \in M$;
2. f is a *submersion* if $(df)_p$ is surjective for all $p \in M$;
3. f is an *embedding* if it is an immersion and $f: M \rightarrow f(M)$ is a homeomorphism;
4. If $M \subset N$ and the inclusion map $i: M \rightarrow N$ is an embedding, then M is a *submanifold* of N .

Remark 1.45.

- An immersion may not be injective (e.g., $f: \mathbb{R} \rightarrow \mathbb{S}^1$, $f(\theta) = e^{\sqrt{-1}\theta}$);
- A submersion may not be surjective (e.g., $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \arctan(x)$).

1.3.2 Tangent Bundle

Consider the collection of tangent spaces over each point of a manifold:

$$TM = \bigcup_{p \in M} T_p M = \{(p, v_p) \mid p \in M, v_p \in T_p M\}.$$

There is a natural projection $\pi: TM \rightarrow M$ sending $v_p \in T_p M$ to p .

Theorem 1.46. Let M be a n -dimensional smooth manifold. Then the tangent bundle TM is a $2n$ -dimensional manifold called the *tangent bundle*, and $\pi: TM \rightarrow M$ is a smooth map.

Proof. Given local coordinates $(x^1, \dots, x^n)$ on a chart (U, φ) , for each $p \in U$ and $v_p \in T_p M$, there are $\alpha^1, \dots, \alpha^n \in \mathbb{R}$ such that:

$$v_p = \sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^i} \Big|_p.$$

This gives a bijection:

$$\tilde{\varphi}: \pi^{-1}(U) \rightarrow \varphi(U) \times \mathbb{R}^n, \quad v_p \mapsto (x^1(p), \dots, x^n(p), \alpha^1, \dots, \alpha^n).$$

The transition functions between charts are smooth, so TM is a manifold. □

Definition 1.47. A smooth map $X: M \rightarrow TM$ is called a *section* of $\pi: TM \rightarrow M$ if $\pi \circ X = \text{id}_M$. It is called a *vector field*, denoted $\mathfrak{X}(M) = C^\infty(M, TM)$.

Lemma 1.48. A vector field $X \in \mathfrak{X}(M)$ gives a map $C^\infty(M) \rightarrow C^\infty(M)$ satisfying:

1. $X(af + bg) = aX(f) + bX(g)$, where $a, b \in \mathbb{R}$;
2. $X(fg) = gX(f) + fX(g)$.

Example 1.49. On $\mathbb{S}^n \subset \mathbb{R}^{n+1}$, we have a vector field:

$$X = \sum_{i=0}^n X^i \frac{\partial}{\partial x^i},$$

where

$$X^i = \begin{cases} (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}) & n = 2k + 1, \\ (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}, 0) & n = 2k + 2. \end{cases}$$

Definition 1.50. For $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, $[X, Y] = XY - YX$ is a vector field called the *Lie bracket*. It satisfies:

1. Bilinearity;
2. Antisymmetry: $[X, Y] = -[Y, X]$;
3. Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

1.3.3 Orientability

There are several equivalent definitions for orientability. Let M be an n -dimensional manifold.

Definition 1.51. An *orientation* on M is an equivalence class of oriented atlases. An atlas $\{U_\alpha, \varphi_\alpha\}$ is oriented if for every pair of charts U_α, φ_α and U_β, φ_β , the transition function:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

has positive Jacobian determinant:

$$\det(D(\varphi_\beta \circ \varphi_\alpha^{-1})(x)) > 0.$$

Definition 1.52. At each point $p \in M$, choose one of the two equivalence classes of ordered bases of $T_p M$, where two bases are equivalent if the change-of-basis matrix has positive determinant. This choice must vary smoothly over M .

Example 1.53.

- Sphere and torus are orientable.
- Möbius strip and Klein bottle are not orientable.
- Real projective space $\mathbb{RP}^n$ is orientable if and only if n is odd.

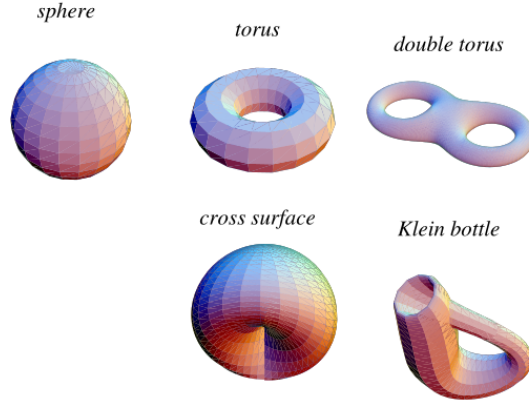


Figure 8: Examples of orientable and non-orientable compact manifolds.

1.3.4 Cotangent bundle

Definition 1.54. The dual space T_p^*M of the tangent space T_pM is called the *cotangent space*. For local coordinates $(x^1, \dots, x^n)$, the basis $(dx^1|_p, \dots, dx^n|_p)$ of T_p^*M satisfies:

$$dx^i|_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i.$$

The *cotangent bundle* is:

$$T^*M = \bigcup_{p \in M} T_p^*M$$

and sections of T^*M are called *differential 1-forms*, denoted $\Omega^1(M) = C^\infty(M, T^*M)$.

It is natural to consider the algebra generated by the basis of T_p^*M over $\mathbb{R}$ satisfying $dx^i \wedge dx^j = -dx^j \wedge dx^i$. The product $\wedge$ is called the *wedge product*.

Definition 1.55. For an n -manifold, we define the *exterior algebra* as follows

$$\Lambda^* T_p^*M = \bigoplus_{k=0}^n \Lambda^k T_p^*M.$$

An element $\omega \in \Lambda^k T_p^*M$ defines an alternating k -linear map $T_pM \times \dots \times T_pM \rightarrow \mathbb{R}$. The bundle

$$\Lambda^k T^*M = \bigcup_p \Lambda^k T_p^*M$$

is a smooth manifold. Its sections are called *differential k -forms*, denoted $\Omega^k(M) = C^\infty(M, \Lambda^k T^*M)$.

Remark 1.56. The notation $\Lambda^* T_p^* M$ denotes the *exterior algebra* of the cotangent space $T_p^* M$. It is defined as the direct sum of all exterior powers $\Lambda^* T_p^* M = \bigoplus_{k=0}^n \Lambda^k T_p^* M$, where $\Lambda^k T_p^* M$ is the space of alternating k -linear forms on $T_p M$, with $\Lambda^0 T_p^* M \cong \mathbb{R}$ and $\Lambda^1 T_p^* M = T_p^* M$. Thus, the exterior algebra collects all differential forms of every degree at the point p into a single graded vector space.

Definition 1.57. M is *orientable* if and only if there exists a nowhere-vanishing differential n -form, i.e., $\omega \in \Omega^n(M)$. Such ω defines an orientation: a basis $(v_1, \dots, v_n)$ of $T_p M$ is positively oriented if:

$$\omega_p(v_1, \dots, v_n) > 0.$$

Definition 1.58. There is a unique linear map $d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ called the *exterior derivative* satisfying:

1. For $f \in C^\infty(M)$, df is the usual differential;
2. $d \circ d = 0$;
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k(M)$.

In local coordinates:

$$d \left(\sum_{i_1 < \dots < i_k} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \right) = \sum_{i_1 < \dots < i_k} d\omega_{i_1 \dots i_k} \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Definition 1.59. For a smooth map $f: M \rightarrow N$, the *pullback* $f^*: \Omega^*(N) \rightarrow \Omega^*(M)$ is defined by:

$$(f^* \omega)(X_1, \dots, X_k) = \omega(df(X_1), \dots, df(X_k)).$$

It satisfies:

1. Linearity;
2. $f^*(\omega \wedge \eta) = f^* \omega \wedge f^* \eta$;
3. $(g \circ f)^* = f^* \circ g^*$;
4. Commutes with d : $df^* = f^* d$.

1.4 De Rham cohomology

1.4.1 De Rham complex

Let us now consider a geometric example in calculus. Let $U \subset \mathbb{R}^n$ be an open set, and $\{x^1, \dots, x^n\}$ be its coordinates. Introduce

$$\Omega^0(U) = \{\text{smooth functions on } U\} = C^\infty(U)$$

$$\Omega^1(U) = \{\text{smooth 1-forms on } U\}.$$

An element $\alpha \in \Omega^1(U)$ can be written as

$$\alpha = \sum_{i=1}^n \alpha_i(x) dx^i \in \Omega^1(U), \quad \alpha_i \in C^\infty(U).$$

The total differential gives a map

$$\begin{aligned} \Omega^0(U) &\xrightarrow{d} \Omega^1(U) \\ f &\mapsto df = \sum_i \partial_i f dx^i, \end{aligned}$$

which can be extended to a cochain complex

$$\Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \rightarrow \dots \xrightarrow{d} \Omega^n(U),$$

where $\Omega^p(U) = \{\text{smooth } p\text{-forms on } U\}$. An element $\xi \in \Omega^p(U)$ takes the form

$$\xi = \sum_{i_1, \dots, i_p} \xi_{i_1 \dots i_p}(x) dx^{i_1} \wedge \dots \wedge dx^{i_p}$$

with $\xi_{i_1 \dots i_p}(x) \in C^\infty(U)$ and the skew-symmetry property

$$dx^i \wedge dx^j = -dx^j \wedge dx^i, \quad \forall i, j$$

In particular, $dx^i \wedge dx^i = 0$. Thus we can also write

$$\Omega^p(U) = \bigoplus_{1 \leq i_1 < \dots < i_p \leq n} C^\infty(U) dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

The de Rham differential $d: \Omega^p(U) \rightarrow \Omega^{p+1}(U)$ is defined by

$$d(\xi) = \sum_{i_1, \dots, i_p, k} \partial_{x^k} \xi_{i_1 \dots i_p} dx^k \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

Thus

$$\Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \rightarrow \dots \rightarrow \Omega^n(U)$$

defines a cochain complex called the de Rham complex and denoted by $(\Omega^*(U), d)$.

Definition 1.60. A p -form $\alpha \in \Omega^p$ is called

- *closed* if $d\alpha = 0$

- *exact* if $\alpha = d\beta$ for some $p-1$ form β .

The quotient space

$$H_{dR}^p(U) = H^p(\Omega^*, d) = \frac{\text{closed } p\text{-forms}}{\text{exact } p\text{-forms}}$$

is called the p -th *de Rham cohomology*.

One important property is that the de Rham differential d is intrinsic: it does not depend on the choice of coordinates. Thus, we can define $(\Omega^*(M), d)$ on a smooth manifold M by gluing the above local construction. Then the de Rham cohomology on manifold $H^p(M) = H_{dR}^p(M)$ is well-defined. We can also consider $(\Omega_c^*(M), d)$, where

$$\Omega_c^*(M) = \{p\text{-form on } M \text{ with compact support}\}.$$

Then we can define de Rham cohomology with compact support

$$H_c^p(M) = H^p(\Omega_c^*(M), d).$$

Remark 1.61. Consider a smooth manifold M .

- The zeroth cohomology group $H^0(M)$ consists of closed 0-forms modulo exact 0-forms. Since every 0-form is just a smooth function and $df = 0$ means f is constant on each connected component, we have $H^0(M) \cong \mathbb{R}^{\#\pi_0(M)}$, where $\pi_0(M)$ denotes the set of connected components of M . Thus H^0 measures the number of connected components.
- The first cohomology group $H^1(M)$ consists of closed 1-forms modulo exact 1-forms. It detects the existence of nontrivial loops in M . For example, on the circle $M = \mathbb{S}^1$ we have $H^1(\mathbb{S}^1) \cong \mathbb{R}$, corresponding to the integral of a closed 1-form around the fundamental loop.
- The second cohomology group $H^2(M)$ classifies closed 2-forms modulo exact 2-forms. On an oriented surface M of genus g , such as the 2-torus $\mathbb{T}^2$, one finds $H^2(\mathbb{T}^2) \cong \mathbb{R}$, generated by the area form. In general, H^2 captures information about “fluxes” or surface integrals, such as the total curvature or electromagnetic flux through a surface.

Example 1.62. Cohomology groups for $\mathbb{S}^1, \mathbb{S}^2, \mathbb{T}^2$.

- H^0 always gives $\mathbb{R}$ because each space is connected.
- H^1 measures independent loops:
 - $\mathbb{S}^1$: one loop $\Rightarrow H^1 \cong \mathbb{R}$.
 - $\mathbb{S}^2$: no nontrivial loops $\Rightarrow H^1 = 0$.
 - $\mathbb{T}^2$: two independent loops $\Rightarrow H^1 \cong \mathbb{R}^2$.

- H^2 measures closed 2-forms (“fluxes”):
 - $\mathbb{S}^1$: no 2-dimensional surfaces $\Rightarrow H^2 = 0$.
 - $\mathbb{S}^2$: oriented area form generates $H^2 \cong \mathbb{R}$.
 - $\mathbb{T}^2$: oriented area form generates $H^2 \cong \mathbb{R}$.

1.5 Integration of differential forms

1.5.1 Integral Theorems

Let M be an oriented smooth n -manifold and $\omega \in \Omega^n(M)$ has compact support. To define $\int_M \omega$, we need partitions of unity.

Theorem 1.63. *Every open cover has a locally finite refinement.*

Theorem 1.64. *A partition of unity subordinate to a locally finite open cover $\{U_\alpha\}$ is a collection $\{\rho_\alpha\} \subset C^\infty(M, [0, 1])$ such that:*

1. $\text{supp}(\rho_\alpha) \subset U_\alpha$;
2. $\sum_\alpha \rho_\alpha = 1$.

In local coordinates, $\rho_\alpha \omega = f_\alpha(x) dx^1 \wedge \cdots \wedge dx^n$. Define:

$$\int_M \rho_\alpha \omega = \int \cdots \int f_\alpha(x) dx^1 \cdots dx^n.$$

Then define:

$$\int_M \omega = \sum_\alpha \int_M \rho_\alpha \omega.$$

This is independent of the choice of cover and partition of unity, and it is a finite summation, as ω has compact support.

Theorem 1.65. *Let M be an oriented n -manifold with boundary. If $\omega \in \Omega^{n-1}(M)$ has compact support, then:*

$$\int_M d\omega = \int_{\partial M} \omega$$

Corollary 1.66. *If M is compact and oriented without boundary, then for any $\omega \in \Omega^{n-1}(M)$:*

$$\int_M d\omega = 0.$$

Corollary 1.67.

- Fundamental Theorem of Calculus:

$$\int_a^b f'(x) dx = f(b) - f(a).$$

- Fundamental Theorem for Line Integrals:

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

- Green's Theorem:

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_{\partial D} P dx + Q dy.$$

- Stokes' Theorem (Vector Calculus):

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}.$$

- Divergence Theorem:

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_{\partial V} \mathbf{F} \cdot d\mathbf{S}.$$

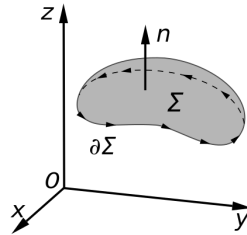


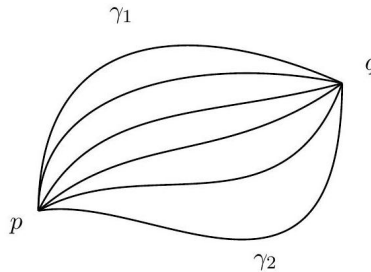
Figure 9: Stokes' theorem with surface Σ its boundary $\partial\Sigma$ and the normal vector n .

Example 1.68. Let $\alpha \in \Omega^1(\mathbb{R}^n)$ and $d\alpha = 0$. Let $p, q \in \mathbb{R}^n$ be two points. Define

$$\int_p^q \alpha = \int_\gamma \alpha$$

by choosing any smooth path γ connecting p and q . It does not depend on the choice of γ since any such path γ_1 can be continuously deformed to another γ_2 . Then

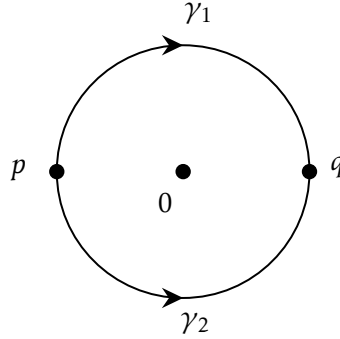
$$\int_{\gamma_1} \alpha = \int_{\gamma_2} \alpha \quad \text{by Stokes' Theorem.}$$



Example 1.69. Consider $\mathcal{U} = \mathbb{R}^2 - \{0\}$ and

$$\alpha = \frac{x dy - y dx}{x^2 + y^2}, \quad d\alpha = 0.$$

We consider two paths from $p = -1$ to $q = 1$ by following the upper and lower semicircle.



Since there is a hole in the middle, γ_1 can not be deformed to γ_2 inside the region $\mathcal{U}$. This can be also seen by computing

$$\int_{\gamma_1} \alpha - \int_{\gamma_2} \alpha = \oint_C \alpha$$

In terms of polar coordinates $x = r \cos \theta, y = r \sin \theta$,

$$\alpha = d\theta, \quad \oint_C \alpha = \int_0^{2\pi} d\theta = 2\pi \neq 0.$$

1.5.2 Cohomology and Integration

Theorem 1.70 (Poincaré lemma).

$$H^p(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & p = 0 \\ 0, & p > 0 \end{cases} \quad H_c^p(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & p = n \\ 0, & p \neq n \end{cases}$$

A generator of $H^0(\mathbb{R}^n)$ is the constant function $f \equiv 1$, and a generator of $H_c^n(\mathbb{R}^n)$ is a compactly supported form $\alpha = f(x) dx^1 \wedge \cdots \wedge dx^n$, where $f \in C_c^\infty(\mathbb{R}^n)$, such that $\int_{\mathbb{R}^n} \alpha = 1$.

Observe that any n -form $\alpha \in \Omega_c^n(\mathbb{R}^n)$ is closed. On the other hand, if $\alpha = d\beta$ for $\beta \in \Omega_c^{n-1}(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} d\beta = \int_{\mathbb{B}} d\beta = \int_{\partial\mathbb{B}} \beta = 0$$

where $\mathbb{B}$ is a large enough ball. The integration map

$$\begin{aligned} \int_{\mathbb{R}^n} : H_c^n(\mathbb{R}^n) &\longrightarrow \mathbb{R} \\ \alpha &\longmapsto \int_{\mathbb{R}^n} \alpha \end{aligned}$$

is the isomorphism described in the Poincaré lemma. Viewed differently, the map

$$\begin{aligned} \Omega_c^n(\mathbb{R}^n) &\longrightarrow H_c^n(\mathbb{R}^n) \stackrel{\text{Poincaré}}{\cong} \mathbb{R} \\ \alpha &\longmapsto [\alpha] \end{aligned}$$

sending a top n -form to its cohomology class determines precisely the integration (up to a rescaling constant). Thus

$$\int_{\mathbb{R}^n} \Longleftrightarrow H_c^n$$

i.e. "integration" can be modeled by a homological operation.

Part II

Riemannian Geometry

2 Riemannian manifold and Levi-Civita connection

2.1 Vector bundle

Definition 2.1. Let M be a smooth manifold. A *real vector bundle* E of rank r on M consists of the following data:

- (1) E is a smooth manifold with surjective map $\pi: E \rightarrow M$, such that
 - (1) For all $x \in M$, fibre E_x is a $\mathbb{R}$ -vector space of dimension r .
 - (2) For all $x \in M$, there exists $x \in U \subseteq M$ and there is a diffeomorphism $\varphi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$ such that

$$\begin{array}{ccc}
 \pi^{-1}(U) & \xrightarrow{\pi} & U \\
 \searrow \varphi & \curvearrowright p_1 & \nearrow \\
 & U \times \mathbb{R}^r & \xrightarrow{p_2} \mathbb{R}^r
 \end{array}$$

and for all $y \in U$, $E_y \xrightarrow{p_2 \circ \varphi} \mathbb{R}^r$ is a $\mathbb{R}$ -vector space isomorphism. The pair (U, φ) is called a trivialization of E over U .

Remark 2.2 (transition functions). Let $(U_\alpha, \varphi_\alpha), (U_\beta, \varphi_\beta)$ be local trivializations. Then $\varphi_\alpha \circ \varphi_\beta^{-1}: (U_\alpha \cap U_\beta) \times \mathbb{R}^r \rightarrow (U_\alpha \cap U_\beta) \times \mathbb{R}^r$ induces a smooth map

$$g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(r, \mathbb{R}),$$

where $g_{\alpha\beta}$ is called transition function. Furthermore, it satisfies

$$\begin{aligned}
 g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} &= \text{id} \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma \\
 g_{\alpha\alpha} &= \text{id} \quad \text{on } U_\alpha.
 \end{aligned}$$

In fact, transition functions contain all information about vector bundle since a vector bundle is locally trivial, so how are these trivial pieces glued together really matters.

Definition 2.3. Let M be a smooth manifold. A (*real*) *vector bundle* E of rank r on M consists of the following data:

- (1) open covering $\{U_\alpha\}$ of M .
- (2) smooth functions $\{g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(r, \mathbb{R})\}$ satisfies cocycle conditions:

$$\begin{aligned}
 g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} &= \text{id} \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma \\
 g_{\alpha\alpha} &= \text{id} \quad \text{on } U_\alpha.
 \end{aligned}$$

Definition 2.4. A vector bundle of rank one is called a *line bundle*.

Example 2.5. Let M be a smooth manifold. Then $M \times \mathbb{R}^r$ is called *trivial vector bundle* of rank r on M .

Definition 2.6. Let M be a smooth manifold and $\pi: E \rightarrow M$ be a vector bundle. A subset $F \subseteq E$ is called a *subbundle* of rank s , if

- (1) For all $x \in M$, $F \cap E_x$ is a subspace of E_x with dimension s .
- (2) $\pi|_F: F \rightarrow M$ is a vector bundle.

Definition 2.7. Let M be a smooth manifold and $\pi: E \rightarrow M$ be a vector bundle. A (*smooth*) *section* of E is a smooth map $s: M \rightarrow E$ such that $\pi \circ s = \text{id}_M$.

Definition 2.8. A section of $\bigotimes^s T^*M \otimes \bigotimes^r TM$ is called an (r, s) -*tensor*.

Example 2.9. A smooth function f is an $(0, 0)$ -tensor.

Example 2.10. A vector field X is an $(0, 1)$ -tensor.

Example 2.11. A 1-form ω is an $(1, 0)$ -tensor.

2.2 Riemannian manifold

2.2.1 Riemannian metric

Definition 2.12.

- (1) A *Riemannian metric* is an $(2, 0)$ -tensor such that for any $x \in M$, $g_x: T_x M \times T_x M \rightarrow \mathbb{R}$ is a positive definite symmetric bilinear form.
- (2) A *Riemannian manifold* (M, g) is a smooth manifold M together with a Riemannian metric g .

Example 2.13. Let $\mathbb{S}^n(R)$ denote n -dimensional sphere with radius R . There is a natural inclusion $f: \mathbb{S}^n(R) \hookrightarrow (\mathbb{R}^{n+1}, g_{\text{can}})$, and we can use f to pullback g_{can} to obtain a metric on $\mathbb{S}^n(R)$, denoted by g . Given a local chart (U, φ, x^i) , we can write

$$f(x^1, \dots, x^n) = \left(x^1, \dots, x^n, \sqrt{R^2 - \sum_{i=1}^n (x^i)^2} \right).$$

For any $\frac{\partial}{\partial x^i}$, we have

$$df \left(\frac{\partial}{\partial x^i} \right) = \frac{\partial f^j}{\partial x^i} \frac{\partial}{\partial x^j} = \frac{\partial}{\partial x^i} - \frac{x^i}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}.$$

Thus for any two $\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}$ we have

$$\begin{aligned} g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) &= g_{\text{can}}\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) \\ &= g_{\text{can}}\left(\frac{\partial}{\partial x^i} - \frac{x^i}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}, \frac{\partial}{\partial x^j} - \frac{x^j}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}\right) \\ &= \delta_{ij} + \frac{x^i x^j}{R^2 - \sum_{i=1}^n (x^i)^2}, \end{aligned}$$

which implies

$$g_{ij} = \delta_{ij} + \frac{x^i x^j}{T^2},$$

where $T^2 = R^2 - \sum_{i=1}^n (x^i)^2$.

Example 2.14. Let $\mathbb{H}^n(R) = \{(x^1, \dots, x^{n-1}, y) \in \mathbb{R}^n \mid y > 0\}$ with Riemannian metric

$$g = R^2 \frac{\delta_{ij} dx^i \otimes dx^j + dy \otimes dy}{y^2}.$$

It is called *hyperbolic upper plane*.

Example 2.15. Let $\mathbb{B}^n(R) = \{x \in \mathbb{R}^n \mid |x| < R\}$ with Riemannian metric

$$g = 4R^4 \frac{\delta_{ij} dx^i \otimes dx^j}{(R^2 - |x|^2)^2}.$$

It is called *Poincaré disk*.

Definition 2.16. A Riemannian metric $\langle \cdot, \cdot \rangle$ on a Lie group G is called *left-invariant* if

$$\langle (dL_g)X, (dL_g)Y \rangle = \langle X, Y \rangle$$

holds for arbitrary $g \in G$ and vector fields X, Y .

Remark 2.17. The right-invariant metric can be defined in a similar statement, and a Riemannian metric which is both left-invariant and right-invariant is called a *bi-invariant metric*.

Proposition 2.18. *There is a bijective correspondence between left-invariant metrics on a Lie group G , and inner products on the Lie algebra $\mathfrak{g}$ of G .*

Proof. Let $\langle \cdot, \cdot \rangle_e$ be an inner product on Lie algebra $\mathfrak{g}$. Then we have a Riemannian metric on G defined as follows

$$\langle X_g, Y_g \rangle := \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e,$$

where X, Y are two vector fields on G . It is left-invariant, since

$$\begin{aligned} \langle (dL_h)X_g, (dL_h)Y_g \rangle &= \langle (dL_{(hg)^{-1}}) \circ (dL_h)X_g, (dL_{(hg)^{-1}}) \circ (dL_h)Y_g \rangle_e \\ &= \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e. \end{aligned}$$

Conversely, if we have a left-invariant inner product $\langle \cdot, \cdot \rangle$ on G , then it is clear we have an inner product on $\mathfrak{g}$, by considering its value at identity. Furthermore, these two constructions are inverse to each other. $\square$

Proposition 2.19. *There is a bijective correspondence between bi-invariant metrics on a Lie group G , and Ad-invariant inner products on the Lie algebra $\mathfrak{g}$ of G .*

Proof. Given an Ad-invariant inner product $\langle \cdot, \cdot \rangle_e$ on the Lie algebra $\mathfrak{g}$, by Proposition 2.18, there is a left-invariant metric $\langle \cdot, \cdot \rangle$ on G , it suffices to check it's also right-invariant:

$$\begin{aligned} \langle (dR_h)X_g, (dR_h)Y_g \rangle &= \langle (dL_{(gh)^{-1}}) \circ (dR_h)X_g, (dL_{(gh)^{-1}}) \circ (dR_h)Y_g \rangle_e \\ &= \langle \text{Ad}(h^{-1})(dL_{g^{-1}})X_g, \text{Ad}(h^{-1})(dL_{g^{-1}})Y_g \rangle_e \\ &= \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e. \end{aligned}$$

Conversely, if we start with a bi-invariant metric, then its restriction to the Lie algebra is an Ad-invariant metric, since $\text{Ad}(g)$ is exactly the differential of $L_g \circ R_{g^{-1}}$. $\square$

2.2.2 Riemannian isometry

Definition 2.20. A smooth map $f: (M, g_M) \rightarrow (N, g_N)$ between Riemannian manifolds is called a *local isometry* if for each point $p \in M$ the differential $(df)_p: T_p M \rightarrow T_{f(p)} N$ is a linear isometry.

Definition 2.21. A local isometry $f: (M, g_M) \rightarrow (N, g_N)$ between Riemannian manifolds is called an *isometry* if it is a diffeomorphism.

Proposition 2.22. *Let $f: (M, g_M) \rightarrow (N, g_N)$ be smooth map between Riemannian manifolds. Then the following statements are equivalent.*

- (1) f is a local isometry.
- (2) For each $p \in M$, there are open neighborhoods U of p and V of $f(p)$ such that $f|_U: U \rightarrow V$ is an isometry.

Proof. it is clear (2) implies (1), and the converse follows immediately from inverse function theorem. $\square$

Definition 2.23. The *isometry group* $\text{Iso}(M, g)$ of Riemannian manifold (M, g) is the group consisting of all isometries from (M, g) to itself.

Proposition 2.24 ([Pet06, Proposition 5.6.1]). *Let $f: (M, g_M) \rightarrow (N, g_N)$ be a local isometry.*

- (1) f maps geodesics to geodesics.
- (2) $f \circ \exp_p = \exp_{f(p)} \circ (df)_p$ holds on V_p .
- (3) f is distance decreasing.
- (4) If f is an isometry, then f is distance preserving.

Theorem 2.25 ([MS39]). Let (M, g) be a Riemannian manifold and $G = \text{Iso}(M, g)$. Then

- (1) G is a Lie group with respect to compact-open topology.
- (2) For each $p \in M$, the isotropy group G_p is compact.
- (3) G is compact if M is compact.

2.3 Motivation of connection

Given a vector bundle E on a smooth manifold M , the motivation of connection on E arises from how to take the "derivative" of a section s of E in a given direction. It is quite natural to ask such a question, since when we learn calculus, we learn how to take the derivative of a smooth function $f: M \rightarrow \mathbb{R}^m$, to obtain a 1-form.

If we change our language in a more fancy way, any smooth function $f: M \rightarrow \mathbb{R}^m$ can be thought as a section of the trivial vector bundle $M \times \mathbb{R}^m$ as follows:

$$x \mapsto (x, f(x)),$$

and its derivative df can be thought a section of $T^*M \otimes (M \times \mathbb{R}^m)$. Then taking derivatives can be seen as the following operator:

$$\nabla: C^\infty(M, M \times \mathbb{R}^m) \rightarrow C^\infty(M, T^*M \otimes (M \times \mathbb{R}^m)).$$

This motivates us why we define a connection on a vector bundle as follows:

Definition 2.26. Let E be a vector bundle on a smooth manifold M . A *connection* ∇ on a vector bundle E is a $\mathbb{R}$ -linear operator

$$\nabla: C^\infty(M, E) \rightarrow C^\infty(M, T^*M \otimes E)$$

satisfying the Leibniz rule $\nabla(fs) = df \otimes s + f\nabla s$, where $f \in C^\infty(M)$ and $s \in C^\infty(M, E)$.

Remark 2.27 (local form). Let $\{e_\alpha\}$ be a local frame of E . For any local section $s = s^\alpha e_\alpha$ of E , the Leibniz rule implies

$$\nabla(s^\alpha e_\alpha) = ds^\alpha \otimes e_\alpha + s^\alpha \nabla e_\alpha.$$

Thus, the connection ∇ is determined by $\nabla e_\alpha = \omega_\alpha^\beta \otimes e_\beta$, where ω_α^β are 1-forms. For simplicity, we always omit the tensor notion $\otimes$ if there is no ambiguity.

The collection of 1-forms $\{\omega_\alpha^\beta\}$ forms a matrix ω , with the lower index is the row index and the upper index is the column index.

Remark 2.28. Let $\{e_\alpha\}$ be a local frame of vector bundle E and ω be the matrix of connection 1-forms. If we write $\omega_\alpha^\beta = \Gamma_{i\alpha}^\beta dx^i$, then $\Gamma_{i\alpha}^\beta$ is called the *Christoffel symbol*.

Let X be a vector field which is locally written as $X = X^i \frac{\partial}{\partial x^i}$ and s be a section of E which is locally written as $s^\alpha e_\alpha$. Then

$$\begin{aligned}\nabla_X s &= \nabla_{X^i \frac{\partial}{\partial x^i}} s^\alpha e_\alpha \\ &= X^i \nabla_{\frac{\partial}{\partial x^i}} s^\alpha e_\alpha \\ &= X^i s^\alpha \nabla_{\frac{\partial}{\partial x^i}} e_\alpha + X^i \frac{\partial s^\alpha}{\partial x^i} e_\alpha \\ &= \left(X^i s^\alpha \Gamma_{i\alpha}^\beta + X(s^\beta) \right) e_\beta,\end{aligned}$$

which is called the *covariant derivative* of s with respect to X .

2.4 Levi-Civita connection

2.4.1 Compatibility with metric

Now consider a vector bundle E with a metric g , which can be locally written as $g_{\alpha\beta} e^\alpha \otimes e^\beta$. If there is a connection ∇ on E , it is natural to ask it to be compatible with metric.

Definition 2.29. A connection ∇ on a vector bundle E is *compatible with the metric* g if for any two sections s, t of E , we have

$$dg(s, t) = g(\nabla s, t) + g(s, \nabla t).$$

Remark 2.30 (local form). Let $\{e_\alpha\}$ be a local frame of E . A direct computation shows

$$\begin{aligned}dg_{\alpha\beta} &= dg(e_\alpha, e_\beta) \\ &= g(\nabla e_\alpha, e_\beta) + g(e_\alpha, \nabla e_\beta) \\ &= \omega_\alpha^\gamma g_{\gamma\beta} + g_{\alpha\gamma} \omega_\beta^\gamma\end{aligned}$$

for all α, β , which is equivalent to

$$\frac{\partial}{\partial x^i} g_{\alpha\beta} = \Gamma_{i\alpha}^\gamma g_{\gamma\beta} + \Gamma_{i\beta}^\gamma g_{\alpha\gamma}$$

for all i, α, β . In matrix notation we have

$$dg = \omega g + g \omega^t.$$

2.4.2 Torsion-freeness

Now let's choose our vector bundle E to be the tangent bundle of a Riemannian manifold (M, g) .

Definition 2.31. A connection ∇ of TM is *torsion-free* if

$$\nabla_X Y - \nabla_Y X = [X, Y],$$

where X, Y are vector fields.

Remark 2.32 (local form). If we choose $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}$, then

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} - \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i} = (\Gamma_{ij}^k - \Gamma_{ji}^k) \frac{\partial}{\partial x^k} = 0,$$

which is equivalent to Γ_{ij}^k is symmetric in i and j .

2.4.3 Levi-Civita connection

There are infinitely many connection on the tangent bundle of a Riemannian manifold, but an interesting thing is that there is a unique torsion-free connection which is compatible with the Riemannian metric.

Note that compatibility with Riemannian metric g implies

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

$$Yg(Z, X) = g(\nabla_Y Z, X) + g(Z, \nabla_Y X)$$

$$Zg(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y),$$

where X, Y and Z are vector fields. Adding the first two equations, subtracting the third one, and using the torsion-free condition, we will see

$$Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) = g([X, Z], Y) + g([Y, Z], X) + g([X, Y], Z) + 2g(Z, \nabla_Y X),$$

and thus

$$g(Z, \nabla_Y X) = \frac{1}{2}(Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) - g([X, Z], Y) - g([Y, Z], X) - g([X, Y], Z)),$$

which implies $\nabla_X Y$ is uniquely determined. The above formula is also called the *Koszul formula*.

Remark 2.33 (local form). Firstly, compatibility implies

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^\ell g_{\ell j} + \Gamma_{kj}^\ell g_{i\ell}.$$

By permuting i, j, k we obtain the following two equations:

$$\frac{\partial g_{jk}}{\partial x^i} = \Gamma_{ij}^\ell g_{\ell k} + \Gamma_{ik}^\ell g_{j\ell}$$

$$\frac{\partial g_{ki}}{\partial x^j} = \Gamma_{jk}^\ell g_{\ell i} + \Gamma_{ji}^\ell g_{k\ell}.$$

By the symmetry of Γ_{ij}^ℓ in i, j and the symmetry of g_{ij} , we have

$$2\Gamma_{ij}^\ell g_{\ell k} = \frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k}.$$

If we use (g^{ij}) to denote the inverse matrix of (g_{ij}) , then we have

$$\Gamma_{ij}^\ell = \frac{1}{2} g^{k\ell} \left(\frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k} \right),$$

which implies the Christoffel symbol is completely determined by the Riemannian metric and its first order partial derivatives.

3 Tensor computation

3.1 Induced connection on tensor

Let M be a smooth manifold and ∇ be a connection on the tangent bundle TM . In this section, we will introduce how to construct a connection on a (r, s) -tensor.

Firstly, let's see how to induce a connection on T^*M . Since there is a natural non-degenerate pairing $(-, -)$ between TM and T^*M , the *induced connection* on T^*M should respect this natural pairing. To be precise, for any $X \in C^\infty(M, TM)$ and $\omega \in C^\infty(M, T^*M)$, the induced connection on T^*M is defined by the following equality

$$d(\omega, X) = (\nabla^{T^*M} \omega, X) + (\omega, \nabla X).$$

Suppose $\{\frac{\partial}{\partial x^i}\}$ is a local frame of TM with dual frame $\{dx^i\}$ of T^*M . Then

$$0 = ((\omega^{T^*M})_k^i dx^k, \frac{\partial}{\partial x^j}) + (dx^i, (\omega)_j^k \frac{\partial}{\partial x^k}),$$

that is,

$$\nabla dx^k = -\Gamma_{ij}^k dx^i \otimes dx^j,$$

where Γ_{ij}^k is the Christoffel symbol of ∇^{TM} .

Remark 3.1 (another viewpoint for torsion-freeness). For any section ω of T^*M , there is a natural 2-form obtained from taking the exterior derivative $d\omega$, and note that $\wedge^2 T^*M$ is just the skew-symmetrization of $T^*M \otimes T^*M$, so it is natural to require the skew-symmetrization of $\nabla \omega$ is $d\omega$. If we check this locally, it suffices to require the skew-symmetrization of ∇dx^k is zero, that is,

$$-\Gamma_{ij}^k dx^i \wedge dx^j = 0.$$

In other words, connection ∇ is torsion-free if and only if the skew-symmetrization of $\nabla \omega$ is $d\omega$.

Definition 3.2. Given a (r, s) -tensor T .

(1) The *covariant derivative* ∇T is a $(r+1, s)$ -tensor, which is defined by

$$\begin{aligned}\nabla T(X_1, X_2, \dots, X_{r+1}, \omega_1, \dots, \omega_s) := & XT(X_1, \dots, X_r, \omega_1, \dots, \omega_s) \\ & - \sum_{k=1}^r T(X_1, \dots, \nabla_{X_1} X_k, \dots, X_r, \omega_1, \dots, \omega_s) \\ & - \sum_{\ell=1}^s T(X_1, \dots, X_r, \omega_1, \dots, \nabla_{X_1} \omega_\ell, \dots, \omega_s),\end{aligned}$$

where $X_1, \dots, X_{r+1}$ are vector fields and $\omega_1, \dots, \omega_s$ are 1-forms.

(2) For a vector field X , the *covariant derivative of T with respect to vector field X* , denoted by $\nabla_X T$ is a (r, s) -tensor, defined by

$$\nabla_X T(X_1, \dots, X_r, \omega_1, \dots, \omega_s),$$

where $X_1, \dots, X_r$ are vector fields and $\omega_1, \dots, \omega_s$ are 1-forms.

Remark 3.3 (local form). If we write a (r, s) -tensor T locally as

$$T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}},$$

and $(r+1, s)$ -tensor ∇T locally as

$$\nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^i \otimes dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}.$$

Then by definition we have

$$\nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} = \frac{\partial T_{i_1 \dots i_r}^{j_1 \dots j_s}}{\partial x^i} - \sum_{k=1}^r \Gamma_{ii_k}^q T_{i_1 \dots i_{k-1} q i_{k+1} \dots i_r}^{j_1 \dots j_s} + \sum_{\ell=1}^s \Gamma_{iq}^{j_\ell} T_{i_1 \dots i_r}^{j_1 \dots j_{\ell-1} q j_{\ell+1} \dots j_s}.$$

Definition 3.4. For a (r, s) -tensor T , the *second covariant derivative* is defined as $\nabla^2 T := \nabla(\nabla T)$, which is a $(r+2, s)$ -tensor.

Given vector fields X, Y and a (r, s) -tensor T , firstly we consider its covariant derivative with respect to direction Y , it gives a (r, s) -tensor $\nabla_Y T$, and then we consider the covariant derivative of $\nabla_Y T$ with respect to the direction X , it gives a (r, s) -tensor $\nabla_X \nabla_Y T$.

Since both $\nabla_{X,Y}^2 T$ and $\nabla_X \nabla_Y T$ are (r, s) -tensors, a natural question is what is the relation between them? The answer is they're not the same, and the main reason is that the former is linear over smooth functions in Y , while the latter is not. Let's start with an easy example:

Example 3.5. Given $(0, 0)$ -tensor f , that is, a smooth function f .

(1) The covariant derivative ∇f is an $(1, 0)$ -tensor, which is given by

$$\nabla f = \nabla_i f dx^i = \frac{\partial f}{\partial x^i} dx^i.$$

In other words, ∇f coincides with df .

- (2) The second covariant derivative $\nabla^2 f$ is called the *Hessian* of f , and usually it is denoted by $\text{Hess } f$. More explicitly, we have

$$\begin{aligned}\text{Hess } f &= \nabla(\nabla_i f \, dx^i) \\ &= \frac{\partial \nabla_i f}{\partial x^k} dx^k \otimes dx^i - \nabla_i f \Gamma_{kj}^i dx^k \otimes dx^j \\ &= \left(\frac{\partial^2 f}{\partial x^k \partial x^i} - \Gamma_{ki}^j \frac{\partial f}{\partial x^j} \right) dx^k \otimes dx^i,\end{aligned}$$

$$\text{that is, } \nabla_{k,i}^2 f = \frac{\partial^2 f}{\partial x^k \partial x^i} - \Gamma_{ki}^j \frac{\partial f}{\partial x^j}.$$

This shows that

$$\nabla_{k,i}^2 f = \nabla_k \nabla_i f - \Gamma_{ki}^j \frac{\partial f}{\partial x^j}.$$

The following proposition shows the relation:

Proposition 3.6. *For any (r, s) -tensor T and vector fields X, Y ,*

$$\nabla_{X,Y}^2 T = \nabla_X \nabla_Y T - \nabla_{(\nabla_X Y)} T.$$

Proof. Let $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ be a local frame of TM and $\{dx^1, \dots, dx^n\}$ be the dual frame of T^*M . Suppose

$$\nabla^2 T = \nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^k \otimes dx^i \otimes dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}$$

and

$$\nabla_k \nabla_i T = \nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}.$$

It suffices to prove

$$\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} = \nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} - \Gamma_{ki}^j \nabla_j T_{i_1 \dots i_r}^{j_1 \dots j_s}.$$

A direct computation shows

$$\begin{aligned}
\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} &= \nabla^2 T \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&= \nabla_{\frac{\partial}{\partial x^k}} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&= \underbrace{\frac{\partial}{\partial x^k} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part I}} \\
&\quad - \underbrace{\nabla T \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part II}} \\
&\quad - \underbrace{\sum_{\ell=1}^s \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, \nabla_{\frac{\partial}{\partial x^k}} dx^{j_\ell}, \dots, dx^{j_s} \right)}_{\text{part III}} \\
&\quad - \underbrace{\sum_{m=1}^r \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^{i_m}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part IV}}.
\end{aligned}$$

Note that Part II is $-\Gamma_{ki}^j \nabla_j T_{i_1 \dots i_r}^{j_1 \dots j_s}$. On the other hand, the covariant derivative $\nabla_i T$ is defined by

$$\nabla_i T = \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right),$$

and thus by definition

$$\begin{aligned}
\nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} &= \frac{\partial}{\partial x^k} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&\quad - \sum_{\ell=1}^s \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, \nabla_{\frac{\partial}{\partial x^k}} dx^{j_\ell}, \dots, dx^{j_s} \right) \\
&\quad - \sum_{m=1}^r \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^{i_m}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right).
\end{aligned}$$

This shows $\nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s}$ coincides with Part I + part III + part IV. □

Proposition 3.7. *Let g be a Riemannian metric. Then a connection ∇ is compatible with g if and only if g is parallel, that is, $\nabla g = 0$.*

3.2 Type change of tensor

In this section, we will introduce, how to change a tensor of type (r, s) to another tensor of type $(s - k, r + k)$ for all k such that $s - k \geq 0, r + k \geq 0$, by music isomorphism.

Firstly we have to introduce the induced metric on

Let (M, g) be a Riemannian manifold. Firstly, we have to introduce the induce metric on T^*M . Recall that a Riemannian metric g is an $(2, 0)$ -tensor, locally written as

$$g = g_{ij} dx^i \otimes dx^j.$$

As the inverse of a positive definite symmetric matrix is again a positive definite symmetric matrix, it induces a metric on T^*M , locally given by

$$g^{ij} \frac{\partial}{\partial x^i} \otimes \frac{\partial}{\partial x^j}.$$

For any vector field X , it gives a 1-form by

$$X^\flat: Y \mapsto g(X, Y),$$

where Y is a vector field. Let $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ be a local frame of TM and $\{dx^1, \dots, dx^n\}$ be the dual frame of T^*M . If we write $(\frac{\partial}{\partial x^i})^\flat = A_j dx^j$, then

$$A_j = g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}) = g_{ij}.$$

This shows $(\frac{\partial}{\partial x^i})^\flat = g_{ij} dx^j$ of T^*M .

For any 1-form ω , it can be regarded as a section of $TM = T^{**}M$ as follows

$$\omega^\sharp: \beta \mapsto g(\omega, \beta),$$

If we write $(dx^i)^\sharp = B^j \frac{\partial}{\partial x^j}$, then

$$B^j = g(dx^i, dx^j) = g^{ij}.$$

This shows $(dx^i)^\sharp = g^{ij} \frac{\partial}{\partial x^j}$.

In a summary, we have the so-called *music isomorphisms*, which locally look like

$$\begin{aligned} \flat: C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M) & \sharp: C^\infty(M, T^*M) &\rightarrow C^\infty(M, TM) \\ \frac{\partial}{\partial x^i} &\mapsto g_{ij} dx^j, & dx^j &\mapsto g^{ij} \frac{\partial}{\partial x^i}. \end{aligned}$$

Example 3.8. For a smooth function f , ∇f is an $(1, 0)$ -tensor, locally written as

$$\nabla f = \frac{\partial f}{\partial x^i} dx^i.$$

Then we can change it to a tensor of type $(0, 1)$, that is,

$$\nabla f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}.$$

It is called *gradient* of f . More generally, for a 1-form ω , locally looks like $\omega_i dx^i$. Then we can change it in an $(1,0)$ -tensor, and it locally looks like

$$\omega^\sharp = g^{ij} \omega_i \frac{\partial}{\partial x^j}.$$

An application of type change is, we can use it to define the trace of an arbitrary tensor. Let's see a simple example: For an $(1,1)$ -tensor T , we can define its "trace", since there is a natural isomorphism between $TM \otimes T^*M$ and $\text{End}(TM)$. Thus, we can take its trace in the sense of matrix. To be explicit, if we locally write T as $T = T_j^i \frac{\partial}{\partial x^i} \otimes dx^j$, then trace of T , denoted by $\text{tr}_g T$, is defined as T_i^i .

If T is not of $(1,1)$ -type, then we change it into $(1,1)$ -type and then take trace:

- (1) If $T = T_{ij} dx^i \otimes dx^j$, then $T = g^{ik} T_{ij} dx^j \otimes \frac{\partial}{\partial x^k}$. Thus, $\text{tr}_g T = g^{ij} T_{ij}$.
- (2) If $T = T^{ij} \frac{\partial}{\partial x^i} \otimes \frac{\partial}{\partial x^j}$, then $T = g_{kj} T^{ij} dx^k \otimes \frac{\partial}{\partial x^i}$, that is, $\text{tr}_g T = g_{ij} T^{ij}$.

In general, if a tensor of type (r,s) with $r+s = 2n$, we can change its type into (n,n) and take trace n times to obtain a number. Later we will see we obtain Ricci curvature by taking trace of curvature, and we obtain scalar curvature by taking trace of Ricci curvature.

Remark 3.9. For a smooth function $f: M \rightarrow \mathbb{R}$, $\nabla^2 f$ is an $(2,0)$ -form, locally looks like

$$\nabla_{i,j}^2 f dx^i \otimes dx^j.$$

Its trace looks like

$$\text{tr}_g \nabla^2 f = g^{ij} \nabla_{i,j}^2 f.$$

That's called the *scalar Laplacian* of f , which is denoted by Δf .

Proposition 3.10 (magic formula). *For an $(2,0)$ -tensor T , we have*

$$X(\text{tr}_g T) = g(g, \nabla_X T).$$

Remark 3.11 (local form). The magic formula can be locally written as

$$\nabla_i (g^{jk} T_{jk}) = g^{jk} (\nabla_i T_{jk}),$$

that is, g^{jk} can "pass through" the covariant derivative, and that is why we call it "magic formula".

4 Geodesic and normal coordinate

In this section we always assume (M, g) is a Riemannian manifold equipped with Levi-Civita connection ∇ .

4.1 Geodesic

Definition 4.1. A smooth curve $\gamma: (-\varepsilon, \varepsilon) \rightarrow M$ is called a *geodesic*, if for each local coordinate $\{x^i\}$, it satisfies

$$\frac{d^2\gamma^k}{dt^2} + \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt} \Gamma_{ij}^k \circ \gamma = 0,$$

where $\gamma^i = x^i \circ \gamma$.

Theorem 4.2. For any $p \in M, v \in T_p M$, there exists $\varepsilon > 0$ and a geodesic $\gamma: (-\varepsilon, \varepsilon) \rightarrow M$ such that

$$\gamma(0) = p,$$

$$\gamma'(0) = v.$$

Moreover, any two such geodesics agree on their common domain.

Proof. It follows from standard results about existence and uniqueness in ODEs' theory. $\square$

Remark 4.3. Note that standard result in ODEs' theory only guarantees the short time existence of geodesic. If we use I to denote the maximal interval such that γ can be defined on it, then in general $I \neq \mathbb{R}$.

Notation 4.4. For $v \in T_p M$, γ_v denotes the unique geodesic such that $\gamma(0) = p$ and $\gamma'(0) = v$.

Lemma 4.5. For each $p \in M, v \in T_p M$ and $c, t \in \mathbb{R}$, one has

$$\gamma_{cv}(t) = \gamma_v(ct),$$

whenever either side is defined.

Proof. It is clear by uniqueness. $\square$

Definition 4.6. For any $p \in M$, V_p is a subspace of $T_p M$ defined by

$$V_p := \{v \in T_p M \mid \gamma_v(1) \text{ is defined}\}.$$

Remark 4.7. From Lemma 4.5, we know that $v \in V_p$ if $|v| < \varepsilon$ for sufficiently small $\varepsilon > 0$.

4.2 Normal coordinate

4.2.1 Exponential map and geodesical ball

Definition 4.8. For $p \in M$, the *exponential map* at point p is the map

$$\begin{aligned}\exp_p: V_p &\rightarrow M \\ v &\mapsto \gamma_v(1).\end{aligned}$$

Theorem 4.9. The exponential map $\exp_p$ is a local diffeomorphism at $0 \in T_p M$, that is, there exists a neighborhood V of $0 \in T_p M$ which is mapped diffeomorphically to a neighborhood U of $p \in M$.

Proof. Note that

$$(\mathrm{d}\exp_p)_0: T_0(T_p M) \rightarrow T_p M$$

and if we identify $T_0(T_p M)$ with $T_p M$ by using $0 + tv \mapsto v$, then $(\mathrm{d}\exp_p)_0$ is a linear map from $T_p M$ to itself. By inverse function theorem, it suffices to check $(\mathrm{d}\exp_p)_0$ is an isomorphism. For all $v \in T_p M$, a direct computation shows

$$\begin{aligned}(\mathrm{d}\exp_p)_0(v) &\stackrel{(1)}{=} \left. \frac{d}{dt} \right|_{t=0} \exp_p(0 + tv) \\ &= \left. \frac{d}{dt} \right|_{t=0} \gamma_{tv}(1) \\ &= \left. \frac{d}{dt} \right|_{t=0} \gamma_v(t) \\ &= \gamma'_v(0) \\ &= v,\end{aligned}$$

where (1) holds from identification $T_0(T_p M) \cong T_p M$. □

Definition 4.10. For $p \in M$, if $B(0, \delta)$ is a ball in $T_p M$ such that $\exp_p$ is a diffeomorphism on $B(0, \delta)$, then $B(p, \delta) := \exp_p(B(0, \delta))$ is called a *geodesic ball* centered at p .

Theorem 4.11. For each $p \in M$, there exists a local coordinate centered at p , called *normal coordinate*, such that we have the following expressions:

- (1) $g_{ij}(0) = \delta_{ij}$;
- (2) $\Gamma_{ij}^k(0) = 0$.

Proof. Let $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ be an orthonormal basis of $T_p M$ with respect to inner product $g(p)$, and fix the following linear isomorphism

$$\begin{aligned}\Phi: T_p M &\rightarrow \mathbb{R}^n \\ v^i \frac{\partial}{\partial x^i} \Big|_p &\mapsto (v^1, \dots, v^n).\end{aligned}$$

By Theorem 4.9 there exists a neighborhood U of p which is mapped by $\Phi \circ \exp_p^{-1}$ diffeomorphically onto a neighborhood V of $0 \in \mathbb{R}^n$, so $\Phi \circ \exp_p^{-1}: U \rightarrow V$ gives a local coordinate centered at p . This is called the normal coordinate. Given a local coordinate $\varphi: U \rightarrow V$ centered at $p \in M$, we have the following diagram

$$\begin{array}{ccc} T_p M \times T_p M & \longrightarrow & \mathbb{R} \\ (\mathrm{d}\varphi)_p \times (\mathrm{d}\varphi)_p \downarrow & & \downarrow \\ \mathbb{R}^n \times \mathbb{R}^n & \longrightarrow & \mathbb{R}, \end{array}$$

and we can use this diagram to express Riemannian metric and Christoffel symbols as a function defined on $V \subseteq \mathbb{R}^n$.

For (1). Let e_i denotes $(0, \dots, \underbrace{1}_{i\text{-th}}, \dots, 0) \in \mathbb{R}^n$. Then

$$\begin{aligned} g_{ij}(0) &:= \left\langle (\mathrm{d}\exp_p \circ \Phi^{-1})_0 e_i, (\mathrm{d}\exp_p \circ \Phi^{-1})_0 e_j \right\rangle_p \\ &\stackrel{(1)}{=} \left\langle (\mathrm{d}\exp_p)_0 \frac{\partial}{\partial x^i} \Big|_p, (\mathrm{d}\exp_p)_0 \frac{\partial}{\partial x^j} \Big|_p \right\rangle_p \\ &\stackrel{(2)}{=} \left\langle \frac{\partial}{\partial x^i} \Big|_p, \frac{\partial}{\partial x^j} \Big|_p \right\rangle_p \\ &\stackrel{(3)}{=} \delta_{ij}, \end{aligned}$$

where

1. (1) holds from Φ is a linear map, thus $\mathrm{d}\Phi^{-1} = \Phi^{-1}$.
2. (2) holds from Theorem 4.9.
3. (3) holds from $\{\frac{\partial}{\partial x^i} \Big|_p\}$ is an orthonormal basis with respect to $g(p)$.

For (2). For $v = (v^1, \dots, v^n) \in \mathbb{R}^n$, by definition $\gamma(t) = \exp_p(t\Phi^{-1}(v))$ is the geodesic with $\gamma(0) = p$ and $\gamma'(0) = \Phi^{-1}(v)$. In normal coordinate (x^i, U, p) , one has $\Phi \circ \exp_p^{-1}(\gamma(t)) = (tv^1, \dots, tv^n)$. Since the Riemannian metric on V is given by the one on U , that is, $\Phi \circ \exp_p^{-1}$ is an isometry, we have $\Phi \circ \exp_p^{-1}(\gamma(t))$ is a geodesic, and the geodesic equation is given by

$$\Gamma_{ij}^k(tv)v^i v^j = 0.$$

By evaluating this expression at $t = 0$ one has $\Gamma_{ij}^k(0)v^i v^j = 0$ for arbitrary index k , and take $v = e_i + e_j$ to conclude $\Gamma_{ij}^k(0) = 0$ for all i, j, k . $\square$

Remark 4.12. Tensor computation is one of the hallmarks of Riemannian geometry, but the normal coordinate can help you to simplify it. The idea is, if we want to prove an identity of tensors, it suffices to choose a coordinate and check it pointwisely. Thus, the normal coordinate is a good

choice, since by Theorem 4.11, one has the Riemannian metric is standard metric, and Christoffel symbols vanish.

Corollary 4.13. *In the normal coordinate, we have the following identities*

$$\frac{\partial \Gamma_{ij}^k}{\partial x^\ell}(0) + \frac{\partial \Gamma_{\ell i}^k}{\partial x^j}(0) + \frac{\partial \Gamma_{j\ell}^k}{\partial x^i}(0) = 0.$$

Proof. As shown in the proof of Theorem 4.11, in normal coordinate we have

$$0 = \Gamma_{ij}^k(tv)v^i v^j.$$

Then by taking differential with respect to x^ℓ it gives

$$0 = \frac{\partial \Gamma_{ij}^k}{\partial x^\ell}(tx)tx^i x^j,$$

which implies

$$\sum_{\sigma \in S_3} \frac{\partial \Gamma_{\sigma(i)\sigma(j)}^k}{\partial x^{\sigma(\ell)}}(0) = 0.$$

Then the desired result holds from the symmetry of Christoffel symbol in term i, j . □

Corollary 4.14. *In normal coordinate (x^i, U, p) , the Taylor expansion of g_{ij} around zero is*

$$g_{ij}(x) = \delta_{ij} + O(|x|^2).$$

Proof. It follows from

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^\ell(0)g_{\ell j}(0) + \Gamma_{kj}^\ell(0)g_{i\ell}(0) = 0.$$

□

4.3 Hopf-Rinow's Theorem

In this section we will figure out when the exponential map is defined on the whole space $T_p M$.

Definition 4.15. A Riemannian manifold M is *geodesically complete* if for all $p \in M$, the exponential map $\exp_p$ is defined on the whole $T_p M$.

At this stage it is convenient to introduce a distance function on a Riemannian manifold M which is not necessarily geodesic complete as follows: For $p, q \in M$, consider all the piecewise smooth curves joining p and q . Since M is connected, such curves always exist (cover a continuous curve joining p and q by a finite number of coordinates neighborhood and replace each piece contained in a coordinate neighborhood by a smooth one).

Definition 4.16. Let (M, g) be a Riemannian manifold, $p, q \in M$, the *distance* between p and q is defined by the infimum of the lengths of all piecewise smooth curves joining p and q , denoted by $\text{dist}(p, q)$.

Proposition 4.17 ([dC92, Proposition 2.6]). *The topology induced by distance function on M coincides with the original topology on M .*

Theorem 4.18 (Hopf-Rinow). *Let (M, g) be a Riemannian manifold and $p \in M$. The following statements are equivalent.*

- (1) *M is geodesically complete.*
- (2) *The closed and bounded sets of M are compact.*
- (3) *M is complete as a topological space.*

In addition, any of statements above implies that for any $p, q \in M$, there exists a geodesic joining p and q with length $\text{dist}(p, q)$.

Proof. See [dC92, Theorem 2.8]. □

Remark 4.19. Note that (2) is equivalent to (3) is a basic fact in general topology.

Definition 4.20. A Riemannian manifold is *complete*, if it is geodesically complete.

Corollary 4.21. *If M is compact, then it is complete.*

5 Riemannian Curvature

5.1 Curvature form and curvature tensor

5.1.1 Curvature form

Let (M, g) be a Riemannian manifold with connection ∇ of a vector bundle E over M . Now we're going to extend the connection to something called the exterior derivative defined on sections of vector bundle valued k -forms as follows

$$\begin{aligned} d^\nabla: C^\infty(M, \Omega_M^k \otimes E) &\rightarrow C^\infty(M, \Omega_M^{k+1} \otimes E) \\ \omega \otimes e &\mapsto d\omega \otimes e + (-1)^k \omega \wedge \nabla e. \end{aligned}$$

Suppose $\{e_\alpha\}$ is a local frame of E . Then

$$\begin{aligned}
(d^\nabla)^2(s^\alpha e_\alpha) &= d^\nabla(ds^\alpha \otimes e_\alpha + s^\alpha \omega_\alpha^\beta \otimes e_\beta) \\
&= -ds^\alpha \wedge \omega_\alpha^\beta \otimes e_\beta + d(s^\alpha \omega_\alpha^\beta) \otimes e_\beta - s^\alpha \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= s^\alpha (d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta. \\
(d^\nabla)^2(e_\alpha) &= d^\nabla(\omega_\alpha^\beta \otimes e_\beta) \\
&= d\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \nabla e_\beta \\
&= d\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= (d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta.
\end{aligned}$$

This shows $(d^\nabla)^2$ is a global section of $\Omega_M^2 \otimes \text{End } E$, that is, it is compatible with change of basis. Indeed, for two local frames $e, \tilde{e}$ such that $\tilde{e} = ge$, we have

$$\begin{aligned}
g(d^\nabla)^2 e &= (d^\nabla)^2 ge \\
&= (d^\nabla)^2 \tilde{e} \\
&= (d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega}) \tilde{e} \\
&= (d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega}) ge,
\end{aligned}$$

which implies

$$g^{-1}(d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega})g = d\omega - \omega \wedge \omega.$$

In general case, for $s \in C^\infty(M, \Omega_M^k \otimes E)$, locally written as $s = s^\alpha e_\alpha$, where s^α is a k -form, a direct computation also shows

$$(d^\nabla)^2(s^\alpha e_\alpha) = s^\alpha \wedge (d^\nabla)^2(e_\alpha).$$

Definition 5.1. Let E be a vector bundle over M equipped with connection ∇ . Then there exists a section $\Omega \in C^\infty(M, \Omega_M^2 \otimes \text{End } E)$, called *curvature form*, such that

$$(d^\nabla)^2 s = \Omega \wedge s$$

for all $s \in C^\infty(X, \Omega_M^k \otimes E)$.

Remark 5.2 (local form). Suppose $\{e_\alpha\}$ is a local frame of E and write $\Omega = \Omega_\alpha^\beta e_\beta \otimes e^\alpha$, where

$$\Omega_\alpha^\beta = \Omega_{ij\alpha}^\beta dx^i \wedge dx^j.$$

Then $\Omega = d\omega - \omega \wedge \omega$ shows

$$\begin{aligned}
\Omega_{ij\alpha}^\beta dx^i \wedge dx^j &= \Omega_\alpha^\beta \\
&= d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta \\
&= d(\Gamma_{i\alpha}^\beta dx^i) - (\Gamma_{i\alpha}^\gamma dx^i) \wedge (\Gamma_{j\gamma}^\beta dx^j) \\
&= (-\partial_j \Gamma_{i\alpha}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \wedge dx^j.
\end{aligned}$$

In other words, we have $\Omega_{ij\alpha}^\beta = -(\partial_j \Gamma_{i\alpha}^\beta + \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta)$.

Remark 5.3. In physicists' language, a connection is a "field", the curvature is the "strength" of the field, and choosing a local frame is called "fixing the gauge". The reason for these names comes from H. Weyl's work on rewriting Maxwell's equations.

5.1.2 Curvature tensor

Another standard way to define curvature is to define it as follows:

$$R: C^\infty(M, TM) \times C^\infty(M, TM) \times C^\infty(M, E) \rightarrow C^\infty(M, E)$$

$$(X, Y, s) \mapsto R(X, Y)s := \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

It is easy to check R defines a section of $\bigotimes^2 T^*M \otimes \text{End } E$

Remark 5.4 (local form). Let $\{e_\alpha\}$ be a local frame of E with dual frame $\{e^\alpha\}$. Then locally

$$R = R_{ij\alpha}^\beta dx^i \otimes dx^j \otimes e^\alpha \otimes e_\beta.$$

A direct computation shows that

$$\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} e_\alpha = \nabla_{\frac{\partial}{\partial x^i}} (\Gamma_{j\alpha}^\beta e_\beta) = \partial_i \Gamma_{j\alpha}^\beta e_\beta + \Gamma_{j\alpha}^\beta \Gamma_{i\beta}^\gamma e_\gamma = (\partial_i \Gamma_{j\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta) e_\beta.$$

Thus

$$R_\alpha^\beta = (\partial_i \Gamma_{j\alpha}^\beta - \partial_j \Gamma_{i\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \otimes dx^j.$$

In particular, when the curvature form is regarded as a tensor, it coincides with curvature tensor since

$$\begin{aligned} \Omega_{ij\alpha}^\beta dx^i \wedge dx^j &= -(\partial_j \Gamma_{i\alpha}^\beta + \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta)(dx^i \otimes dx^j - dx^j \otimes dx^i) \\ &= (\partial_i \Gamma_{j\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta - \partial_j \Gamma_{i\alpha}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \otimes dx^j \\ &= R_{ij\alpha}^\beta dx^i \otimes dx^j. \end{aligned}$$

5.1.3 Riemannian curvature

Let (M, g) be a Riemannian manifold and ∇ be the Levi-Civita connection on TM . Then the curvature R is a $(3, 1)$ -tensor, which is locally written as

$$R_{ijk}^r dx^i \otimes dx^j \otimes dx^k \otimes \frac{\partial}{\partial x^r}.$$

Now let's give an explicit expression of $R_{ijkl} = g_{lr} R_{ijk}^r$. By definition we have

$$\begin{aligned}
R_{ijkl} &= R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell}\right) \\
&= \left\langle \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k} - \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle \\
&= \partial_i \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \right\rangle \\
&\quad - \left(\partial_j \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \right\rangle \right) \\
&= \underbrace{\partial_i \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \partial_j \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle}_{\text{part I}} \\
&\quad + \underbrace{\left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \right\rangle}_{\text{part II}}.
\end{aligned}$$

For part II, we have

$$g_{rs}(\Gamma_{ik}^r \Gamma_{j\ell}^s - \Gamma_{jk}^r \Gamma_{i\ell}^s).$$

For part I, note that

$$\begin{aligned}
\partial_i(\Gamma_{jk}^r g_{r\ell}) &= \partial_i \left(\frac{1}{2} g^{rs} (\partial_j g_{ks} + \partial_k g_{js} - \partial_s g_{jk}) g_{r\ell} \right) \\
&= \partial_i \left(\frac{1}{2} \delta_\ell^s (\partial_j g_{ks} + \partial_k g_{js} - \partial_s g_{jk}) \right) \\
&= \frac{1}{2} \partial_i (\partial_j g_{k\ell} + \partial_k g_{j\ell} - \partial_\ell g_{jk}).
\end{aligned}$$

Thus we have part I is

$$\partial_i(\Gamma_{jk}^r g_{r\ell}) - \partial_j(\Gamma_{ik}^r g_{r\ell}) = \frac{1}{2} (\partial_i \partial_k g_{j\ell} + \partial_j \partial_\ell g_{ik} - \partial_i \partial_\ell g_{jk} - \partial_j \partial_k g_{i\ell}).$$

So we have an explicit expression for R_{ijkl}

$$R_{ijkl} = \frac{1}{2} (\partial_i \partial_k g_{j\ell} + \partial_j \partial_\ell g_{ik} - \partial_i \partial_\ell g_{jk} - \partial_j \partial_k g_{i\ell}) + g_{rs}(\Gamma_{ik}^r \Gamma_{j\ell}^s - \Gamma_{jk}^r \Gamma_{i\ell}^s).$$

From this expression, we can see in general curvature depends on second order partial derivatives of metric. Furthermore, there are some skew symmetries and symmetries of R_{ijkl} .

1. $R_{ijkl} = -R_{jikl}$;
2. $R_{ijkl} = -R_{ijlk}$;
3. $R_{ijkl} = R_{klij}$.

Proposition 5.5. *In normal coordinate, we have*

$$g_{ij} = \delta_{ij} - \frac{1}{3}R_{ik\ell j}(0)x^k x^\ell + O(|x|^3).$$

Proof. Equivalently, it suffices to prove that

$$\frac{\partial^2 g_{ij}}{\partial x^\ell \partial x^k}(0) = -R_{ik\ell j}(0).$$

Recall that the compatibility of connection and Riemannian metric gives

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^m g_{mj} + \Gamma_{kj}^m g_{mi}.$$

Then by taking partial differential with respect to x^ℓ , evaluating at $x = 0$, it gives

$$\frac{\partial^2 g_{ij}}{\partial x^\ell \partial x^k}(0) = \frac{\partial \Gamma_{ki}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{kj}^m}{\partial x^\ell}(0)g_{mi}(0),$$

since the Christoffel symbols vanish at $x = 0$.

On the other hand, we have

$$\begin{aligned} R_{ijk\ell}(0) &= \left(\frac{\partial \Gamma_{jk}^m}{\partial x^i}(0) - \frac{\partial \Gamma_{ik}^m}{\partial x^j}(0) \right) g_{m\ell}(0) \\ &\stackrel{(1)}{=} - \left(\frac{\partial \Gamma_{ij}^m}{\partial x^k}(0) + \frac{\partial \Gamma_{ki}^m}{\partial x^j}(0) + \frac{\partial \Gamma_{ik}^m}{\partial x^j}(0) \right) g_{m\ell}(0) \\ &= - \left(\frac{\partial \Gamma_{ij}^m}{\partial x^k}(0) + 2 \frac{\partial \Gamma_{ki}^m}{\partial x^j}(0) \right) g_{m\ell}(0), \end{aligned}$$

where (1) holds from Lemma 4.13. Thus we have

$$\begin{aligned} 2R_{ikj\ell}(0)x^k x^\ell &= - \left(R_{ik\ell j}(0) + R_{j\ell ki}(0) \right) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0) + 2 \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0) \right) g_{mj}(0)x^k x^\ell + \left(\frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0) + 2 \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0) \right) g_{mi}(0) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) \right) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) \right) \\ &= 3 \frac{\partial^2 g_{ij}}{\partial x^k \partial x^\ell}(0). \end{aligned}$$

This completes the proof. □

Corollary 5.6. *In Riemannian normal coordinate we have*

1. $g^{ij} = \delta_{ij} + \frac{1}{3}R_{ik\ell j}(0)x^k x^\ell + O(|x|^3)$.
2. $\det(g_{ij}) = 1 - \frac{1}{3}R_{k\ell}x^k x^\ell + O(|x|^3)$.
3. $\sqrt{\det(g_{ij})} = 1 - \frac{1}{6}R_{k\ell}x^k x^\ell + O(|x|^3)$.

Theorem 5.7 (Ricci identity). *Let (M, g) be a Riemannian manifold and T be a (r, s) -tensor, locally written as $T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}$. Then*

$$\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} - \nabla_{i,k}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} = \sum_{\ell=1}^s R_{kiq}^{j_\ell} T_{i_1 \dots i_r}^{j_1 \dots j_{\ell-1} q j_{\ell+1} \dots j_s} - \sum_{m=1}^r R_{kii_m}^q T_{i_1 \dots i_{m-1} q i_{m+1} \dots i_r}^{j_1 \dots j_s}.$$

5.2 Bianchi identities

5.2.1 First Bianchi

Proposition 5.8 (first Bianchi identity). *Let (M, g) be a Riemannian manifold and R be the curvature tensor of Levi-Civita connection. Let X, Y, Z, W be vector fields on M . Then*

$$R(X, Y, Z, W) + R(Y, Z, X, W) + R(Z, X, Y, W) = 0.$$

Proof. It suffices to prove this identity locally. Suppose $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}, Z = \frac{\partial}{\partial x^k}, W = \frac{\partial}{\partial x^\ell}$. Then locally the first Bianchi identity is stated as

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0.$$

In other words, it suffices to prove

$$R_{ijk}^r + R_{jki}^r + R_{kij}^r = 0.$$

Since we have

$$R_{ijk}^r = \underbrace{\partial_i \Gamma_{jk}^r - \partial_j \Gamma_{ik}^r}_{\text{part I}} + \underbrace{\Gamma_{jk}^s \Gamma_{is}^r - \Gamma_{ik}^s \Gamma_{js}^r}_{\text{part II}}.$$

(1) For the first part, if we permuting i, j, k , we have

$$\partial_i \Gamma_{jk}^r - \partial_j \Gamma_{ik}^r + \partial_j \Gamma_{ki}^r - \partial_k \Gamma_{ji}^r + \partial_k \Gamma_{ij}^r - \partial_i \Gamma_{kj}^r = 0,$$

since $\Gamma_{ij}^r = \Gamma_{ji}^r$ by torsion-free.

(2) For the second part, if we permuting i, j, k , we have

$$\Gamma_{jk}^s \Gamma_{is}^r - \Gamma_{ik}^s \Gamma_{js}^r + \Gamma_{ki}^s \Gamma_{js}^r - \Gamma_{ji}^s \Gamma_{ks}^r + \Gamma_{ij}^s \Gamma_{ks}^r - \Gamma_{kj}^s \Gamma_{is}^r = 0$$

by the same reason.

This completes the proof of first Bianchi identity. □

Remark 5.9. If we consider connection on arbitrary vector bundle E , there is no first Bianchi identity, since e_α is just a section of E , not a section of TM , and thus $R(e_\alpha, \cdot)$ or $R(\cdot, e_\alpha)$ is nonsense.

5.2.2 Second Bianchi

Proposition 5.10 (second Bianchi identity). *Let (M, g) be a Riemannian manifold and R be the curvature tensor of Levi-Civita connection. Let X, Y, Z, W be vector fields on M . Then*

$$\nabla_X R(Y, Z, W, T) + \nabla_Y R(Z, X, W, T) + \nabla_Z R(X, Y, W, T) = 0,$$

where $X, Y, Z, W, T \in C^\infty(M, TM)$.

The first approach. Without lose of generality, we may choose the normal coordinate $\{x^i\}$. Let $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}, Z = \frac{\partial}{\partial x^k}, W = \frac{\partial}{\partial x^\ell}, T = \frac{\partial}{\partial x^m}$. Then

$$\begin{aligned} \nabla_X R(Y, Z, W, T) &\stackrel{(1)}{=} \nabla_{\frac{\partial}{\partial x^i}} g \left(\nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell}, \frac{\partial}{\partial x^m} \right) \\ &\stackrel{(2)}{=} g \left(\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell}, \frac{\partial}{\partial x^m} \right), \end{aligned}$$

where

1. (1) holds since $[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}] = 0$;
2. (2) holds from the compatibility of connection and metric, and Christoffel symbols vanish at p in the normal coordinate.

By permuting the index i, j, k , we have

$$\begin{aligned} &\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \\ &+ \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} \\ &+ \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \\ &= R \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} + R \left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} + R \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i} \right) \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \\ &= 0, \end{aligned}$$

where the last equality holds from $\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} = 0$ under the normal coordinate. This completes proof of the second Bianchi identity. $\square$

The second approach. Let Ω be the curvature form of ∇ . Then we claim that

$$d\Omega + \Omega \wedge \omega - \omega \wedge \Omega = 0.$$

Indeed,

$$\begin{aligned} d\Omega + \Omega \wedge \omega - \omega \wedge \Omega &= d(d\omega - \omega \wedge \omega) + (d\omega - \omega \wedge \omega) \wedge \omega - \omega \wedge (d\omega - \omega \wedge \omega) \\ &= d^2\omega - d\omega \wedge \omega + \omega \wedge d\omega + d\omega \wedge \omega - \omega \wedge \omega \wedge \omega - \omega \wedge d\omega + \omega \wedge \omega \wedge \omega \\ &= 0. \end{aligned}$$

If we write this in the local form, it gives

$$(\partial_k \Omega_{ij\alpha}^\beta + \Omega_{ij\alpha}^\gamma \Gamma_{k\gamma}^\beta - \Gamma_{k\alpha}^\gamma \Omega_{ij\gamma}^\beta) dx^k \wedge dx^i \wedge dx^j = 0.$$

By using the relations between curvature tensor and curvature form, this also completes the proof. $\square$

5.3 Other curvatures

5.3.1 Sectional curvature

In this section we're going to introduce sectional curvature, which is used to characterize a two dimensional subspace of tangent space.

Fix $p \in M$ and let x, y are two linearly independent tangent vectors in $T_p M$, then sectional curvature for these two vectors are defined as

$$K_p(x, y) = \frac{R_p(x, y, y, x)}{g_p(x, x)g_p(y, y) - g_p(x, y)^2}.$$

In order to show it is a invariant defined for a two dimensional subspace, we need to check if $\text{span}_{\mathbb{R}}\{x, y\} = \text{span}_{\mathbb{R}}\{z, w\}$, then

$$K_p(x, y) = K_p(z, w).$$

Indeed, if we write

$$\begin{cases} z = ax + by \\ w = cx + dy. \end{cases}$$

Then by symmetry and skew symmetry properties of R we have

$$\begin{aligned} R_p(z, w, w, z) &= R_p(ax + by, cx + dy, cx + dy, ax + by) \\ &= R_p(ax, dy, dy, ax) + R_p(ax, dy, cx, by) + R_p(by, cx, dy, ax) + R_p(by, cx, cx, by) \\ &= a^2 d^2 R_p(x, y, y, x) - abcd R_p(x, y, y, x) - abcd R_p(x, y, y, x) + b^2 c^2 R_p(x, y, y, x) \\ &= (ad - bc)^2 R_p(x, y, y, x). \end{aligned}$$

And by the same computations we have

$$g_p(z, z)g_p(w, w) - g_p(z, w)^2 = (ad - bc)^2 \{g_p(x, x)g_p(y, y) - g_p(x, y)^2\}$$

Thus

$$K_p(x, y) = K_p(z, w).$$

Definition 5.11. The *sectional curvature* $K_p(\sigma)$ for two dimensional subspace $\sigma \subseteq T_p M$ is defined as

$$K_p(\sigma) := K_p(x, y),$$

where $\{x, y\}$ is a basis of σ .

Definition 5.12. A Riemannian manifold (M, g) is called *isotropic*, if for each point $p \in M$, the sectional curvature $K_p(\sigma)$ is independent of σ .

Definition 5.13. A Riemannian manifold (M, g) has *constant sectional curvature*, if $K_p(\sigma)$ is constant for arbitrary $\sigma \subset T_p M, p \in M$.

Remark 5.14. By definition, if a Riemannian manifold has constant sectional curvature, then it must be isotropic; Conversely, if a Riemannian manifold is isotropic and its dimension is ≥ 3 , then it has constant sectional curvature, see Corollary 5.33

Lemma 5.15.

$$-6R(X, Y, Z, W) = \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} \{R(X + sZ, Y + tW, Y + tW, X + sZ) - R(X + sW, Y + tZ, Y + tZ, X + sW)\},$$

where X, Y, Z, W are vector fields.

Proof. It suffices to compute coefficients of st of $R(X + sZ, Y + tW, Y + tW, X + sZ)$ and exchange Z with W to obtain coefficients of st of $R(X + sW, Y + tZ, Y + tZ, X + sW)$.

it is easy to see coefficients of st of $R(X + sZ, Y + tW, Y + tW, X + sZ)$ is

$$R(Z, W, Y, X) + R(Z, Y, W, X) + R(X, W, Y, Z) + R(X, Y, W, Z),$$

so coefficients of st of $R(X + sW, Y + tZ, Y + tZ, X + sW)$ is

$$R(W, Z, Y, X) + R(W, Y, Z, X) + R(X, Z, Y, W) + R(X, Y, Z, W).$$

Thus the right hand of our desired identity is

$$-4R(X, Y, Z, W) - (R(Y, Z, W, X) + R(W, Y, Z, X)) - (R(W, X, Y, Z) + R(W, Y, Z, X)).$$

By first Bianchi identity we have

$$\begin{aligned} R(Y, Z, W, X) + R(W, Y, Z, X) &= R(Y, Z, W, X) + R(Z, X, W, Y) \\ &= R(X, Y, Z, W) \\ R(W, X, Y, Z) + R(W, Y, Z, X) &= R(Y, Z, W, X) + R(Z, X, W, Y) \\ &= R(X, Y, Z, W). \end{aligned}$$

This completes the proof. □

Notation 5.16. For convenience, we use $R_0(X, Y, Z, W)$ to denote

$$R_0(X, Y, Z, W) = g(X, W)g(Y, Z) - g(X, Z)g(Y, W),$$

where X, Y, Z, W are vector fields. Then we can write sectional curvature as

$$K_p(\sigma) = \frac{R(x, y, y, x)}{R_0(x, y, y, x)}.$$

where $\sigma \subset T_p M$ is spanned by x, y .

Proposition 5.17. *A Riemannian manifold has constant sectional curvature K_p at point $p \in M$ if and only if the Riemannian curvature tensor $R_p = K_p R_{0,p}$, where K_p is a constant (depending on p).*

Proof. If $R_p = K_p R_{0,p}$, then for a arbitrary x, y , we have

$$K_p(x, y) = \frac{R_p(x, y, y, x)}{R_{0,p}(x, y, y, x)} = K_p.$$

Conversely, if $K(\sigma)$ is constant at point $p \in M$, that is for arbitrary x, y we have

$$R_p(x, y, y, x) = K_p R_{0,p}(x, y, y, x).$$

If we denote

$$F(s, t) = R_p(x + sz, y + tw, y + tw, x + sz) - R_p(x + sw, y + tz, y + tz, x + sw)$$

$$F_0(s, t) = R_{0,p}(x + sz, y + tw, y + tw, x + sz) - R_{0,p}(x + sw, y + tz, y + tz, x + sw),$$

we still have $F(s, t) = K_p F_0(s, t)$. By Lemma 5.15, we have

$$R_p(x, y, z, w) = -\frac{1}{6} \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} F(s, t),$$

and it is easy to see

$$R_{0,p}(x, y, z, w) = -\frac{1}{6} \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} F_0(s, t).$$

This completes the proof. □

Corollary 5.18. *A Riemannian manifold is isotropic if and only if $R = KR_0$, where K is a smooth function.*

Corollary 5.19. *A Riemannian manifold has constant sectional curvature K if and only if $R = KR_0$, where K is a constant.*

Remark 5.20. An important corollary is that curvature tensor of Riemannian manifold with constant sectional curvature K is quite simple, since

$$R_{ijkl} = K(g_{il}g_{jk} - g_{ik}g_{jl}),$$

that is, curvature is completely determined by zero order partial derivatives of metric, not the second order in general.

Remark 5.21. Let (M, g) be a Riemannian manifold of dimension two and $\{e_1, e_2\}$ be a basis of $T_p M$. Then

$$K_p = K_p(e_1, e_2) = \frac{R(e_1, e_2, e_2, e_1)}{|e_1|^2 |e_2|^2 - |g(e_1, e_2)|^2}$$

is called the *Gauss curvature* in the theory of surface.

5.3.2 Ricci curvature and scalar curvature

Definition 5.22. For a Riemannian manifold (M, g) , the *Ricci curvature* is defined to be

$$\text{Ric}(X, Y) := \text{tr}_g(Z \mapsto R(Z, X)Y),$$

where X, Y are vector fields.

Remark 5.23 (local form). The trace of above endomorphism is exactly R_{ijk}^i , and it can be written as

$$g^{i\ell} R_{ijk\ell}.$$

In other words, Ricci curvature tensor is the contracted tensor of curvature with respect to the first and fourth index.

Definition 5.24. For a point $p \in M$, and $x \in T_p M$, the *Ricci curvature in the direction x* is defined as

$$\text{Ric}_p(x) := \text{Ric}_p(x, x).$$

Remark 5.25. For $x \in T_p M$, we can write it as $x = x^i e_i$, where $\{e_1, \dots, e_n\}$ is a basis of $T_p M$, then

$$\text{Ric}_p(x) = R_{jk} x^j x^k.$$

Definition 5.26. For a Riemannian manifold (M, g) , the *scalar curvature S* at $p \in M$ is defined as

$$S(p) := \sum_{i=1}^n \text{Ric}_p(e_i),$$

where $\{e_1, \dots, e_n\}$ is an orthonormal basis of $T_p M$.

Remark 5.27 (local form). Locally we have

$$S = g^{jk} R_{jk}.$$

Proposition 5.28 (contracted Bianchi identity).

$$g^{jk} \nabla_k R_{ij} = \frac{1}{2} \nabla_i S,$$

where R_{ij} is Ricci curvature and S is scalar curvature.

Proof. Direct computation shows

$$\begin{aligned}
g^{jk}\nabla_k R_{ij} &= g^{jk}\nabla_k g^{pq} R_{pijq} \\
&= g^{jk} g^{pq} \nabla_k R_{pijq} \\
&= g^{jk} g^{pq} (-\nabla_p R_{ikjq} - \nabla_i R_{kpjq}) \\
&= -g^{pq} \nabla_p R_{iq} + \nabla_i S \\
&= -g^{jk} \nabla_k R_{ij} + \nabla_i S.
\end{aligned}$$

This completes the proof. □

Proposition 5.29. *The scalar curvature S at $p \in M$ is given by*

$$S(p) = \frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} \text{Ric}_p(x) d\mathbb{S}^{n-1},$$

where α_n is the volume of n -dimension unit ball in $\mathbb{R}^n$ and $d\mathbb{S}^{n-1}$ is the area elements in $\mathbb{S}^{n-1}$.

Proof. Choose an orthonormal basis $\{e_1, \dots, e_n\}$ in $T_p M$ and write $x = x^i e_i$, then

$$\text{Ric}_p(x) = \text{Ric}_p(x^i e_i) = (x^i)^2 \text{Ric}_p(e_i).$$

Since $|x| = 1$, then the vector $\mu = (x^1, \dots, x^n)$ is a unit normal vector on $\mathbb{S}^{n-1}$. Denoting $V = (x^1 \text{Ric}_p(e_1), \dots, x^n \text{Ric}_p(e_n))$, then Stokes theorem implies

$$\begin{aligned}
\frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} (x^i)^2 \text{Ric}_p(e_i) d\mathbb{S}^{n-1} &= \frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} \langle V, \mu \rangle d\mathbb{S}^{n-1} \\
&= \frac{1}{\alpha_n} \int_{B^n} \text{div } V d\mathbb{B}^n \\
&= \text{div } V \\
&= \sum_{i=1}^n \text{Ric}_p(e_i) \\
&= S(p),
\end{aligned}$$

where B^n is unit ball in $T_p M$ with $\partial B^n = \mathbb{S}^{n-1}$. □

Theorem 5.30. *Let (M, g) be a Riemannian manifold, then for all $p \in M$ and r sufficiently small, the volume of the geodesic ball $B(p, r)$ is*

$$\text{vol}(B(p, r)) = \alpha_n r^n \left(1 - \frac{S(p)}{6(n+2)} r^2 + O(r^3) \right),$$

where α_n is the volume of n -dimension unit ball in $\mathbb{R}^n$.

Proof. Note that we have

$$\sqrt{\det(g_{ij})} = \delta_{ij} - \frac{1}{6}R_{jk}(p)x^j x^k + O(|x|^3).$$

Directly computation shows

$$\begin{aligned} \text{Vol}(B(p, r)) &= \int_0^r \int_{\mathbb{S}^{n-1}(t)} \sqrt{\det g} dS dt \\ &= \int_0^r \int_{\mathbb{S}^{n-1}(t)} \left(1 - \frac{1}{6}\text{Ric}_p(x) + O(|x|^3)\right) dS dt \\ &= \alpha_n r^n - \frac{\alpha_n}{6} \int_0^r t^{n+1} dt + O(r^{n+3}) \\ &= \alpha_n r^n - \frac{\alpha_n S(p) r^{n+2}}{6(n+2)} + O(r^{n+3}) \\ &= \alpha_n r^n \left(1 - \frac{S(p)}{6(n+2)} r^2 + O(r^3)\right), \end{aligned}$$

where we use the fact $\alpha_n = \omega_{n-1}/n$. □

5.4 Basic models

5.4.1 Einstein manifold

Definition 5.31. A Riemannian manifold (M, g) is called *Einstein manifold*, if its Ricci curvature satisfies $R_{ij} = \lambda g_{ij}$ for some $\lambda \in \mathbb{R}$.

Lemma 5.32 (Schur's lemma). *Let (M, g) be a Riemannian manifold with $\dim M \geq 3$, suppose $R_{ij} = f g_{ij}$, where f is a smooth function, then (M, g) is an Einstein manifold.*

Proof. If $R_{ij} = f g_{ij}$, then contracted Bianchi identity shows

$$\frac{n}{2} \nabla_i f = g^{jk} \nabla_k f g_{ij} = \nabla_i f$$

for arbitrary i , which implies f is constant, since $n \geq 3$. □

Corollary 5.33. *For a Riemannian manifold (M, g) with $\dim M \geq 3$, it is isotropic if and only if it has constant sectional curvature.*

Proof. By Remark 5.14, it suffices to show if M is isotropic then it has constant sectional curvature. If M is isotropic, then there exists a smooth function K such that

$$R_{ijk\ell} = K(g_{i\ell}g_{jk} - g_{ik}g_{j\ell}).$$

Consider its Ricci curvature, that is

$$R_{jk} = (n-1)K g_{jk}.$$

Then Schur's lemma implies $(n-1)K$ is constant, that is, K is constant. □

Proposition 5.34. *Let (M, g) be an Einstein 3-manifold, then (M, g) has constant sectional curvature.*

Proof. For arbitrary point $p \in M$, without loss of generality we consider normal coordinate, that is $g_{ij} = \delta_{ij}$. Then

$$R_{11} = g^{ij} R_{i11j} = R_{2112} + R_{3113} = \lambda.$$

Similarly we have

$$R_{1221} + R_{3223} = \lambda, \quad R_{1331} + R_{2332} = \lambda.$$

Thus we can conclude

$$R_{1221} = R_{1331} = R_{2332} = \frac{\lambda}{2},$$

that is,

$$R_{ijkl} = \frac{\lambda}{2}(\delta_{il}\delta_{jk} - \delta_{ik}\delta_{jl}).$$

This shows (M, g) has constant sectional curvature $\lambda/2$. □

Remark 5.35. In fact, it is a special case of Ricci curvature controls curvature. For a n -dimensional Riemannian manifold, it is easy to see R_{jk} has $n(n+1)/2$ independent components. But for R_{ijkl} , this counting problem becomes a little bit complicated, it has

$$\frac{n^2(n^2 - 1)}{12}$$

independent components. Indeed, since R_{ijkl} is skew symmetric in ij and $k\ell$, this means that these pair of indices can take

$$m = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Also, R_{ijkl} is symmetric when you swap ij with $k\ell$, which means there would be

$$\frac{m(m+1)}{2} = \frac{n^4 - 2n^3 + 3n^2 - 2n}{8}$$

choices. However, these are not independent, since there is first Bianchi identity

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0,$$

and it provides

$$\binom{n}{4} = \frac{n^4 - 6n^3 + 11n^2 - 6n}{24}$$

relations between these components, thus the number of independent components of R_{ijkl} is

$$\frac{n^4 - 2n^3 + 3n^2 - 2n}{8} - \frac{n^4 - 6n^3 + 11n^2 - 6n}{24} = \frac{n^4 - n^2}{12} = \frac{n^2(n^2 - 1)}{12}.$$

Therefore curvature is fully determined by the Ricci curvature if and only if

$$\frac{n^2(n^2 - 1)}{12} \leq \frac{n(n+1)}{2},$$

or in other words, $n \leq 3$.

5.4.2 Lie group with invariant metric

Lemma 5.36. *Let G be a Lie group equipped with left-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for all left-invariant vector fields X, Y, Z ,*

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (\langle [X, Y], Z \rangle + \langle [X, Z], Y \rangle + \langle [Y, Z], X \rangle).$$

Proof. Recall that

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (Y \langle Z, X \rangle + Z \langle X, Y \rangle - X \langle Y, Z \rangle - \langle [Y, X], Z \rangle - \langle [Z, X], Y \rangle - \langle [Y, Z], X \rangle).$$

But $Y \langle Z, X \rangle = Z \langle X, Y \rangle = X \langle Y, Z \rangle = 0$ since both metric and X, Y, Z are left-invariant, that is,

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (\langle [X, Y], Z \rangle + \langle [X, Z], Y \rangle + \langle [Y, Z], X \rangle).$$

Now set $Y = Z$ to conclude. □

Proposition 5.37. *Let G be a Lie group equipped with bi-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for all left-invariant vector fields X, Y, Z ,*

$$\langle [X, Y], Z \rangle = \langle X, [Y, Z] \rangle.$$

Proof. Let y_t be the flow of Y , then

$$[X, Y] = \lim_{t \rightarrow 0} \frac{1}{t} ((dy_t)(X) - X).$$

On the other hand, since Y is left-invariant, that is $L_g \circ y_t = y_t \circ L_g$, giving

$$y_t(g) = y_t(L_g(e)) = L_g y_t(e) = g y_t(e) = R_{y_t(e)}(g).$$

Thus $dy_t = dR_{y_t(e)}$ and

$$[X, Y] = \lim_{t \rightarrow 0} \frac{1}{t} ((dR_{y_t(e)})(X) - X).$$

Note that the metric is bi-invariant, thus

$$\langle X, Z \rangle = \langle (dR_{y_t(e)}) \circ (dL_{y_t^{-1}(e)}) X, (dR_{y_t(e)}) \circ (dL_{y_t^{-1}(e)}) Z \rangle = \langle (dR_{y_t(e)}) X, (dR_{y_t(e)}) Z \rangle.$$

Differentiating the expression above with respect to t and setting $t = 0$ we conclude

$$0 = \langle [X, Y], Z \rangle + \langle X, [Z, Y] \rangle.$$

This completes the proof. □

Theorem 5.38. *Let G be a Lie group equipped with bi-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for every left-invariant vector field X on G , then $\nabla_X X = 0$.*

Proof. From Lemma 5.36, we have

$$\langle Y, \nabla_X X \rangle = \langle X, [Y, X] \rangle,$$

From Proposition 5.37, we have

$$\langle X, [Y, X] \rangle = \langle [X, Y], X \rangle = -\langle X, [Y, X] \rangle,$$

that is, $\langle Y, \nabla_X X \rangle = 0$ for arbitrary vector field Y , which implies $\nabla_X X = 0$. $\square$

Corollary 5.39. *The assumptions are as above. If X, Y are left-invariant vector fields, then $\nabla_X Y = \frac{1}{2}[X, Y]$.*

Proof. Note that

$$\begin{aligned} 0 &= \nabla_{X+Y}(X+Y) \\ &= \nabla_X Y + \nabla_Y X + \nabla_X X + \nabla_Y Y \\ &= \nabla_X Y + \nabla_Y X \\ &= 2\nabla_X Y - [X, Y]. \end{aligned}$$

Division by two finally yields

$$\nabla_X Y = \frac{1}{2}[X, Y].$$

$\square$

Corollary 5.40. *The assumptions are as above. If X, Y, Z are left-invariant vector fields, then $R(X, Y)Z = -\frac{1}{4}[[X, Y], Z]$.*

Proof. Directly from $\nabla_X Y = \frac{1}{2}[X, Y]$ and Jacobi's identity. $\square$

Corollary 5.41. *The assumptions are as above. If X, Y are left-invariant vector fields which are orthogonal, and σ is the plane generated by X and Y . Then*

$$K(\sigma) = \frac{1}{4}||[X, Y]||^2.$$

Proof.

$$\begin{aligned} K(\sigma) &= \frac{\langle R(X, Y)Y, X \rangle}{||X||^2 ||Y||^2 - \langle X, Y \rangle^2} \\ &= \frac{\langle -\frac{1}{4}[[X, Y], Y], X \rangle}{||X||^2 ||Y||^2} \\ &= -\frac{1}{4} \langle [[X, Y], Y], X \rangle \\ &= -\frac{1}{4} \langle [X, Y], [Y, X] \rangle \\ &= \frac{1}{4} ||[X, Y]||^2. \end{aligned}$$

$\square$

Remark 5.42. Therefore, sectional curvature of a Lie group with bi-invariant metric is non-negative. Furthermore, if the center of Lie algebra $\mathfrak{g}$ is trivial, then the sectional curvature is positive.

Theorem 5.43. *Let G be a Lie group equipped with bi-invariant metric, the geodesics on G are precisely the integral curves of left-invariant vector fields.*

Proof. Let $X \in \mathfrak{g}$ be a left-invariant vector field, and $\gamma : \mathbb{R} \rightarrow G$ its integral curve. Then

$$\nabla_{\frac{d}{dt}} \gamma_* \left(\frac{d}{dt} \right) = \nabla_{\gamma_* \left(\frac{d}{dt} \right)} \gamma_* \left(\frac{d}{dt} \right) = \nabla_X X = 0,$$

which implies integral curves of left-invariant vector fields are geodesics. Furthermore, since geodesics are unique, we have geodesics are precisely integral curves of left-invariant vector fields. $\square$

Corollary 5.44. *The exponential map for the Lie group coincides with the exponential map of the Levi-Civita connection with respect to bi-invariant metric.*

6 Hodge theory on Riemannian manifold

In this section, we always assume (M, g) is a compact oriented Riemannian n -manifold, as we need to consider integrations on M .

6.1 Inner product on differential forms

Before we introduce the Hodge theory on (M, g) , let's recall some basic facts about differential k -forms. For a k -form φ , locally it can be written as

$$\varphi = \sum_{1 \leq i_1 < \dots < i_k \leq n} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

where $\varphi_{i_1 \dots i_k} := \varphi\left(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}}\right)$ is a skew-symmetric function. Equivalently, we have

$$\varphi = \frac{1}{k!} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Notation 6.1. For convenience, we use $\varphi_I dx^I$ to denote the summation $\varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$.

Definition 6.2. Let φ, ψ be two k -forms. The *metric pairing* $\langle -, - \rangle$ on Ω_M^k is defined as

$$\langle \varphi, \psi \rangle := \frac{1}{k!} g(\varphi, \psi),$$

where g is induced metric on $\Omega_M^k \subseteq \bigotimes^k T^*M$.

Lemma 6.3. For $\varphi = \varphi_I dx^I, \psi = \psi_J dx^J$, we have

$$\langle \varphi, \psi \rangle = \varphi_I \psi_J g^{IJ},$$

where

$$g^{IJ} = \frac{1}{k!} g(dx^I, dx^J) = \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

Proof. It suffices to show

$$g(dx^I, dx^J) = k! \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

By definition one has

$$dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_{\sigma \in S_k} (-1)^{|\sigma|} dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}}.$$

Then

$$\begin{aligned} g(dx^I, dx^J) &= \sum_{\sigma, \tau} (-1)^{|\sigma|} (-1)^{|\tau|} g(dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}}, dx^{j_{\tau(1)}} \otimes \dots \otimes dx^{j_{\tau(k)}}) \\ &= \sum_{\sigma, \tau} (-1)^{|\sigma|} (-1)^{|\tau|} g^{i_{\sigma(1)} j_{\tau(1)}} \dots g^{i_{\sigma(k)} j_{\tau(k)}} \\ &= \sum_{\sigma, \tau} (-1)^{|\sigma \tau^{-1}|} g^{i_{\sigma \tau^{-1}(1)} j_1} \dots g^{i_{\sigma \tau^{-1}(k)} j_k} \\ &= \sum_{\sigma} \sum_{\rho} (-1)^{|\rho|} g^{i_{\rho(1)} j_1} \dots g^{i_{\rho(k)} j_k} \\ &= \sum_{\sigma} \det(g^{i_p j_q}) \\ &= k! \det(g^{i_p j_q}). \end{aligned}$$

□

Corollary 6.4. For two k -forms φ, ψ , locally written as

$$\begin{aligned} \varphi &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ \psi &= \sum_{1 \leq j_1 < \dots < j_k \leq n} \psi_{j_1 \dots j_k} dx^{j_1} \wedge \dots \wedge dx^{j_k} \end{aligned}$$

with φ_I, ψ_J is skew-symmetric. Then

$$\langle \varphi, \psi \rangle = \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n \\ 1 \leq j_1 < \dots < j_k \leq n}} \varphi_{i_1 \dots i_k} \psi_{j_1 \dots j_k} \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

Example 6.5. Let φ, ψ be two 2-forms, locally written as

$$\begin{aligned}\varphi &= \varphi_{i_1 i_2} dx^{i_1} \wedge dx^{i_2} \\ \psi &= \psi_{j_1 j_2} dx^{j_1} \wedge dx^{j_2},\end{aligned}$$

where $i_1 < i_2, j_1 < j_2$. Then

$$\begin{aligned}\langle \varphi, \psi \rangle &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} g(dx^{i_1} \wedge dx^{i_2}, dx^{j_1} \wedge dx^{j_2}) \\ &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} g(dx^{i_1} \otimes dx^{i_2} - dx^{i_2} \otimes dx^{i_1}, dx^{j_1} \otimes dx^{j_2} - dx^{j_2} \otimes dx^{j_1}) \\ &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} (g^{i_1 j_1} g^{i_2 j_2} - g^{i_1 j_2} g^{i_2 j_1} - g^{i_2 j_1} g^{i_1 j_2} + g^{i_2 j_2} g^{i_1 j_1}) \\ &= \varphi_{i_1 i_2} \psi_{j_1 j_2} (g^{i_1 j_1} g^{i_2 j_2} - g^{i_1 j_2} g^{i_2 j_1}) \\ &= \varphi_{i_1 i_2} \psi_{j_1 j_2} \det \begin{pmatrix} g^{i_1 j_1} & g^{i_1 j_2} \\ g^{i_2 j_1} & g^{i_2 j_2} \end{pmatrix}.\end{aligned}$$

Definition 6.6. The *volume form* vol is an n -form such that $\langle \text{vol}, \text{vol} \rangle = 1$.

Remark 6.7 (local form). The volume form is locally given by $\sqrt{\det g} dx^1 \wedge \dots \wedge dx^n$.

Definition 6.8. For two k -forms φ, ψ , the *inner product* is defined as

$$(\varphi, \psi) := \int_M \langle \varphi, \psi \rangle \text{vol}.$$

Definition 6.9. For a k -form φ and a $(k+1)$ -form ψ , the *formal adjoint* of d is an operator $d^* : C^\infty(M, \Omega_M^{k+1}) \rightarrow C^\infty(M, \Omega_M^k)$ such that

$$(d\varphi, \psi) = (\varphi, d^*\psi).$$

Remark 6.10. There is no guarantee for existence, but later we will see such d^* do exist, and give an explicit formula.

Definition 6.11. The *Laplace-Beltrami operator* $\Delta_g : C^\infty(M, \Omega_M^k) \rightarrow C^\infty(M, \Omega_M^k)$ is defined as

$$\Delta_g = dd^* + d^*d.$$

Definition 6.12. A k -form α is called *harmonic*, if $\Delta_g \alpha = 0$.

Notation 6.13. The space of all harmonic forms is denoted by $\mathcal{H}^k(M)$.

Lemma 6.14. A k -form α is harmonic if and only if $d\alpha = 0$ and $d^*\alpha = 0$.

Proof. Note that

$$\begin{aligned}(\alpha, \Delta\alpha) &= (\alpha, dd^*\alpha) + (\alpha, d^*d\alpha) \\ &= \|d^*\alpha\|^2 + \|d\alpha\|^2.\end{aligned}$$

□

6.2 Hodge star operator

Although we have defined an inner product on Ω_M^k , it is still quite complicated to compute. However, inner product on Ω_M^k is independent of the choice of local frame, so we can use normal coordinate to give a local frame, and define the Hodge star operator on it, which will give us an effective method to compute.

6.2.1 Baby case

Let V be an $\mathbb{R}$ -vector space with inner product $\langle -, - \rangle$, and $\{e_1, \dots, e_n\}$ is a basis of V . For any $0 \leq k \leq n$, there is a natural basis of $\bigwedge^k V$, consisting of $\{e_I := e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < \dots < i_k \leq n\}$.

With respect to this basis, the induced metric on $\bigwedge^k V$ can be written as

$$\langle e_{i_1} \wedge \dots \wedge e_{i_k}, e_{j_1} \wedge \dots \wedge e_{j_k} \rangle = \det \begin{pmatrix} \langle e_{i_1}, e_{j_1} \rangle & \dots & \langle e_{i_1}, e_{j_k} \rangle \\ \vdots & & \vdots \\ \langle e_{i_k}, e_{j_1} \rangle & \dots & \langle e_{i_k}, e_{j_k} \rangle \end{pmatrix}.$$

It is clear if $\{e_1, \dots, e_n\}$ is an orthonormal basis of V , then $\{e_I\}$ is an orthonormal basis of $\bigwedge^k V$. From now on, we assume $\{e_I\}$ is an orthonormal basis of $\bigwedge^k V$ with respect to inner product induced from the inner product on V .

Definition 6.15. The *Hodge star operator* is a $\mathbb{R}$ -linear operator defined as

$$\begin{aligned} \star: \bigwedge^k V &\rightarrow \bigwedge^{n-k} V \\ e_I &\mapsto \text{sign}(I, I^c) e_{I^c}, \end{aligned}$$

where I^c is $\{1, 2, \dots, n\} \setminus I = \{i'_1, \dots, i'_{n-k}\}$ and $\text{sign}(I, I^c)$ is the sign of the permutation $(i_1, \dots, i_k, i'_1, \dots, i'_{n-k})$.

Example 6.16. It is clear $\star 1 = \text{vol}$ and $\star \text{vol} = 1$.

Exercise 6.17. Let $I \subseteq \{1, 2, \dots, n\}$ be an index set such that $|I| = k$. Then

$$\text{sign}(I, I^c) \text{sign}(I^c, I) = (-1)^{k(n-k)}.$$

Proposition 6.18.

$$\star^2 = (-1)^{k(n-k)} \text{id}$$

holds on $\bigwedge^k V$.

Proof. It suffices to check on basis e_I as follows

$$\begin{aligned} \star^2 e_I &= \star(\text{sign}(I, I^c) e_{I^c}) \\ &= \text{sign}(I, I^c) \text{sign}(I^c, I) e_I \\ &= (-1)^{k(n-k)} e_I. \end{aligned}$$

□

Proposition 6.19. For $u \in \bigwedge^k V, v \in \bigwedge^{n-k} V$, we have

$$\star(u \wedge v) = (-1)^{k(n-k)} \langle u, \star v \rangle.$$

Proof. It suffices to check on basis. Without loss of generality, we may assume $u = e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$ and $v = e_{I^c}$. A direct computation shows

$$\begin{aligned} \star(e_I \wedge e_{I^c}) &= \star(\text{sign}(I, I^c) \text{vol}) \\ &= \text{sign}(I, I^c). \\ \langle e_I, \star e_{I^c} \rangle &= \langle e_I, \text{sign}(I, I^c) e_I \rangle \\ &= \text{sign}(I, I^c) \langle e_I, e_I \rangle \\ &= \text{sign}(I, I^c). \end{aligned}$$

This completes the proof. □

Corollary 6.20. For $u, v \in \bigwedge^k V$, we have

- (1) $u \wedge \star v = v \wedge \star u = \langle u, v \rangle \text{vol}$.
- (2) $\langle \star u, \star v \rangle = \langle u, v \rangle$, that is, Hodge star is a linear isometry.

Proof. For (1).

$$\star(u \wedge \star v) = (-1)^{k(n-k)} \langle u, \star^2 v \rangle = \langle u, v \rangle,$$

which implies $u \wedge \star v = \langle u, v \rangle \text{vol}$. Since $\langle u, v \rangle = \langle v, u \rangle$, we obtain $u \wedge \star v = v \wedge \star u$.

For (2).

$$\begin{aligned} \langle \star u, \star v \rangle &= (-1)^{k(n-k)} \star(\star u \wedge v) \\ &= (-1)^{2k(n-k)} \star(v \wedge \star u) \\ &= (-1)^{3k(n-k)} \langle v \wedge \star^2 u \rangle \\ &= (-1)^{4k(n-k)} \langle v, u \rangle \\ &= \langle u, v \rangle. \end{aligned}$$

□

Corollary 6.21. For $u \in \bigwedge^k V, v \in \bigwedge^{n-k} V$, we have

$$\langle u, \star v \rangle = (-1)^{k(n-k)} \langle \star u, v \rangle.$$

Proof.

$$\langle u, \star v \rangle = \langle \star u, \star^2 v \rangle = (-1)^{k(n-k)} \langle \star u, v \rangle.$$

□

Remark 6.22. This corollary implies the adjoint operator of $\star$ is $(-1)^{k(n-k)} \star$.

6.2.2 General case

Proposition 6.23. *Let (M, g) be an oriented compact Riemannian manifold. Then for each $k = 0, 1, \dots, n$, there is a unique $\mathbb{R}$ -linear map*

$$\star: C^\infty(M, \Omega_M^k) \rightarrow C^\infty(M, \Omega_M^{n-k})$$

such that

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle \text{vol}$$

for all $\omega, \eta \in C^\infty(M, \Omega_M^k)$.

Proof. Firstly let's prove the uniqueness. Suppose $\star_1$ and $\star_2$ are $\mathbb{R}$ -linear maps from $C^\infty(M, \Omega_M^k)$ to $C^\infty(M, \Omega_M^{n-k})$ satisfying desired property. Then for any $\eta \in C^\infty(M, \Omega_M^k)$, we have

$$\omega \wedge \star_1 \eta = \langle \omega, \eta \rangle \text{vol} = \omega \wedge \star_2 \eta$$

holds for all $\omega \in C^\infty(M, \Omega_M^k)$, that is, $\omega \wedge (\star_1 \eta - \star_2 \eta) = 0$ for all $\omega \in C^\infty(M, \Omega_M^k)$. This implies that $\star_1 \eta = \star_2 \eta$, and thus $\star_1 = \star_2$ as η is arbitrary.

Now let's define the Hodge star operator pointwisely, that is, for any k -form η and any $p \in M$, we define a vector $(\star \eta)_p \in \Omega_{M,p}^k$ such that

$$\omega_p \wedge (\star \eta)_p = \langle \omega_p, \eta_p \rangle \text{vol}_p$$

holds for any $\omega_p \in \Omega_{M,p}^k$. For each point $p \in M$, consider the local frame $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ of TM given by normal coordinate, with dual frame $\{dx^1, \dots, dx^n\}$. Then $\{dx_p^I\}$ forms an orthonormal basis of $\Omega_{M,p}^k$, and we define the Hodge star operator as

$$\star(dx_p^I) := \text{sign}(I, I^c) dx_p^{I^c}.$$

By Corollary 6.20, we know that it satisfies the desired properties. This completes the proof. $\square$

Theorem 6.24.

- (1) $\star 1 = \text{vol}, \star \text{vol} = 1$.
- (2) $\star^2 = (-1)^{k(n-k)}$ on k -forms.
- (3) If u is a k -form and v a $(n-k)$ -form. Then

$$\begin{aligned} \star(u \wedge v) &= (-1)^{k(n-k)} \langle u, \star v \rangle, \\ \langle u, \star v \rangle &= (-1)^{k(n-k)} \langle \star u, v \rangle. \end{aligned}$$

- (4) For any two k -forms u, v . Then

$$\begin{aligned} u \wedge \star v &= v \wedge \star u = \langle u, v \rangle \text{vol} = \langle v, u \rangle \text{vol}, \\ \langle \star u, \star v \rangle &= \langle u, v \rangle. \end{aligned}$$

(5) $d^* = (-1)^{nk+n+1} \star d \star$ on k -forms.

Proof. It suffices to check (5), as other cases follows from the baby case. Take any $(k-1)$ -form α and k -form β , we need to show

$$(d\alpha, \beta) = (\alpha, d^*\beta),$$

that is, to show

$$\int_M d\alpha \wedge \star \beta = \int_M \alpha \wedge \star d^*\beta.$$

By Stokes theorem and Leibniz rule we have

$$0 = \int_M d(\alpha \wedge \star \beta) = \int_M d\alpha \wedge \star \beta + (-1)^{k-1} \int_M \alpha \wedge d \star \beta.$$

Since $\star^2 = (-1)^{(n-k+1)(k-1)}$ on $(n-k+1)$ -forms, we have

$$(-1)^{k-1} \int_M \alpha \wedge d \star \beta = (-1)^{k-1+(n-k+1)(k-1)} \int_M \alpha \wedge \star^2 d \star \beta.$$

Therefore

$$\begin{aligned} (d\alpha, \beta) &= \int_M d\alpha \wedge \star \beta \\ &= (-1)^{k+(n-k+1)(k-1)} \int_M \alpha \wedge \star \star d \star \beta \\ &= (-1)^{nk+k+1} \int_M \alpha \wedge \star (\star d \star \beta), \end{aligned}$$

which implies

$$d^*\beta = (-1)^{nk+k+1} \star d \star \beta.$$

□

Remark 6.25. For differential k -forms φ and ψ , (4) allows us to give a new expression for inner product by

$$(\varphi, \psi) := \int_M \langle \varphi, \psi \rangle \text{vol} = \int_M \varphi \wedge \star \psi.$$

6.3 Computations of adjoint operator

Lemma 6.26 (Jacobi's formula). *For a differentiable function $(a_{ij}(t))$ which is valued in $\text{GL}(n, \mathbb{R})$, we have*

$$\frac{d}{dt} \det(a_{ij}(t)) = \det(a_{ij}(t)) a^{ij}(t) \frac{da_{ij}(t)}{dt},$$

where $(a^{ij}(t))$ is the inverse matrix of $(a_{ij}(t))$.

Lemma 6.27. *Let (M, g) be a Riemannian manifold. For any two vector fields X, Y , one has*

$$\nabla_Y \circ \iota_X = \iota_X \circ \nabla_Y + \iota_{\nabla_Y X}.$$

Proof. Let ω be a $(k+1)$ -form. Then for vector fields $Y_1, \dots, Y_k$, a direct computation shows

$$\begin{aligned}\nabla_Y \circ \iota_X \omega(Y_1, \dots, Y_k) &= \nabla_Y \omega(X, Y_1, \dots, Y_k) \\ &= Y \omega(X, Y_1, \dots, Y_k) - \omega(\nabla_X Y, Y_1, \dots, Y_k) \\ &\quad - \sum_{i=1}^k \omega(X, Y_1, \dots, Y_{i-1}, \nabla_X Y_i, Y_{i+1}, \dots, Y_k) \\ &= (\iota_X \circ \nabla_Y \omega + \iota_{\nabla_X Y} \omega)(Y_1, \dots, Y_k).\end{aligned}$$

□

Lemma 6.28. *Let (M, g) be a compact Riemannian manifold. Then*

$$\langle dx^i \wedge \alpha, \beta \rangle = \langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle,$$

where $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ is a local frame of TM , and α, β are forms with appropriate degrees.

Proof. It suffices to check with respect to normal coordinate. If we locally write $\alpha = \alpha_I dx^I$ and $\beta = \beta_J dx^J$, it suffices to check the case there is no dx^i in dx^I and there is dx^i in dx^J , since other cases are trivial. Suppose $|I| = k$ and $|J| = k+1$, and dx^i in the m -th position of dx^J . Then

$$\langle \alpha, \iota_{\frac{\partial}{\partial x^i}} \beta \rangle = (-1)^{m+2} \alpha_I \beta_J \det G,$$

where G is a $k \times k$ matrix. By definition if we write

$$\langle dx^i \wedge \alpha, \beta \rangle = \alpha_I \beta_J \det G',$$

where G' is a $(k+1) \times (k+1)$ matrix. It is clear $\det G' = (-1)^{m+2} \det G$ by expansion of $\det G'$ by the first row. □

Definition 6.29. For any vector field X , its *divergence* $\operatorname{div} X$ is defined as $\operatorname{tr} \nabla X$.

Remark 6.30 (local form). If we locally write X as $X^i \frac{\partial}{\partial x^i}$, then

$$\operatorname{div} X = \frac{\partial X^i}{\partial x^i} + \Gamma_{ij}^j X^i.$$

Lemma 6.31.

$$\operatorname{div} X \operatorname{vol} = \mathcal{L}_X \operatorname{vol}.$$

Proof. Cartan's magic formula shows that

$$\mathcal{L}_X = \iota_X \circ d + d \circ \iota_X.$$

So

$$\begin{aligned}
\mathcal{L}_X \text{vol} &= (\iota_X \circ d + d \circ \iota_X) \text{vol} \\
&= d \circ \iota_X \text{vol} \\
&= d((-1)^{i-1} X^i \sqrt{\det g} dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n) \\
&= \frac{1}{\sqrt{\det g}} \frac{\partial(X^i \sqrt{\det g})}{\partial x^i} \text{vol} \\
&= \frac{1}{\sqrt{\det g}} \left(\frac{\partial X^i}{\partial x^i} \sqrt{\det g} + X^i \frac{\partial \sqrt{\det g}}{\partial x^i} \right) \text{vol} \\
&= \left(\frac{\partial X^i}{\partial x^i} + X^i \frac{\partial \log \sqrt{\det g}}{\partial x^i} \right) \text{vol} \\
&= \left(\frac{\partial X^i}{\partial x^i} + \frac{1}{2} X^i \frac{\partial \log \det g}{\partial x^i} \right) \text{vol}.
\end{aligned}$$

Note that Jacobi's formula says

$$\frac{\partial \log \det g}{\partial x^i} = \frac{1}{\det g} \frac{\partial \det g}{\partial x^i} = g^{jk} \frac{\partial g_{jk}}{\partial x^i} = g^{jk} (\Gamma_{ij}^\ell g_{\ell k} + \Gamma_{ik}^\ell g_{j\ell}) = 2\Gamma_{ij}^j.$$

Thus

$$\begin{aligned}
\mathcal{L}_X \text{vol} &= \left(\frac{\partial X^i}{\partial x^i} + \frac{1}{2} X^i \frac{\partial \log \det g}{\partial x^i} \right) \text{vol} \\
&= \left(\frac{\partial X^i}{\partial x^i} + \Gamma_{ij}^j X^i \right) \text{vol} \\
&= \nabla_i X^i \text{vol}.
\end{aligned}$$

□

Remark 6.32. From the proof, we can say there is the following formula for divergence of a vector field X written as $X^i \frac{\partial}{\partial x^i}$, one has

$$\text{div} X = \frac{1}{\sqrt{\det g}} \frac{\partial}{\partial x^i} (\sqrt{\det g} X^i).$$

Proposition 6.33. *Let (M, g) be an oriented compact Riemannian manifold equipped with Levi-Civita connection ∇ . Then*

$$d^* = -g^{ij} \iota_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}},$$

where $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ is a local frame of TM .

Proof. A direct computation shows

$$\begin{aligned}
0 &= \int_M d(\alpha \wedge \star \beta) \\
&= \int_M \mathcal{L}_{\frac{\partial}{\partial x^i}} (dx^i \wedge \alpha \wedge \star \beta) \\
&= \int_M \mathcal{L}_{\frac{\partial}{\partial x^i}} (\langle dx^i \wedge \alpha, \beta \rangle \text{vol}) \\
&\stackrel{(1)}{=} \int_M \mathcal{L}_{\frac{\partial}{\partial x^i}} (\langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle \text{vol}) \\
&\stackrel{(2)}{=} \int_M (\langle \nabla_i \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle + \langle \alpha, \nabla_i (g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta) \rangle + \frac{1}{2} g^{pq} \frac{\partial g_{pq}}{\partial x^i} \langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle) \text{vol} \\
&= \int_M (\langle dx^i \wedge \nabla_i \alpha, \beta \rangle + \langle \alpha, \frac{\partial g^{ij}}{\partial x^i} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle + \langle \alpha, g^{ij} \nabla_i (\iota_{\frac{\partial}{\partial x^j}} \beta) \rangle + \frac{1}{2} g^{pq} \frac{\partial g_{pq}}{\partial x^i} \langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle) \text{vol} \\
&\stackrel{(3)}{=} \int_M (\langle dx^i \wedge \nabla_i \alpha, \beta \rangle + \langle \alpha, \frac{\partial g^{ij}}{\partial x^i} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle + \langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} (\nabla_i \beta) \rangle + \langle \alpha, g^{i\ell} \Gamma_{i\ell}^j \iota_{\frac{\partial}{\partial x^j}} \beta \rangle \\
&\quad + \frac{1}{2} g^{pq} \frac{\partial g_{pq}}{\partial x^i} \langle \alpha, g^{ij} \iota_{\frac{\partial}{\partial x^j}} \beta \rangle) \text{vol},
\end{aligned}$$

where

1. (1) holds from Lemma 6.28.
2. (2) holds from Lemma 6.31.
3. (3) holds from Lemma 6.27.

It is easy to see

$$\frac{\partial g^{ij}}{\partial x^i} + g^{i\ell} \Gamma_{i\ell}^j + \frac{1}{2} g^{ij} g^{pq} \frac{\partial g_{pq}}{\partial x^i} = 0,$$

since

$$\Gamma_{ij}^\ell = \frac{1}{2} g^{k\ell} \left(\frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k} \right).$$

□

In the following examples, we always compute with respect to normal coordinate.

Example 6.34. For a 1-form ω locally written as $\omega_i dx^i$. Then

$$\begin{aligned}
d^* \omega &= -g^{ij} \iota_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} (\omega_k dx^k) \\
&= -g^{ij} \iota_{\frac{\partial}{\partial x^i}} \left(\frac{\partial \omega_k}{\partial x^j} dx^k - \omega_k \Gamma_{j\ell}^k dx^\ell \right) \\
&= -g^{ij} \left(\frac{\partial \omega_i}{\partial x^j} - \omega_k \Gamma_{ji}^k \right)
\end{aligned}$$

Example 6.35. For a smooth function f . Then

$$\begin{aligned}
\Delta_g f &= (dd^* + d^*d)f \\
&= d^*df \\
&= d^*\left(\frac{\partial f}{\partial x^i} dx^i\right) \\
&= -g^{ij} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \frac{\partial f}{\partial x^k} \Gamma_{ji}^k \right)
\end{aligned}$$

This shows the Laplace-Beltrami operator differs a sign with scalar Laplacian.

Remark 6.36. Unless otherwise specified, we use Δ to denote the scalar Laplacian and Δ_g to denote the Laplace-Beltrami operator.

Corollary 6.37 (divergence theorem).

$$\int_M \operatorname{div} X \operatorname{vol} = 0.$$

Proposition 6.38. Let f be a smooth function on M . Then for any smooth function φ , one has

$$-\int_M \langle \nabla \varphi, \nabla f \rangle \operatorname{vol} = \int_M \Delta \varphi \cdot f \operatorname{vol}.$$

Proof. The idea is to prove that

$$\operatorname{div}(f \nabla \varphi) = \langle \nabla \varphi, \nabla f \rangle + f \Delta \varphi.$$

Then the divergence theorem completes the proof.

A direct computation shows

$$\begin{aligned}
\nabla(f \nabla \varphi) &= \nabla\left(f \frac{\partial \varphi}{\partial x^i} g^{ij} \frac{\partial}{\partial x^j}\right) \\
&= g^{ij} \nabla\left(f \frac{\partial \varphi}{\partial x^i} \frac{\partial}{\partial x^j}\right) \\
&= g^{ij} \frac{\partial f}{\partial x^k} \frac{\partial \varphi}{\partial x^i} dx^k \otimes \frac{\partial}{\partial x^j} + \underbrace{g^{ij} \frac{\partial^2 \varphi}{\partial x^k \partial x^i} dx^k \otimes \frac{\partial}{\partial x^j} + g^{ij} f \frac{\partial \varphi}{\partial x^i} \Gamma_{kj}^\ell dx^k \otimes \frac{\partial}{\partial x^\ell}}_{\text{part II}}.
\end{aligned}$$

Thus

$$\operatorname{div}(f \nabla \varphi) = \underbrace{g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial \varphi}{\partial x^i}}_{\text{part I}} + \underbrace{g^{ij} \frac{\partial^2 \varphi}{\partial x^j \partial x^i} + g^{ij} f \frac{\partial \varphi}{\partial x^i} \Gamma_{kj}^k}_{\text{part II}}$$

We have the following observations:

(1) Part I equals

$$\begin{aligned}
g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial \varphi}{\partial x^i} &= \left\langle \frac{\partial f}{\partial x^j} dx^j, \frac{\partial \varphi}{\partial x^i} dx^i \right\rangle \\
&= \langle \nabla f, \nabla \varphi \rangle.
\end{aligned}$$

(2) It is clear part II equals $f\Delta\varphi$.

□

6.4 Conformal Laplacian

For a smooth function u , according to Example 6.34 and Remark 6.32, we can write $\Delta_g u$ as follows

$$\begin{aligned}\Delta_g u &= d^* du \\ &= -\operatorname{div}(g^{ij} \frac{\partial u}{\partial x^i} \frac{\partial}{\partial x^j}) \\ &= -\frac{1}{\sqrt{\det g}} \frac{\partial}{\partial x^j} (\sqrt{\det g} g^{ij} \frac{\partial u}{\partial x^i}).\end{aligned}$$

Thus the Laplace-Beltrami Δ_g with respect to g is

$$\Delta_g = -\frac{1}{\sqrt{\det g}} \frac{\partial}{\partial x^j} (\sqrt{\det g} g^{ij} \frac{\partial}{\partial x^i}).$$

If we consider conformal transformation of metric given by $\tilde{g} = e^{2f} g$ for some smooth function f , we have

$$\begin{aligned}\tilde{g}_{ij} &= e^{2f} g_{ij} \\ \tilde{g}^{ij} &= e^{-2f} g^{ij} \\ \det \tilde{g} &= e^{2nf} \det g \\ \sqrt{\det \tilde{g}} &= e^{nf} \sqrt{\det g}.\end{aligned}$$

Thus

$$\begin{aligned}\Delta_{\tilde{g}} &= -\frac{1}{e^{nf} \sqrt{\det g}} \frac{\partial}{\partial x^j} (e^{nf} \sqrt{\det g} e^{-2f} g^{ij} \frac{\partial}{\partial x^i}) \\ &= -\frac{e^{-nf}}{\sqrt{\det g}} \frac{\partial}{\partial x^j} (e^{(n-2)f} \sqrt{\det g} g^{ij} \frac{\partial}{\partial x^i}) \\ &= -\frac{e^{-2f}}{\sqrt{\det g}} \frac{\partial}{\partial x^j} (\sqrt{\det g} g^{ij} \frac{\partial}{\partial x^i}) - \frac{(n-2)e^{-2f}}{\sqrt{\det g}} \frac{\partial f}{\partial x^j} \sqrt{\det g} g^{ij} \frac{\partial}{\partial x^i} \\ &= -e^{-2f} \Delta_g - (n-2)e^{-2f} g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial}{\partial x^i}.\end{aligned}$$

This shows

$$\Delta_{\tilde{g}} = -e^{-2f} \Delta_g$$

when $n = 2$. In this case, the Laplace-Beltrami Laplacian is called conformal invariant. However, it fails in higher dimensional cases.

In the following section, we assume $n > 3$ and introduce the so-called conformal Laplacian, which is defined as

$$L: C^\infty(M) \rightarrow C^\infty(M)$$

$$u \mapsto -\frac{4(n-1)}{n-2} \Delta_g u + S u,$$

where S is the scalar curvature. Let's show

$$\widetilde{L}u = e^{-\frac{n+2}{2}f} L(e^{\frac{n-2}{2}f} u),$$

where $\widetilde{L}$ is the conformal Laplacian after conformal transformation. In order to show this, we divide computations into several parts:

(1)

$$\begin{aligned} \nabla^2(e^{\frac{n-2}{2}f} u) &= \nabla \left(\frac{n-2}{2} e^{\frac{n-2}{2}f} \frac{\partial f}{\partial x^i} u dx^i + e^{\frac{n-2}{2}f} \frac{\partial u}{\partial x^i} dx^i \right) \\ &= e^{\frac{n-2}{2}f} \nabla^2 u + \frac{n-2}{2} e^{\frac{n-2}{2}f} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^i} dx^i \otimes dx^j \\ &\quad + \left(\frac{(n-2)^2}{4} e^{\frac{n-2}{2}f} u \frac{\partial f}{\partial x^j} \frac{\partial f}{\partial x^i} + \frac{n-2}{2} e^{\frac{n-2}{2}f} \frac{\partial f}{\partial x^i} \frac{\partial u}{\partial x^j} \right) dx^i \otimes dx^j + \frac{n-2}{2} e^{\frac{n-2}{2}f} u \nabla^2 f. \end{aligned}$$

(2)

$$\begin{aligned} \Delta_g(e^{\frac{n-2}{2}f} u) &= \text{tr}_g \nabla^2(e^{\frac{n-2}{2}f} u) \\ &= e^{\frac{n-2}{2}f} \Delta_g u + \frac{n-2}{2} e^{\frac{n-2}{2}f} g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^i} \\ &\quad + g^{ij} \left(\frac{(n-2)^2}{4} e^{\frac{n-2}{2}f} u \frac{\partial f}{\partial x^j} \frac{\partial f}{\partial x^i} + \frac{n-2}{2} e^{\frac{n-2}{2}f} \frac{\partial f}{\partial x^i} \frac{\partial u}{\partial x^j} \right) + \frac{n-2}{2} e^{\frac{n-2}{2}f} u \Delta_g f. \end{aligned}$$

(3)

$$\begin{aligned} e^{-\frac{n+2}{2}f} L(e^{\frac{n-2}{2}f} u) &= -\frac{4(n-1)}{n-2} e^{-2f} \Delta_g u - 4(n-1) e^{-2f} g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^i} \\ &\quad - g^{ij} (n-2)(n-1) e^{-2f} u \frac{\partial f}{\partial x^j} \frac{\partial f}{\partial x^i} - 2(n-1) e^{-2f} u \Delta_g f + e^{-2f} S u \\ &= -\frac{4(n-1)}{n-2} e^{-2f} \Delta_g u - 4(n-1) e^{-2f} g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^i} \\ &\quad - (n-2)(n-1) e^{-2f} u |df|^2 - 2(n-1) e^{-2f} u \Delta_g f + e^{-2f} S u. \end{aligned}$$

(4)

$$-\frac{4(n-1)}{n-2} \Delta_{\widetilde{g}} u = -\frac{4(n-1)}{n-2} e^{-2f} \Delta_g u - 4(n-1) e^{-2f} g^{ij} \frac{\partial f}{\partial x^j} \frac{\partial u}{\partial x^i}.$$

(5) Note that

$$\widetilde{S} = e^{-2f} S - 2(n-1) e^{-2f} \Delta_g f - (n-2)(n-1) e^{-2f} |df|^2.$$

This completes the computation. In particular, in (2) if we take $u = 1$ we have

$$-\frac{4(n-1)}{n-2} \Delta_g(e^{\frac{n-2}{2}f}) = -(n-2)(n-1) e^{\frac{n-2}{2}f} |df|^2 - 2(n-1) e^{\frac{n-2}{2}f} \Delta_g f.$$

Thus we have

$$\widetilde{S} = e^{-\frac{n+2}{2}f} \left(-\frac{4(n-1)}{n-2} \Delta_g e^{\frac{n-2}{2}f} + S e^{\frac{n-2}{2}f} \right) = e^{-\frac{n+2}{2}f} L(e^{\frac{n-2}{2}f}).$$

So if we put $e^{2f} = \varphi^{\frac{4}{n-2}}$, we have

$$\widetilde{S} = \varphi^{-\frac{n+2}{n-2}} L\varphi.$$

So it is clear g is conformal to $\widetilde{g}$ with constant scalar curvature λ if and only if φ is a smooth positive solution to the Yamabe equation

$$L\varphi = \lambda \varphi^{\frac{n+2}{n-2}}.$$

6.5 Hodge theorem and corollaries

Theorem 6.39 (Hodge theorem). *Let $\Delta_g : C^\infty(M, \Omega_M^k) \rightarrow C^\infty(M, \Omega_M^k)$ be the Laplace operator. Then*

- (1) $\dim_{\mathbb{R}} \mathcal{H}^k(M) < \infty$.
- (2) *There is an orthogonal decomposition*

$$C^\infty(M, \Omega_M^k) = \mathcal{H}^k(M) \perp \text{im } \Delta_g.$$

Corollary 6.40. *There is the following more explicit orthogonal decomposition*

$$C^\infty(M, \Omega_M^k) = \mathcal{H}^k(M) \oplus d(C^\infty(M, \Omega_M^{k-1})) \oplus d^*(C^\infty(M, \Omega_M^{k+1})).$$

Proof. For a $(k-1)$ -form α and a $(k+1)$ -form β , we have

$$(d\alpha, d^*\beta) = (d^2\alpha, \beta) = 0.$$

This shows that $d(C^\infty(M, \Omega_M^{k-1}))$ is orthogonal to $d^*(C^\infty(M, \Omega_M^{k+1}))$, and completes the proof. □

Corollary 6.41.

$$\ker d = \mathcal{H}^k(M) \oplus d(C^\infty(M, \Omega_M^{k-1})),$$

$$\ker d^* = \mathcal{H}^k(M) \oplus d^*(C^\infty(M, \Omega_M^{k+1})).$$

Proof. It is clear from above corollary. □

Corollary 6.42. *The natural map $\mathcal{H}^k(M) \rightarrow H^k(M, \mathbb{R})$ is an isomorphism. In other words, every element in $H^k(M, \mathbb{R})$ is represented by a unique harmonic form.*

Proof. It is clear from above corollary. □

Corollary 6.43 (Poincaré duality). *For any $k = 0, 1, \dots, n$, we have $H^k(M, \mathbb{R}) \cong H^{n-k}(M, \mathbb{R})$.*

Proof. By Corollary 6.42, it suffices to show the Hodge star $\star$ maps harmonic forms to harmonic forms, since $\star$ maps k -forms to $n - k$ -forms and $\star^2 = (-1)^{k(n-k)} \text{id}$ on k -forms. By Lemma 6.14, we just need to show $d\star\alpha = d^*\star\alpha = 0$ for a harmonic form α . A direct computation shows

$$d\star\alpha = (-1)^{\bullet_1} \star\star d\star\alpha = (-1)^{\bullet_2} \star d^*\alpha = 0,$$

$$d^*\star\alpha = (-1)^{\bullet_3} \star d\star\star\alpha = (-1)^{\bullet_4} \star d\alpha = 0.$$

Here we use $\bullet, \bullet'$ to denote the power of (-1) , since it is not necessary for us to know what exactly it is. □

Remark 6.44. In fact, above corollary follows from the following identity

$$\Delta_g \circ \star = \star \circ \Delta_g.$$

7 Bochner's technique

7.1 Bochner formula

Let (M, g) be a Riemannian manifold with Levi-Civita connection ∇ . In Example 3.5 and Remark 3.9 we have already seen the Hessian of a smooth function f and its scalar Laplacian locally can be written as follows:

$$\begin{aligned}\text{Hess } f &= \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right) dx^i \otimes dx^j \\ \Delta f &= g^{ij} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right).\end{aligned}$$

Proposition 7.1 (Bochner formula). *Let $f: (M, g) \rightarrow \mathbb{R}$ be a smooth function. Then*

$$\frac{1}{2} \Delta |\nabla f|^2 = |\text{Hess } f|^2 + \text{Ric}(\nabla f, \nabla f) + g(\nabla \Delta f, \nabla f).$$

Proof. Note that

$$\begin{aligned}|\nabla f|^2 &= g(\nabla f, \nabla f) \\ &= g(\nabla_i f dx^i, \nabla_j f dx^j) \\ &= g^{ij} \nabla_i f \nabla_j f.\end{aligned}$$

In the following computation we may use the normal coordinate. Then in this case

$$\begin{aligned}\frac{1}{2} \Delta |\nabla f|^2 &\stackrel{(1)}{=} \frac{1}{2} g^{k\ell} \nabla_k \nabla_\ell (g^{ij} \nabla_i f \nabla_j f) \\ &\stackrel{(2)}{=} \frac{1}{2} g^{k\ell} g^{ij} \nabla_k \nabla_\ell (\nabla_i f \nabla_j f) \\ &\stackrel{(3)}{=} g^{k\ell} g^{ij} \nabla_\ell \nabla_i f \cdot \nabla_k \nabla_j f + g^{k\ell} g^{ij} \nabla_k \nabla_\ell \nabla_i f \cdot \nabla_j f \\ &= |\text{Hess } f|^2 + g^{k\ell} g^{ij} \nabla_k \nabla_\ell \nabla_i f \cdot \nabla_j f,\end{aligned}$$

where

1. (1) holds from in normal coordinate $\Delta f = g^{ij} \nabla_i \nabla_j f$.
2. (2) holds from Proposition 3.10, that is magic formula.
3. (3) holds from the following direct computation

$$\begin{aligned}\nabla_k \nabla_\ell (\nabla_i f \nabla_j f) &= \nabla_k (\nabla_\ell \nabla_i f \cdot \nabla_j f + \nabla_i f \cdot \nabla_\ell \nabla_j f) \\ &= \nabla_k \nabla_\ell \nabla_i f \cdot \nabla_j f + \nabla_\ell \nabla_i f \cdot \nabla_k \nabla_j f + \nabla_k \nabla_i f \cdot \nabla_\ell \nabla_j f + \nabla_i f \cdot \nabla_k \nabla_\ell \nabla_j f \\ &= 2 \nabla_\ell \nabla_i f \cdot \nabla_k \nabla_j f + 2 \nabla_k \nabla_\ell \nabla_i f \cdot \nabla_j f.\end{aligned}$$

Then the following computation completes the proof:

$$\begin{aligned}
g^{k\ell} g^{ij} \nabla_k \nabla_\ell \nabla_i f \cdot \nabla_j f &\stackrel{(4)}{=} g^{k\ell} g^{ij} \nabla_k \nabla_i \nabla_\ell f \cdot \nabla_j f \\
&\stackrel{(5)}{=} g^{k\ell} g^{ij} (\nabla_i \nabla_k \nabla_\ell f - R_{kil}^s \nabla_s f) \cdot \nabla_j f \\
&= g^{ij} \nabla_i (g^{k\ell} \nabla_k \nabla_\ell f) \cdot \nabla_j f + g^{ij} R_i^s \nabla_s f \cdot \nabla_j f \\
&= g^{ij} \nabla_i (\Delta f) \cdot \nabla_j f + \text{Ric}(\nabla f, \nabla f) \\
&= g(\nabla \Delta f, \nabla f) + \text{Ric}(\nabla f, \nabla f),
\end{aligned}$$

where

1. (4) holds from symmetry of Hessian.
2. (5) holds from Theorem 5.7, that is, Ricci identity.

□

7.2 Obstruction to the existence of Killing fields

Definition 7.2. A vector field X on a Riemannian manifold (M, g) is called a *Killing field*, if $\mathcal{L}_X g = 0$.

Proposition 7.3. X is a Killing field if and only if flows generated by X acts as isometries.

Proof. If ϕ_t is the flow generated by X , then by definition of Lie derivative one has

$$\mathcal{L}_X g = \left. \frac{d}{dt} \right|_{t=0} (\phi_t)^* g.$$

Thus X is a Killing vector field if and only if its flow acts as isometries.

□

Theorem 7.4. The following statements are equivalent:

- (1) X is a Killing field.
- (2) For any two vector fields Y, Z , we have

$$\langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle = 0.$$

Proof. Note that

$$\begin{aligned}
\mathcal{L}_X \langle Y, Z \rangle &= X \langle Y, Z \rangle - \langle \mathcal{L}_X Y, Z \rangle - \langle Y, \mathcal{L}_X Z \rangle \\
&= \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle - \langle [X, Y], Z \rangle - \langle Y, [X, Z] \rangle \\
&= \langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle.
\end{aligned}$$

□

Remark 7.5. For (2) locally we have

$$g_{kj} \nabla_i X^j = -g_{ij} \nabla_k X^j.$$

Thus X is a Killing vector if and only if ∇X is a skew-symmetric $(1, 1)$ -tensor, that is $\nabla_i X^j$ is skew-symmetric in i, j .

Corollary 7.6. *If X is a Killing field, then for arbitrary vector field Y one has*

$$\langle \nabla_Y X, Y \rangle = 0.$$

Proof. Set $Y = Z$ in $\langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle = 0$ to conclude. □

Corollary 7.7. *If X is parallel, then X is Killing.*

Proof. A zero matrix must be skew-symmetric. □

Corollary 7.8. *If X is Killing, then $\operatorname{div} X = \nabla_i X^i = 0$.*

Proof. The trace of a skew-symmetric matrix is zero. □

Lemma 7.9 (Bochner formula for Killing field). *Let X be a Killing field, and $f = \frac{1}{2}|X|^2$. Then*

- (1) $\nabla f = -\nabla_X X$.
- (2) $\operatorname{Hess} f(Y, Y) = \langle \nabla_Y X, \nabla_Y X \rangle - R(Y, X, X, Y)$ holds for any vector field Y .
- (3) $\Delta f = |\nabla X|^2 - \operatorname{Ric}(X, X)$.

Proof. For (1). By direct computation we have

$$\begin{aligned} \nabla f &= \langle \nabla X, X \rangle \\ &= \langle \nabla_k X^i dx^k \otimes \frac{\partial}{\partial x^i}, X^j \frac{\partial}{\partial x^j} \rangle \\ &= g_{ij} X^j \nabla_k X^i dx^k \\ &\stackrel{\text{I}}{=} -g_{ik} X^j \nabla_j X^i dx^k. \\ \nabla_X X &= X^j \nabla_j X^i \frac{\partial}{\partial x^i} \\ &= g_{ik} X^j \nabla_j X^i dx^k, \end{aligned}$$

where I holds from skew-symmetry of ∇X .

For (2). By direct computation we have

$$\begin{aligned} \operatorname{Hess} f(Y, Y) &= \frac{1}{2} Y^i Y^j \nabla_i \nabla_j (g_{k\ell} X^k X^\ell) \\ &= Y^i Y^j g_{k\ell} (\nabla_i X^k \nabla_j X^\ell + \nabla_i \nabla_j X^k \cdot X^\ell) \\ &= \langle \nabla_Y X, \nabla_Y X \rangle + Y^i Y^j g_{k\ell} \nabla_i \nabla_j X^k \cdot X^\ell, \end{aligned}$$

and

$$\begin{aligned}
Y^i Y^j g_{k\ell} \nabla_i \nabla_j X^k \cdot X^\ell &= -Y^i Y^j g_{kj} \nabla_i \nabla_\ell X^k \cdot X^\ell \\
&\stackrel{\text{II}}{=} -Y^i Y^j g_{kj} X^\ell (\nabla_\ell \nabla_i X^k + R_{ilm}^k X^m) \\
&= -Y^i Y^j X^\ell X^m R_{i\ell m j} \\
&= -R(Y, X, X, Y),
\end{aligned}$$

where (II) holds from $g_{kj} X^\ell \nabla_\ell \nabla_i X^k = 0$, since this expression is skew symmetric in i, j .

(3) holds from (2) directly. □

Theorem 7.10 (Bochner). *Let (M, g) be a compact, oriented Riemannian manifold.*

- (1) *If $\text{Ric}(g) \leq 0$, then every Killing field is parallel.*
- (2) *If $\text{Ric}(g) \leq 0$ and $\text{Ric}(g) < 0$ at some point, then there is no non-trivial Killing field.*

Proof. For (1). Let X be a Killing field and set $f = \frac{1}{2}|X|^2$. Then

$$\begin{aligned}
0 &= \int_M \Delta f \, \text{vol} \\
&= \int_M (|\nabla X|^2 - \text{Ric}(X, X)) \, \text{vol} \\
&\geq \int_M |\nabla X|^2 \, \text{vol} \\
&\geq 0.
\end{aligned}$$

Thus $|\nabla X| \equiv 0$, that is X is parallel.

For (2). From proof of (1) one can see if $\text{Ric}(g) \leq 0$ and X is a Killing field, then

$$\int_M \text{Ric}(X, X) = 0,$$

which implies $\text{Ric}(X, X) \equiv 0$. If $\text{Ric}(g) < 0$ at some point $p \in M$, then $X_p = 0$, and thus $X \equiv 0$, since it is parallel. □

7.3 Obstruction to the existence of harmonic 1-forms

In some sense, the Killing field is “dual” to harmonic 1-form. Let’s explain this in more detail.

Lemma 7.11. *For a harmonic 1-form α , locally written as $\alpha_i dx^i$, we have*

$$\begin{aligned}
\nabla_i \alpha_j &= \nabla_j \alpha_i, \\
g^{ij} \nabla_j \alpha_i &= 0.
\end{aligned}$$

Proof. Recall α is harmonic if and only if

$$\begin{aligned}
d\alpha &= 0, \\
d^* \alpha &= 0.
\end{aligned}$$

It is clear

$$d(\alpha_j dx^j) = \nabla_i \alpha_j dx^i \wedge dx^j = 0$$

implies $\nabla_i \alpha_j = \nabla_j \alpha_i$. Similarly, explicit expression for d^* implies the second identity. $\square$

Remark 7.12. Recall that the Killing field condition implies $g_{ij} \nabla_k X^j$ is skew-symmetric in i, k , we can see both Killing field and harmonic 1-form implies some (skew)symmetries.

Lemma 7.13. *If α is a harmonic 1-form, then*

$$\frac{1}{2} \Delta |\alpha|^2 = |\nabla \alpha|^2 + \text{Ric}(\alpha^\sharp, \alpha^\sharp).$$

Proof. Routine computation as follows:

$$\begin{aligned} \frac{1}{2} \Delta |\alpha|_g^2 &= \frac{1}{2} g^{k\ell} \nabla_k \nabla_\ell (g^{ij} \alpha_i \alpha_j) \\ &= |\nabla \alpha|^2 + g^{k\ell} g^{ij} \nabla_k \nabla_\ell \alpha_i \cdot \alpha_j \\ &= |\nabla \alpha|^2 + g^{k\ell} g^{ij} \nabla_k \nabla_i \alpha_l \cdot \alpha_j \\ &= |\nabla \alpha|^2 + g^{k\ell} g^{ij} (\nabla_i \nabla_k \alpha_l - R_{kil}^s \alpha_s) \alpha_j \\ &= |\Delta \alpha|^2 - g^{k\ell} g^{ij} R_{kil}^s \alpha_s \cdot \alpha_j \\ &= |\Delta \alpha|^2 + \text{Ric}(\alpha^\sharp, \alpha^\sharp). \end{aligned}$$

$\square$

Theorem 7.14 (Bochner). *Let (M, g) be a compact, oriented Riemannian manifold.*

- (1) *If $\text{Ric}(g) \geq 0$, then every harmonic 1-form is parallel.*
- (2) *If $\text{Ric}(g) \geq 0$ and $\text{Ric}(g) > 0$ at some point, then there is no non-trivial harmonic 1-form.*

Proof. The same as before. $\square$

Corollary 7.15. *Let (M, g) be a compact, oriented Riemannian manifold with $\text{Ric}(g) \geq 0$ and $\text{Ric}(g) > 0$ at some point. Then $b_1(M) = 0$.*

Proof. It is clear from above theorem and Corollary 6.42. $\square$

Remark 7.16. It is a kind of vanishing theorem. In geometry, "positivity" may cause "vanishing", which is an important philosophy.

Part III

Complex Geometry

8 Complex manifold

8.1 First definitions, properties and examples

Definition 8.1. A *holomorphic atlas* on a smooth manifold is an atlas $\{(U_\alpha, \varphi_\alpha)\}$ of the form $\varphi_\alpha : U_\alpha \cong \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ such that transition functions $\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_{\alpha\beta}) \rightarrow \varphi_\alpha(U_{\alpha\beta})$ are holomorphic functions. Furthermore,

- (1) the pair $(U_\alpha, \varphi_\alpha)$ is called a *holomorphic chart*.
- (2) two holomorphic atlases are called *equivalent*, if the union of them is a holomorphic atlas.

Definition 8.2. A *complex n -manifold* X is a smooth $2n$ -manifold admitting an equivalence class of holomorphic atlases.

Remark 8.3. A complex manifold is called connected, compact, simply-connected and so on, if its underlying real manifold has this property.

Definition 8.4. Let X be a complex n -manifold and $Y \subseteq X$ be a smooth manifold of (real) dimension $2k$. Then Y is a *complex submanifold* if there exists a holomorphic atlas $\{(U_i, \varphi_i)\}$ of X such that $\varphi_i : U_i \cap Y \cong \varphi_i(U_i) \cap \mathbb{C}^k$.

Definition 8.5. Let X, Y be complex manifolds. A continuous map $f : X \rightarrow Y$ is a *holomorphic map* if for any holomorphic charts (U, φ) and (U', φ') of X and Y respectively, the map $\varphi' \circ f \circ \varphi^{-1} : \varphi(f^{-1}(U') \cap U) \rightarrow \varphi'(U')$ is holomorphic.

Definition 8.6. Let X, Y be two complex manifolds. X, Y are called *biholomorphic*, if there exists a holomorphic homeomorphism $f : X \rightarrow Y$.

Definition 8.7. A *holomorphic function* on complex manifold X is a holomorphic map $f : X \rightarrow \mathbb{C}$.

Notation 8.8. We always use $\mathcal{O}_X$ to denote the sheaf of holomorphic functions on complex manifold X , and use $\Gamma(U, \mathcal{O}_X)$ to denote the set of sections over open subset $U \subseteq X$.

Proposition 8.9. Let X be a compact connected complex manifold. Then $\Gamma(X, \mathcal{O}_X) = \mathbb{C}$.

Proof. It's clear from maximum principle. □

Definition 8.10. A *meromorphic function* on a complex manifold X is a map $f : X \rightarrow \coprod_{x \in X} Q(\mathcal{O}_{X,x})$ which associates to any $x \in X$ an element $f_x \in Q(\mathcal{O}_{X,x})$ such that for any $x_0 \in X$ there exists a neighborhood $x_0 \in U \subseteq X$ and two holomorphic functions $g, h : U \rightarrow \mathbb{C}$ with $f_x = g/h$ for all $x \in U$.

Notation 8.11. We always use $\mathcal{K}_X$ to denote the sheaf of meromorphic functions on complex manifold X , and use $K(X)$ to denote $\Gamma(X, \mathcal{K}_X)$.

Example 8.12 (affine space). The n -dimensional complex plane $\mathbb{C}^n$ is a complex manifold.

Example 8.13 (complex tori). If V is a complex vector space of dimension n and $\Gamma \subseteq V$ is a free abelian, discrete subgroup of order $2n$, then $X = V/\Gamma$ is a complex manifold, which is called *complex tori*.

Remark 8.14. The underlying manifolds of complex tori with different Γ are not very interesting since they are all diffeomorphic to $(S^1)^{2n}$. However, if you pick two lattices Γ_1, Γ_2 randomly, then $\mathbb{C}^n/\Gamma_1$ and $\mathbb{C}^n/\Gamma_2$ will not be biholomorphic to each other.

Example 8.15 (projective space). The projective space $\mathbb{CP}^n$ is a complex manifold. Indeed, atlas are given by $U_i = \{[z] \in \mathbb{CP}^n \mid z^i \neq 0\}$, $0 \leq i \leq n$, and φ_i is defined as

$$\begin{aligned} \varphi_i: U_i &\rightarrow \mathbb{C}^n \\ [z] &\mapsto \left(\frac{z_0}{z^i}, \dots, \frac{\widehat{z^i}}{z^i}, \dots, \frac{z_n}{z^i} \right) \end{aligned}$$

The transition functions are calculated as follows: For $i < j$

$$\varphi_i \circ \varphi_j^{-1}: (u_1, \dots, u_n) \mapsto \left(\frac{u_1}{u_i}, \dots, \frac{\widehat{u_i}}{u_i}, \dots, \frac{u_{j-1}}{u_i}, \frac{1}{u_i}, \frac{u_{j+1}}{u_i}, \dots, \frac{u_n}{u_i} \right)$$

It's holomorphic on $U_i \cap U_j$.

Remark 8.16. $\mathbb{CP}^n$ is compact since $\mathbb{CP}^n$ is diffeomorphic to S^{2n+1}/S^1 , which is called Hopf fibration.

Definition 8.17. A complex manifold X is called *projective* if X is biholomorphic to a closed complex submanifold of some projective space $\mathbb{CP}^N$.

Example 8.18 (Grassmannian manifold). The Grassmannian manifold

$$Gr(k, n+1) = \{k\text{-dimensional subspace of } \mathbb{C}^{n+1}\}$$

Now we're going to show $Gr(k, n+1)$ is a manifold of dimension $k(n+1-k)$. Any $W \in Gr(k, n+1)$ is generated by the rows of a $k \times (n+1)$ matrix A of rank k . Let us denote the set of these matrices by $M_{k,n+1}$, which is an open subset of the set of all $k \times (n+1)$ matrices. The latter space is a complex manifold which is canonically isomorphic to $\mathbb{C}^{k(n+1)}$. Thus we obtain a natural surjection $\pi: M_{k,n+1} \rightarrow Gr(k, n+1)$, which is the quotient by the natural action of $GL(k, \mathbb{C})$ on $M_{k,n+1}$.

Let us fix an ordering $\{B_1, \dots, B_m\}$ of all $k \times k$ -minors of matrices $A \in M_{k,n+1}$. Define an open covering $Gr(k, n+1) = \bigcup_{i=1}^m U_i$, where U_i is the open subset $\{\pi(A) \mid \det(B_i) \neq 0\}$. Note that U_i is well-defined since if $\pi(A) = \pi(A')$, then A and A' differs an action of $GL(k, \mathbb{C})$, so $\det(B_i) \neq 0$ if and only if $\det(B'_i) \neq 0$. So without lose of generality, we may assume A is of form (B_i, C_i) , where C_i is a $k \times (n+1-k)$ matrix. Then the map $\varphi_i: U_i \rightarrow \mathbb{C}^{k(n+1-k)}$, given by $\pi(A) \rightarrow B_i^{-1}C_i$ is well-defined, and $\{(U_i, \varphi_i)\}$ will give atlas of $Gr(k, n+1)$, since all operations are matrix operation, thus they're holomorphic. This shows $Gr(k, n+1)$ is a complex manifold with dimension $k(n+1-k)$.

Remark 8.19. If V is a complex vector space of dimension $n + 1$, then $Gr(k, V)$ is defined as the set consisting of all k -dimensional subspaces of V , which is biholomorphic to $Gr(k, n + 1)$.

Example 8.20. Let V be a complex vector space of dimension $n + 1$, then

$$\Phi: Gr(k, V) \hookrightarrow \mathbb{CP}\left(\bigwedge^k V\right)$$

defined by $W \subseteq V$ with basis $w_1, \dots, w_k$ is mapped to $[w_1 \wedge \dots \wedge w_k]$, is called *Plücker embedding*. It's well-defined, thanks to the following lemma.

Lemma 8.21. Let W be a complex vector space of dimension k , and $\mathcal{B}_1 = \{w_1, \dots, w_k\}$ and $\mathcal{B}_2 = \{v_1, \dots, v_k\}$ are two basis for W . Then $v_1 \wedge \dots \wedge v_k = \lambda w_1 \wedge \dots \wedge w_k$ for some $\lambda \in \mathbb{C}^*$.

Proof. If we express $w_j = a_{1j}v_1 + \dots + a_{kj}v_k$, then direct computation shows that

$$\begin{aligned} w_1 \wedge \dots \wedge w_k &= (a_{11}v_1 + \dots + a_{k1}v_k) \wedge \dots \wedge (a_{1k}v_1 + \dots + a_{kk}v_k) \\ &= \sum_{\sigma \in S_k} \text{sign}(\sigma) a_{1\sigma(1)} \dots a_{k\sigma(k)} v_1 \wedge \dots \wedge v_k \\ &= \lambda v_1 \wedge \dots \wedge v_k \end{aligned}$$

Note that λ is exactly the determinant of the change of basis matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$. □

Exercise 8.22. Check that Plücker embedding is injective. In particular, Grassmannian manifold is projective.

8.2 Holomorphic (co)tangent bundle

Definition 8.23. Let X be a complex n -manifold.

1. A *complex vector bundle* of rank r on X is defined by an open covering $\{U_\alpha\}$ together with a collection of smooth functions $\{g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{C})\}$ satisfying cocycle conditions;
2. A *holomorphic vector bundle* of rank r on X is defined by an open covering $\{U_\alpha\}$ together with a collection of holomorphic functions $\{g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{C})\}$ satisfying cocycle conditions.

Definition 8.24. Let X be a complex n -manifold with an atlas $\{U_\alpha, \varphi_\alpha: U_\alpha \rightarrow V_\alpha \subseteq \mathbb{C}^n\}$, where $V_\alpha \subseteq \mathbb{C}^n$ is an open subset with coordinates $z_\alpha = (z_\alpha^1, \dots, z_\alpha^n)$. Then *holomorphic tangent bundle* TX is given by holomorphic transition functions

$$\begin{aligned} g_{\alpha\beta}: U_\alpha \cap U_\beta &\rightarrow GL(n, \mathbb{C}) \\ z &\mapsto J_{\mathbb{C}}(\varphi_\alpha \circ \varphi_\beta^{-1})(\varphi_\beta(z)), \end{aligned}$$

where $J_{\mathbb{C}}(\varphi_\alpha \circ \varphi_\beta^{-1})$ is the matrix of partial derivatives of $\varphi_\alpha \circ \varphi_\beta^{-1}$ with respect to z_β .

Let X be a complex n -manifold and $\{z^i = x^i + \sqrt{-1}y^i\}_{1 \leq i \leq n}$ be a local coordinate of X . Then $\{x^1, \dots, x^n, y^1, \dots, y^n\}$ gives a local coordinate of its underlying real $2n$ -manifold. For convenience, we denote the real tangent bundle of the underlying real manifold by $T_{\mathbb{R}}X$, and $T_{\mathbb{C}}X := T_{\mathbb{R}}X \otimes_{\mathbb{R}} \mathbb{C}$ is called *complexified tangent bundle*. There is an almost complex structure J on $T_{\mathbb{R}}X$ given by

$$\begin{aligned} J\left(\frac{\partial}{\partial x^i}\right) &= \frac{\partial}{\partial y^i}, \\ J\left(\frac{\partial}{\partial y^i}\right) &= -\frac{\partial}{\partial x^i}. \end{aligned}$$

The eigenspace decomposition of complexified tangent bundle $T_{\mathbb{C}}X$ with respect to J is denoted as $T_{\mathbb{C}}X = T^{1,0}X \oplus T^{0,1}X$. The subspace $T^{1,0}X$ is called the holomorphic tangent bundle and $T^{0,1}X$ is called the antiholomorphic tangent bundle.

Moreover, the local frame of $T^{1,0}X$ and $T^{0,1}X$ is given as follows:

- (1) $\{\frac{\partial}{\partial z^i} := \frac{1}{2}(\frac{\partial}{\partial x^i} - \sqrt{-1}\frac{\partial}{\partial y^i})\}$ is a local frame of $T^{1,0}X$.
- (2) $\{\frac{\partial}{\partial \bar{z}^i} := \frac{1}{2}(\frac{\partial}{\partial x^i} + \sqrt{-1}\frac{\partial}{\partial y^i})\}$ is a local frame of $T^{0,1}X$.

Exercise 8.25. Prove that $(T_{\mathbb{R}}X, J)$ is isomorphic to $T^{1,0}X$ as a complex vector bundle.

On the other hand, there is also an almost complex structure on the dual bundle of $T_{\mathbb{R}}X$, which is denoted by $\Omega_{X, \mathbb{R}}^1$, and the eigenspace decomposition of complexified cotangent bundle is denoted by

$$\Omega_{X, \mathbb{C}}^1 = \Omega_{X, \mathbb{R}}^1 \otimes \mathbb{C} = \Omega_X^{1,0} \oplus \Omega_X^{0,1}.$$

It induces a decomposition of complexified k -forms

$$\Omega_{X, \mathbb{C}}^k = \bigwedge^k \Omega_{X, \mathbb{C}}^1 = \bigoplus_{p+q=k} \Omega_X^{p,q},$$

where $\Omega_X^{p,q} = \bigwedge^p \Omega_X^{1,0} \otimes \bigwedge^q \Omega_X^{0,1}$.

Definition 8.26. A k -form ω of type (p, q) is a smooth section of $\Omega_X^{p,q}$, that is

$$\omega \in C^\infty(X, \Omega_X^{p,q}) \subseteq C^\infty(X, \Omega_{X, \mathbb{C}}^k).$$

Remark 8.27 (local form). Suppose $\{z^1, \dots, z^n\}$ is a local coordinate of X , and denote $z^i = x^i + \sqrt{-1}y^i$. Then $\{dx^1, \dots, dx^n, dy^1, \dots, dy^n\}$ gives a local frame of $\Omega_{X, \mathbb{R}}^1$, and the induced almost complex structure is given by

$$\begin{aligned} J^*(dx^i)\left(\frac{\partial}{\partial x^i}\right) &= dx^i\left(J\left(\frac{\partial}{\partial x^i}\right)\right) = dx^i\left(\frac{\partial}{\partial y^i}\right) = 0, \\ J^*(dx^i)\left(\frac{\partial}{\partial y^i}\right) &= dx^i\left(J\left(\frac{\partial}{\partial y^i}\right)\right) = dx^i\left(-\frac{\partial}{\partial x^i}\right) = -1. \end{aligned}$$

In other words, we have

$$\begin{aligned} J^*(dx^i) &= -dy^i \\ J^*(dy^i) &= dx^i. \end{aligned}$$

Similarly we have

- (1) $\{dz^i := dx^i + \sqrt{-1}dy^i\}$ is a local frame of $\Omega_X^{1,0}$.
- (2) $\{d\bar{z}^i := dx^i - \sqrt{-1}dy^i\}$ is a local frame of $\Omega_X^{0,1}$.

For a k -form, locally it looks like

$$\sum_{\substack{|I|=p, |J|=q \\ p+q=k}} f_{IJ} dz^I \wedge d\bar{z}^J,$$

where f_{IJ} are $\mathbb{C}$ -valued smooth functions, and a k -form is a (p, q) -form if and only if locally it looks like

$$\sum_{|I|=p, |J|=q} f_{IJ} dz^I \wedge d\bar{z}^J.$$

Exercise 8.28. For $\mathbb{C}^n \cong \mathbb{R}^{2n}$ with coordinate $z^i = x^i + \sqrt{-1}y^i$, one has

$$dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n = \left(\frac{\sqrt{-1}}{2}\right)^n dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n.$$

8.3 Dolbeault operators

Let X be a complex manifold. By considering the $\mathbb{C}$ -linear extension of the exterior derivative, we have the following differential operator

$$d: C^\infty(X, \Omega_{X, \mathbb{C}}^k) \rightarrow C^\infty(X, \Omega_{X, \mathbb{C}}^{k+1}).$$

In this section we study how to decompose this differential operator with respect to bidegree structure of $C^\infty(X, \Omega_{X, \mathbb{C}}^k)$.

Example 8.29. For a smooth function α on X , locally we have

$$\begin{aligned} d\alpha &= \frac{\partial \alpha}{\partial x^i} dx^i + \frac{\partial \alpha}{\partial y^i} dy^i \\ &= \frac{1}{2} \left(\frac{\partial \alpha}{\partial x^i} - \sqrt{-1} \frac{\partial \alpha}{\partial y^i} \right) dz^i + \frac{1}{2} \left(\frac{\partial \alpha}{\partial x^i} + \sqrt{-1} \frac{\partial \alpha}{\partial y^i} \right) d\bar{z}^i \\ &= \frac{\partial \alpha}{\partial z^i} dz^i + \frac{\partial \alpha}{\partial \bar{z}^i} d\bar{z}^i. \end{aligned}$$

If we denote

$$\begin{aligned} \partial \alpha &= \frac{\partial \alpha}{\partial z^i} dz^i \\ \bar{\partial} \alpha &= \frac{\partial \alpha}{\partial \bar{z}^i} d\bar{z}^i, \end{aligned}$$

then $d\alpha = \partial\alpha + \bar{\partial}\alpha$, where $\partial\alpha \in C^\infty(X, \Omega_X^{1,0})$ and $\bar{\partial}\alpha \in C^\infty(X, \Omega_X^{0,1})$. In general case, for $\alpha \in C^\infty(X, \Omega_X^{p,q})$, locally looks like

$$\alpha = \sum_{|I|=p, |J|=q} \alpha_{IJ} dz^I \wedge d\bar{z}^J$$

then

$$d\alpha = \sum_{|I|=p, |J|=q} \frac{\partial \alpha_{IJ}}{\partial z^\ell} dz^\ell \wedge dz^I \wedge d\bar{z}^J + \sum_{|I|=p, |J|=q} \frac{\partial \alpha_{IJ}}{\partial \bar{z}^\ell} d\bar{z}^\ell \wedge dz^I \wedge \bar{z}^J.$$

Thus one can define

$$\begin{aligned} \partial: C^\infty(X, \Omega_X^{p,q}) &\rightarrow C^\infty(X, \Omega_X^{p+1,q}) \\ \bar{\partial}: C^\infty(X, \Omega_X^{p,q}) &\rightarrow C^\infty(X, \Omega_X^{p,q+1}) \end{aligned}$$

such that $d = \partial + \bar{\partial}$.

Proposition 8.30.

(1)

$$\partial(\alpha \wedge \beta) = \partial\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge \partial\beta.$$

(2)

$$\partial^2 = \bar{\partial}^2 = 0, \quad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

According to (2) of Proposition 8.30 there is following cochain complex

$$0 \rightarrow C^\infty(X, \Omega_X^{p,0}) \xrightarrow{\bar{\partial}} C^\infty(X, \Omega_X^{p,1}) \xrightarrow{\bar{\partial}} \dots \xrightarrow{\bar{\partial}} C^\infty(X, \Omega_X^{p,n}) \rightarrow 0. \quad (1)$$

Definition 8.31 (Dolbeault cohomology).

$$H^{p,q}(X) := H_{\bar{\partial}}^q(C^\infty(X, \Omega_X^{p,\bullet})).$$

Remark 8.32. Since we have the decomposition $C^\infty(X, \Omega_{X,\mathbb{C}}^k) = \bigoplus_{p+q=k} C^\infty(X, \Omega_X^{p,q})$, and $H^k(X, \mathbb{C})$ computes the cohomology of the cochain complex

$$0 \rightarrow C^\infty(X, \Omega_{X,\mathbb{C}}^0) \xrightarrow{d} C^\infty(X, \Omega_{X,\mathbb{C}}^1) \xrightarrow{d} \dots \xrightarrow{d} C^\infty(X, \Omega_{X,\mathbb{C}}^n) \rightarrow 0.$$

A natural question is do we have the following decomposition

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X).$$

In fact, such decomposition holds for compact Kähler manifold, which is called the Hodge decomposition.

Example 8.33. Note that

$$H^{p,0}(X) = \{\alpha \in C^\infty(X, \Omega_X^{p,0}) \mid \bar{\partial}\alpha = 0\}.$$

For $\alpha \in C^\infty(X, \Omega_X^{p,0})$ locally written as $\alpha = \sum_{|I|=p} \alpha_I dz^I$, one has

$$\bar{\partial}\alpha = \sum_{|I|=p} \frac{\partial \alpha_I}{\partial \bar{z}^\ell} d\bar{z}^\ell \wedge dz^I = 0 \iff \frac{\partial \alpha_I}{\partial \bar{z}^\ell} = 0,$$

which implies α_I is a holomorphic function. This shows $H^{p,0}(X) = \Gamma(X, \Omega_X^p)$.

Now here is a natural question: what does Dolbeault cohomology compute? In the story of smooth manifolds, de Rham cohomology calculates the sheaf cohomology of a constant sheaf, which is due to the Poincaré lemma. For complex manifolds, the Dolbeault cohomology $H^{p,q}(X)$ determines the q -th sheaf cohomology of Ω_X^p , which is based on the following lemma.

Proposition 8.34 ($\bar{\partial}$ -Poincaré lemma). *Let B be an sufficiently small open disc in $\mathbb{C}^n$. If $\alpha \in C^\infty(B, \Omega_X^{p,q})$ is $\bar{\partial}$ -closed and $q > 0$, then there exists $\beta \in C^\infty(B, \Omega_X^{p,q-1})$ such that $\alpha = \bar{\partial}\beta$.*

Proof. [Huy05, Corollary 1.3.9]. □

Proposition 8.35 (functorial). *Let $f: X \rightarrow Y$ be a holomorphic map between complex manifolds with pullback*

$$f^*: C^\infty(Y, \Omega_{Y,\mathbb{C}}^k) \rightarrow C^\infty(X, \Omega_{X,\mathbb{C}}^k).$$

Then

$$f^*: C^\infty(Y, \Omega_{Y,\mathbb{C}}^{p,q}) \rightarrow C^\infty(X, \Omega_{X,\mathbb{C}}^{p,q})$$

and it induces

$$f^*: H^{p,q}(Y) \rightarrow H^{p,q}(X).$$

Example 8.36 (Dolbeault cohomology of a holomorphic vector bundle). For a holomorphic vector bundle $E \rightarrow X$, we can also define

$$\bar{\partial}_E: C^\infty(X, \Omega_X^{0,q} \otimes E) \rightarrow C^\infty(X, \Omega_X^{0,q+1} \otimes E)$$

satisfies $\bar{\partial}_E^2 = 0$. Let us elaborate this construction: Since any global section is glued together by local sections, we just need to define $\bar{\partial}_E$ for local sections and check it is well-defined under the change of local chart. We can choose a local holomorphic frame $\{e^1, \dots, e^n\}$ for E on U , so any section $s \in C^\infty(U, \Omega_X^{0,q} \otimes E)$ locally can be written as $s = s^i \otimes e_i$ with $s^i \in C^\infty(U, \Omega_X^{0,q})$. Then we can define

$$\bar{\partial}_E(s) = \bar{\partial}s^i \otimes e^i$$

It's clear that this definition is independent of the choice of local chart since the transition functions are holomorphic and $\bar{\partial}$ kills them. Furthermore, $\bar{\partial}_E^2 = 0$ holds since $\bar{\partial}^2 = 0$. Thus we can construct a cochain complex and define its cohomology, denoted by

$$H^q(X, E) = H_{\bar{\partial}_E}^q(C^\infty(X, \Omega_X^{0,\bullet} \otimes E))$$

and similarly $H^q(X, E)$ computes the q -th sheaf cohomology of E .

9 Chern connection and Chern curvature

9.1 Connections on complex vector bundle

Let X be a complex manifold and $\pi: E \rightarrow X$ be a complex vector bundle.

9.1.1 Basic definitions

Definition 9.1. A *connection* on E is a $\mathbb{C}$ -linear operator

$$\nabla: C^\infty(X, E) \rightarrow C^\infty(X, \Omega_{X,\mathbb{C}}^1 \otimes E)$$

satisfying the Leibniz rule

$$\nabla(fs) = df \otimes s + f \nabla s$$

for $f \in C^\infty(X)$ and $s \in C^\infty(X, E)$.

Remark 9.2 (connection form). Let $\{e_\alpha\}$ be a local frame of E . Then any section s of E can be written as $s = s^\alpha e_\alpha$, and

$$\begin{aligned} \nabla(s^\alpha e_\alpha) &= ds^\alpha e_\alpha + s^\alpha \nabla s_\alpha \\ &= ds^\alpha e_\alpha + s^\alpha \omega_\alpha^\beta e_\beta, \end{aligned}$$

where ω_α^β are 1-forms, which is called connection 1-form. In terms of Christoffel symbol, one has

$$\omega_\alpha^\beta = \Gamma_{i\alpha}^\beta dz^i + \Gamma_{i\alpha}^\beta d\bar{z}^i.$$

9.1.2 Curvature form

The connection can be extended to exterior derivative which is defined on sections of vector bundle valued k -forms as follows:

$$\begin{aligned} d^\nabla: C^\infty(X, \Omega_{X,\mathbb{C}}^k \otimes E) &\rightarrow C^\infty(X, \Omega_{X,\mathbb{C}}^{k+1} \otimes E) \\ \omega \otimes s &\mapsto d\omega \otimes s + (-1)^k \omega \wedge \nabla s. \end{aligned}$$

Definition 9.3. Let E be a complex vector bundle over a complex manifold X equipped with connection ∇ . There exists a section $\Theta \in C^\infty(X, \Omega_{X,\mathbb{C}}^2 \otimes \text{End } E)$, called *curvature form*, such that

$$(d^\nabla)^2 s = \Theta \wedge s$$

for all $s \in C^\infty(X, \Omega_{X,\mathbb{C}}^k \otimes E)$.

Remark 9.4 (local form). Let $\{e_\alpha\}$ be a local frame of E . The curvature form Θ can be written as

$$\Theta = \Theta_{ij\alpha}^\beta dz^i \wedge dz^j \otimes e^\alpha \otimes e_\beta + \Theta_{i\bar{j}\alpha}^\beta dz^i \wedge d\bar{z}^j \otimes e^\alpha \otimes e_\beta + \Theta_{\bar{i}j\alpha}^\beta d\bar{z}^i \wedge dz^j \otimes e^\alpha \otimes e_\beta,$$

where $\Theta_{ij\alpha}^\beta$, $\Theta_{i\bar{j}\alpha}^\beta$ and $\Theta_{\bar{i}j\alpha}^\beta$ can be expressed in terms of Christoffel symbols.

9.2 Chern connection

In this section, we will introduce additional structures on a complex vector bundle on a complex manifold X . These structures include Hermitian metrics and holomorphic structures. We will explore connections that align harmoniously with these structures, similar to our approach in Riemannian geometry. By doing so, we will obtain the Chern connection, which runs parallel to the Levi-Civita connection.

9.2.1 Compatibility with Hermitian metric

Definition 9.5. Let E be a complex vector bundle. A *Hermitian metric* h on E is a smooth section of $E^* \otimes \bar{E}^*$.

Remark 9.6 (local form). Let $\{e_\alpha\}$ be a local frame of E . Then a Hermitian metric is determined by a positive definite Hermitian matrix $(h_{\alpha\bar{\beta}})$, that is

$$h = h_{\alpha\bar{\beta}} e^\alpha \otimes \bar{e}^\beta,$$

where $h_{\alpha\bar{\beta}} = h(e_\alpha, \bar{e}_\beta)$.

Definition 9.7. A complex vector bundle E together with a Hermitian metric h is called a *Hermitian vector bundle* (E, h) .

Remark 9.8. Let (L, h) be a Hermitian line bundle. Suppose $\{g_{\alpha\beta}\}$ is transition function of L with respect to open covering $\{U_\alpha\}$. A Hermitian metric h is locally given by h_α such that $h_\alpha = |g_{\alpha\beta}|^{-2} h_\beta$. If we write $h_\alpha = e^{-2\varphi_\alpha}$, where φ_α is called *metric weight* over U_α , such that

$$\varphi_\alpha = \varphi_\beta + \log |g_{\alpha\beta}|.$$

In other words, a Hermitian metric on a line bundle is equivalent to a collection of metric weights.

Definition 9.9. For a Hermitian vector bundle (E, h) over complex manifold X , there is a *sesquilinear map*

$$C^\infty(X, \Omega_{X, \mathbb{C}}^k \otimes E) \times C^\infty(X, \Omega_{X, \mathbb{C}}^\ell \otimes E) \rightarrow C^\infty(X, \Omega_{X, \mathbb{C}}^{k+\ell})$$

$$(s, t) \mapsto \{s, t\},$$

locally given by

$$\{s^\alpha e_\alpha, t^\beta e_\beta\} = h_{\alpha\bar{\beta}} s^\alpha \wedge \overline{t^\beta}.$$

Definition 9.10. A connection ∇ on a Hermitian vector bundle (E, h) is called a *metric connection*, if

$$d\langle s, t \rangle = \{\nabla s, t\} + \{s, \nabla t\},$$

where s, t are sections of E .

Remark 9.11 (local form). If $\{e_\alpha\}$ is a local frame of E , then

$$\begin{aligned} dh_{\alpha\bar{\beta}} &= d\langle e_\alpha, \bar{e}_\beta \rangle \\ &= \{\nabla e_\alpha, \bar{e}_\beta\} + \{e_\alpha, \nabla \bar{e}_\beta\} \\ &= \omega_\alpha^\gamma h_{\gamma\bar{\beta}} + \overline{\omega_\beta^\gamma} h_{\alpha\bar{\gamma}}. \end{aligned}$$

In terms of matrix notation, we have

$$dh = \omega h + h \bar{\omega}^T$$

In particular, if we take $\{e_\alpha\}$ to be orthogonal local frame of E with respect to h , we will find $\omega + \bar{\omega}^T = 0$, that is ω is skew-Hermitian matrix.

Proposition 9.12. Let (E, h) be a Hermitian vector bundle over a complex manifold X . A connection ∇ is a metric connection if and only if

$$d\{s, t\} = \{\nabla s, t\} + (-1)^p \{s, \nabla t\},$$

where $s \in C^\infty(X, \Omega_{X, \mathbb{C}}^p \otimes E)$ and $t \in C^\infty(X, \Omega_{X, \mathbb{C}}^q \otimes E)$.

Proposition 9.13. Let (E, h) be a Hermitian vector bundle equipped with connection ∇^E . Then ∇^E is a metric connection if and only if $\nabla^{E^* \otimes \bar{E}} h = 0$.

9.2.2 Compatibility with holomorphic structure

For a complex manifold X , we have decomposition

$$\Omega_{X, \mathbb{C}}^1 = \Omega_X^{1,0} \oplus \Omega_X^{0,1}.$$

Let $E \rightarrow X$ be a complex vector bundle with connection ∇ . Then we can decompose $\nabla = \nabla^{1,0} + \nabla^{0,1}$ by composing the projection as follows

$$\begin{array}{ccc}
& & C^\infty(X, \Omega_X^{1,0} \otimes E) \\
& \nearrow & \\
C^\infty(X, E) & \xrightarrow{\nabla} & C^\infty(X, \Omega_{X,\mathbb{C}}^1 \otimes E) \\
& \searrow & \\
& & C^\infty(X, \Omega_X^{0,1} \otimes E).
\end{array}$$

If we write $\nabla = d + \omega$ locally, then

$$\nabla^{1,0} = \partial + \omega^{1,0}$$

$$\nabla^{0,1} = \bar{\partial} + \omega^{0,1}.$$

Definition 9.14. A connection ∇ on a holomorphic vector bundle E over a complex manifold X is said to be compatible with holomorphic structure if $\nabla^{0,1} = \bar{\partial}_E$.

Remark 9.15 (local form). Let $\{e_\alpha\}$ be a holomorphic local form of E , and denote

$$\nabla e_\alpha = (\Gamma_{i\alpha}^\beta dz^i + \bar{\Gamma}_{i\alpha}^\beta d\bar{z}^i) e_\beta,$$

that is,

$$\nabla^{0,1} e_\alpha = \bar{\Gamma}_{i\alpha}^\beta e_\beta d\bar{z}^i$$

But since $\{e_\alpha\}$ is holomorphic, that is $\bar{\partial}_E e_\alpha = 0$, which implies ∇ is compatible with holomorphic structure if and only if $\bar{\Gamma}_{i\alpha}^\beta = 0$.

9.2.3 Chern connection

Theorem 9.16. Let X be a complex manifold and (E, h) be a Hermitian holomorphic vector bundle. Then there exists a unique metric connection called Chern connection such that it's compatible with holomorphic structure.

Proof. If metric connection ∇ is compatible with complex structure, then the following three equations are equivalent

$$dh = \omega h + h \bar{\omega}^t$$

$$\partial h = \omega h$$

$$\bar{\partial} h = h \bar{\omega}^t,$$

since ω is a $(1,0)$ -valued matrix. This shows the Chern connection is uniquely determined by $\omega = (\partial h)h^{-1}$. $\square$

Remark 9.17 (local form). The Chern connection is locally determined by

$$\frac{\partial h_{\alpha\bar{\beta}}}{\partial z^i} = \Gamma_{i\alpha}^\gamma h_{\gamma\bar{\beta}}.$$

Definition 9.18. Let X be a complex manifold and (E, h) be a Hermitian holomorphic vector bundle. The *Chern curvature* Θ_h of (E, h) is defined as the curvature of Chern connection with respect to h .

Corollary 9.19. Let X be a complex manifold and (E, h) a Hermitian holomorphic vector bundle equipped with Chern connection ∇ locally given by ω . Then

- (1) $\partial\omega = \omega \wedge \omega$.
- (2) $\Theta_h = \bar{\partial}\omega$.
- (3) $\bar{\partial}\Theta_h = 0$.

Proof. For (1). Since $\omega = (\partial h)h^{-1}$, then directly computation shows

$$\begin{aligned}\partial\omega &= -\partial h \wedge \partial(h^{-1}) \\ &= -\partial h \wedge (-h^{-1} \partial h h^{-1}) \\ &= (\partial h)h^{-1} \wedge (\partial h)h^{-1} \\ &= \omega \wedge \omega.\end{aligned}$$

For (2). Θ_h locally looks like

$$\Theta_h = d\omega - \omega \wedge \omega = d\omega - \partial\omega = \bar{\partial}\omega.$$

For (3). It's clear from (2). □

Remark 9.20 (local form). The Chern curvature can be expressed in terms of Christoffel symbol as follows

$$\Theta_h = \Theta_{i\bar{j}\alpha}^{\gamma} dz^i \wedge d\bar{z}^j \otimes e^{\alpha} \otimes e_{\gamma},$$

where $\Theta_{i\bar{j}\alpha}^{\gamma} = -\frac{\partial \Gamma_{i\alpha}^{\gamma}}{\partial \bar{z}^j}$. In other type one has

$$\begin{aligned}\Theta_{i\bar{j}\alpha\bar{\beta}} &= h_{\gamma\bar{\beta}} \Theta_{i\bar{j}\alpha}^{\gamma} \\ &= -h_{\gamma\bar{\beta}} \partial_{\bar{j}} (h^{\gamma\bar{\delta}} \frac{\partial h_{\alpha\bar{\delta}}}{\partial z^i}) \\ &= -\frac{\partial^2 h_{\alpha\bar{\beta}}}{\partial z^i \partial \bar{z}^j} + h^{\gamma\bar{\delta}} \frac{\partial h_{\alpha\bar{\delta}}}{\partial z^i} \frac{\partial h_{\gamma\bar{\beta}}}{\partial \bar{z}^j}.\end{aligned}$$

9.2.4 Useful formulas of Chern connection

Let X be a complex manifold and (E, h) be a Hermitian holomorphic vector bundle over X . Let ∇ be Chern connection determined by Christoffel symbol $\Gamma_{i\alpha}^{\beta}$ on (E, h) with curvature Θ_h . Suppose $\{z^i\}$ is local coordinate of X , $\{e_{\alpha}\}$ is the local frame of E , and $\{e^{\alpha}\}$ and $\{\bar{e}_{\alpha}\}$ denote local frames of E^* and $\bar{E}$ respectively.

Proposition 9.21.

$$\Gamma_{i\alpha}^\beta = h^{\beta\bar{\gamma}} \frac{\partial h_{\alpha\bar{\gamma}}}{\partial z^i}.$$

Proof. See Remark 9.17. □

Proposition 9.22.

$$\begin{aligned}\Theta_{i\bar{j}\alpha}^\gamma &= -\frac{\partial \Gamma_{i\alpha}^\gamma}{\partial \bar{z}^j} \\ \Theta_{i\bar{j}\alpha\bar{\beta}} &= -\frac{\partial^2 h_{\alpha\bar{\beta}}}{\partial z^i \partial \bar{z}^j} + h^{\gamma\bar{\delta}} \frac{\partial h_{\alpha\bar{\delta}}}{\partial z^i} \frac{\partial h_{\gamma\bar{\beta}}}{\partial \bar{z}^j}.\end{aligned}$$

Proof. See Remark 9.20. □

Proposition 9.23.

$$\begin{aligned}\nabla_{\frac{\partial}{\partial z^i}} e_\alpha &= \Gamma_{i\alpha}^\beta e_\beta, & \nabla_{\frac{\partial}{\partial \bar{z}^i}} e_\alpha &= 0, & \nabla_{\frac{\partial}{\partial z^i}} \bar{e}_\alpha &= 0, & \nabla_{\frac{\partial}{\partial \bar{z}^i}} \bar{e}_\alpha &= \bar{\Gamma}_{i\alpha}^{\bar{\beta}} \bar{e}_{\bar{\beta}}, \\ \nabla_{\frac{\partial}{\partial z^i}} e^\alpha &= -\Gamma_{i\beta}^\alpha e^\beta, & \nabla_{\frac{\partial}{\partial \bar{z}^i}} e^\alpha &= 0, & \nabla_{\frac{\partial}{\partial z^i}} \bar{e}^\alpha &= 0, & \nabla_{\frac{\partial}{\partial \bar{z}^i}} \bar{e}^\alpha &= -\bar{\Gamma}_{i\beta}^{\bar{\alpha}} \bar{e}^{\bar{\beta}}.\end{aligned}$$

Proof. It suffices to show the first two equalities, and others can be obtained from taking conjugates and dualities. The first one holds from definition of Christoffel symbol, and $\bar{\Gamma}_{i\alpha}^\beta = 0$ holds from the Remark 9.15. □

Corollary 9.24.

(1) For $s \in C^\infty(X, E)$, locally written as $s = s^\alpha e_\alpha$, one has

$$\begin{aligned}\nabla_{\frac{\partial}{\partial z^i}} s &= \left(\frac{\partial s^\beta}{\partial z^i} + s^\alpha \Gamma_{i\alpha}^\beta \right) e_\beta, \\ \nabla_{\frac{\partial}{\partial \bar{z}^i}} s &= \frac{\partial s^\beta}{\partial \bar{z}^i} e_\beta.\end{aligned}$$

(2) For $s \in C^\infty(X, \bar{E})$, locally written as $s = s^{\bar{\alpha}} \bar{e}_\alpha$, one has

$$\begin{aligned}\nabla_{\frac{\partial}{\partial z^i}} s &= \frac{\partial s^{\bar{\beta}}}{\partial z^i} \bar{e}_\beta, \\ \nabla_{\frac{\partial}{\partial \bar{z}^i}} s &= \left(\frac{\partial s^{\bar{\beta}}}{\partial \bar{z}^i} + s^{\bar{\alpha}} \bar{\Gamma}_{i\alpha}^{\bar{\beta}} \right) \bar{e}_\beta.\end{aligned}$$

(3) For $s \in C^\infty(X, E^*)$, locally written as $s = s_\alpha e^\alpha$, one has

$$\begin{aligned}\nabla_{\frac{\partial}{\partial z^i}} s &= \left(\frac{\partial s_\beta}{\partial z^i} - s_\alpha \Gamma_{i\beta}^\alpha \right) e^\beta, \\ \nabla_{\frac{\partial}{\partial \bar{z}^i}} s &= \frac{\partial s_\beta}{\partial \bar{z}^i} e^\beta.\end{aligned}$$

(4) For $s \in C^\infty(X, \bar{E}^*)$, locally written as $s = s_{\bar{\alpha}} \bar{e}^\alpha$, one has

$$\begin{aligned}\nabla_{\frac{\partial}{\partial z^i}} s &= \frac{\partial s_{\bar{\beta}}}{\partial z^i} \bar{e}^\beta, \\ \nabla_{\frac{\partial}{\partial \bar{z}^i}} s &= \left(\frac{\partial s_{\bar{\beta}}}{\partial \bar{z}^i} - s_{\bar{\alpha}} \Gamma_{i\bar{\beta}}^{\bar{\gamma}} \right) \bar{e}^\beta.\end{aligned}$$

Proposition 9.25 (Ricci identity). For $s \in C^\infty(X, E)$, locally written as $s = s^\alpha e_\alpha$, one has

$$\nabla_{\frac{\partial}{\partial z^i}} \nabla_{\frac{\partial}{\partial \bar{z}^j}} s^\beta - \nabla_{\frac{\partial}{\partial \bar{z}^j}} \nabla_{\frac{\partial}{\partial z^i}} s^\beta = \Theta_{i\bar{j}\alpha}^\beta s^\alpha.$$

9.3 First Chern class

9.3.1 First Chern class of complex line bundle

Let $\pi: X \rightarrow L$ be a complex line bundle with connection ∇ over a complex manifold X . Then the curvature Θ is a global section of $\Omega_{X, \mathbb{C}}^2$ since $\text{End } L$ is trivial bundle. Furthermore, the curvature Θ locally looks like $d\omega$ since for line bundle case we have $\omega \wedge \omega = 0$. An immediate consequence is $d\Theta = 0$, that is, Θ gives a cohomology class

$$[\Theta] \in H^2(X, \mathbb{C})$$

Definition 9.26. Let L be a complex line bundle over complex manifold X equipped with connection ∇ . The *first Chern class* of L is defined as

$$c_1(L) := \left[\frac{\sqrt{-1}}{2\pi} \Theta \right] \in H^2(X, \mathbb{C}),$$

where Θ is the curvature of ∇ .

Proposition 9.27. The first Chern class $c_1(L) \in H^2(X, \mathbb{C})$ is independent of the choice of connection.

Proof. Let $\tilde{\nabla}$ be another connection which is locally given by $\tilde{\omega}$. Then for section s of $\Omega_{X, \mathbb{C}}^k \otimes L$, one has

$$\begin{aligned}(\nabla - \tilde{\nabla})s &= (ds + \omega \wedge s) - (ds + \tilde{\omega} \wedge s) \\ &= (\omega - \tilde{\omega}) \wedge s\end{aligned}$$

Note that $\omega - \tilde{\omega}$ is a global section of $\Omega_{X, \mathbb{C}}^1$, so $\Theta - \tilde{\Theta}$ is exact. □

Proposition 9.28. Let $\pi: L \rightarrow X$ be a complex line bundle over a complex manifold X . Then $c_1(L) \in H^2(X, \mathbb{R})$.

Proof. Let h be a Hermitian metric on L and ∇ be a metric connection with respect to h . Then locally we have

$$\bar{\omega} = -\omega.$$

Thus

$$\frac{\sqrt{-1}}{2\pi} \Theta = -\frac{\sqrt{-1}}{2\pi} \bar{\Theta} = -\frac{\sqrt{-1}}{2\pi} d\bar{\omega} = \frac{\sqrt{-1}}{2\pi} d\omega = \frac{\sqrt{-1}}{2\pi} \Theta.$$

□

Remark 9.29. Here are two facts about first Chern class we cannot prove for now:

- (1) $c_1(L) \in H^2(X, \mathbb{Z})$.
- (2) L is determined by $c_1(L)$.

Definition 9.30. Let E be a complex vector bundle over complex manifold X . The *first Chern class* of E is defined to be the first Chern class of $\det E$.

9.3.2 First Chern class of Hermitian holomorphic line bundle

Let X be a complex manifold, (L, h) a Hermitian holomorphic line bundle, and ∇ be the Chern connection of (L, h) with Chern curvature Θ_h . Then by Proposition 9.28 and Corollary 9.19, we have

$$\left[\frac{\sqrt{-1}}{2\pi} \Theta_h \right] \in H^2(X, \mathbb{R}) \cap H^{1,1}(X).$$

Remark 9.31 (local form). Suppose Hermitian metric h is given by metric weight $\{\varphi_\alpha\}$, that is locally $h = e^{-2\varphi_\alpha}$. Then direct computation shows first Chern class is locally given by

$$\frac{\sqrt{-1}}{2\pi} \Theta_h = \frac{\sqrt{-1}}{\pi} \partial \bar{\partial} \varphi_\alpha$$

Proposition 9.32. $\left[\frac{\sqrt{-1}}{2\pi} \Theta_h \right] \in H^{1,1}(X)$ is independent of h .

Proof. Note that any two metric on a line bundle differ a smooth function which is positive everywhere, so if h and h' are two different metrics, we can write $\|e(z)\|_{h'} = e^f \|e(z)\|_h$ for some globally defined smooth function f . So by Remark 9.31, we have the difference of first Chern classes coming from different metrics is $\frac{\sqrt{-1}}{\pi} \partial \bar{\partial} f$, and it's trivial in $H^{1,1}(X)$ since f is globally defined. □

10 Kähler geometry

10.1 Hermitian manifold and Riemannian manifold

A Hermitian manifold is a complex manifold X together with a Hermitian metric h on holomorphic tangent bundle TX . One way to construct a Hermitian metric on TX is to consider special Riemannian metric on the underlying real manifold.

Suppose $\{z^i = x^i + \sqrt{-1}x^I\}$ is a local coordinate of X , where $1 \leq i \leq n$ and $n+1 \leq I = i+n \leq 2n$. Then $\{x^i, x^I\}$ gives a local coordinate of underlying real manifold of X , and there is a natural almost complex structure J on $T_{\mathbb{R}}X$ which is given by

$$\begin{aligned} J\left(\frac{\partial}{\partial x^i}\right) &= \frac{\partial}{\partial x^I}, \\ J\left(\frac{\partial}{\partial x^I}\right) &= -\frac{\partial}{\partial x^i}. \end{aligned}$$

Definition 10.1. A Riemannian metric g is called compatible with almost complex structure, if

$$g(V, W) = g(JV, JW)$$

for all $V, W \in C^\infty(X, T_{\mathbb{R}}X)$.

Proposition 10.2. Let g be a Riemannian metric which is compatible with J . If g is locally given by

$$g = g_{ij}dx^i \otimes dx^j + g_{iJ}dx^i \otimes dx^J + g_{IJ}dx^I \otimes dx^J,$$

then

$$\begin{aligned} g_{ij} &= g_{IJ}, \\ g_{iJ} &= g_{Ji} = -g_{JI} = -g_{Ij}. \end{aligned}$$

Proof. It follows from direct computation. □

Notation 10.3.

$$\begin{pmatrix} g^{i\ell} & g^{iL} \\ g^{I\ell} & g^{IL} \end{pmatrix} \begin{pmatrix} g_{\ell j} & g_{\ell J} \\ g_{Lj} & g_{LJ} \end{pmatrix} = I_{2n}.$$

In other words,

$$\begin{aligned} g^{i\ell}g_{\ell j} + g^{iL}g_{Lj} &= \delta_j^i, \\ g^{i\ell}g_{\ell J} + g^{iL}g_{LJ} &= 0. \end{aligned}$$

Let g be a Riemannian metric on underlying real manifold which is compatible with J and $g_{\mathbb{C}}$ be the $\mathbb{C}$ -linear extension of g . Then a direct computation shows

$$\begin{aligned} g_{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial z^j}\right) &= 0, \\ g_{\mathbb{C}}\left(\frac{\partial}{\partial \bar{z}^i}, \frac{\partial}{\partial \bar{z}^j}\right) &= 0. \end{aligned}$$

If we denote

$$h_{i\bar{j}} := g_{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right) = \frac{1}{2}(g_{ij} + \sqrt{-1}g_{i\bar{j}}), \quad (2)$$

then $H = (h_{i\bar{j}})_{n \times n}$ is a positive definite Hermitian matrix since g is a positive definite symmetric matrix. Thus h gives a Hermitian metric on $T^{1,0}X$, which makes X a Hermitian manifold. Conversely, if h is a Hermitian metric on $T^{1,0}X$, there is a Riemannian metric on $T_{\mathbb{R}}X$ given by

$$g_{ij} = 2\operatorname{Re}h_{i\bar{j}},$$

$$g_{i\bar{j}} = 2\operatorname{Im}h_{i\bar{j}},$$

$$g_{\bar{i}j} = -g_{ij},$$

$$g_{I\bar{J}} = g_{ij}.$$

In the following part of this section, g always denotes a Riemannian metric on the underlying real manifold which is compatible with almost complex structure J , and h be the Hermitian metric corresponding to g .

Proposition 10.4.

$$\sqrt{\det g} = 2^n \det h.$$

Proof. Direct computation shows

$$\begin{aligned} \det g &= \det \begin{pmatrix} (g_{ij})_{n \times n} & (g_{i\bar{j}})_{n \times n} \\ (g_{\bar{i}j})_{n \times n} & (g_{I\bar{J}})_{n \times n} \end{pmatrix} \\ &\stackrel{(1)}{=} \det \begin{pmatrix} (g_{ij})_{n \times n} & (g_{i\bar{j}})_{n \times n} \\ (-g_{ij})_{n \times n} & (g_{i\bar{j}})_{n \times n} \end{pmatrix} \\ &= \det \begin{pmatrix} (g_{ij} + \sqrt{-1}g_{i\bar{j}})_{n \times n} & (g_{i\bar{j}})_{n \times n} \\ (-g_{ij} + \sqrt{-1}g_{i\bar{j}})_{n \times n} & (g_{i\bar{j}})_{n \times n} \end{pmatrix} \\ &= \det \begin{pmatrix} (g_{ij} + \sqrt{-1}g_{i\bar{j}})_{n \times n} & (g_{i\bar{j}})_{n \times n} \\ O & (g_{ij} + \sqrt{-1}g_{i\bar{j}})_{n \times n} \end{pmatrix} \\ &= (2^n \det h)^2, \end{aligned}$$

where (1) holds from Proposition 10.2. □

Notation 10.5. $h^{i\bar{j}}$ is defined by the (i, j) -entry of $(H^{-1})^T$, that is, $h^{i\bar{\ell}}h_{j\bar{\ell}} = \delta_j^i$, and $h^{\bar{i}j} := \overline{h^{i\bar{j}}}$.

Proposition 10.6.

$$h^{i\bar{j}} = 2(g^{ij} - \sqrt{-1}g^{i\bar{j}}).$$

Proof. Direct computation shows

$$\begin{aligned}
h^{i\bar{\ell}} h_{j\bar{\ell}} &= (g^{i\ell} - \sqrt{-1}g^{iL})(g_{j\ell} + \sqrt{-1}g_{jL}) \\
&= g^{i\ell}g_{j\ell} + g^{iL}g_{jL} + \sqrt{-1}(g^{i\ell}g_{jL} - g^{iL}g_{j\ell}) \\
&= g^{i\ell}g_{\ell j} + g^{iL}g_{Lj} - \sqrt{-1}(g^{i\ell}g_{\ell j} + g^{iL}g_{Lj}) \\
&= \delta_j^i.
\end{aligned}$$

□

Definition 10.7. The *fundamental form* ω of g is defined as

$$\omega(V, W) := g_{\mathbb{C}}(JV, W),$$

where $V, W \in C^\infty(X, T_{\mathbb{C}}X)$.

Proposition 10.8.

$$\omega = \sqrt{-1}h_{i\bar{j}}dz^i \wedge d\bar{z}^j.$$

In particular, ω is a real $(1, 1)$ -form.

Proof. Direct computation shows

$$\begin{aligned}
\omega\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right) &= g_{\mathbb{C}}\left(J\left(\frac{\partial}{\partial z^i}\right), \frac{\partial}{\partial \bar{z}^j}\right) = \sqrt{-1}g_{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right) = 0, \\
\omega\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial z^j}\right) &= g_{\mathbb{C}}\left(J\left(\frac{\partial}{\partial z^i}\right), \frac{\partial}{\partial z^j}\right) = \sqrt{-1}g_{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right) = \sqrt{-1}h_{i\bar{j}}, \\
\omega\left(\frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial z^i}\right) &= g_{\mathbb{C}}\left(J\left(\frac{\partial}{\partial \bar{z}^j}\right), \frac{\partial}{\partial z^i}\right) = -\sqrt{-1}g_{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right) = -\sqrt{-1}h_{i\bar{j}}, \\
\omega\left(\frac{\partial}{\partial \bar{z}^i}, \frac{\partial}{\partial \bar{z}^j}\right) &= g_{\mathbb{C}}\left(J\left(\frac{\partial}{\partial \bar{z}^i}\right), \frac{\partial}{\partial \bar{z}^j}\right) = -\sqrt{-1}g_{\mathbb{C}}\left(\frac{\partial}{\partial \bar{z}^i}, \frac{\partial}{\partial \bar{z}^j}\right) = 0.
\end{aligned}$$

This shows

$$\begin{aligned}
\omega &= \sqrt{-1}h_{i\bar{j}}dz^i \otimes d\bar{z}^j - \sqrt{-1}h_{i\bar{j}}d\bar{z}^i \otimes dz^j \\
&= \sqrt{-1}h_{i\bar{j}}dz^i \otimes d\bar{z}^j - \sqrt{-1}h_{j\bar{i}}d\bar{z}^j \otimes dz^i \\
&= \sqrt{-1}h_{i\bar{j}}dz^i \wedge d\bar{z}^j,
\end{aligned}$$

where the last step holds since h is Hermitian, that is $h_{j\bar{i}} = h_{i\bar{j}}$.

□

Proposition 10.9.

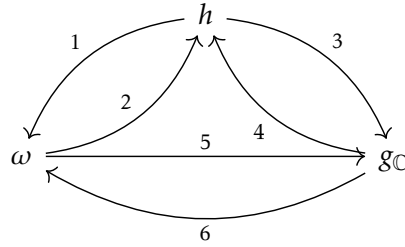
$$2h + \sqrt{-1}\omega = g_{\mathbb{C}}.$$

Proof. Direct computation shows

$$\begin{aligned}
h - \frac{1}{2}g_{\mathbb{C}} &= h_{i\bar{j}}dz^i \otimes d\bar{z}^j - \frac{1}{2}(h_{i\bar{j}}dz^i \otimes d\bar{z}^j + h_{i\bar{j}}d\bar{z}^i \otimes dz^j) \\
&= \frac{1}{2}h_{i\bar{j}}dz^i \otimes d\bar{z}^j - \frac{1}{2}h_{i\bar{j}}d\bar{z}^i \otimes dz^j \\
&= \frac{1}{2}h_{i\bar{j}}dz^i \wedge d\bar{z}^j \\
&= -\frac{\sqrt{-1}}{2}\omega.
\end{aligned}$$

□

Remark 10.10. In fact, Hermitian metric h , Riemannian metric g on underlying real manifold which is compatible with almost complex structure and fundamental form ω are the same things on a complex manifold X , and any of them gives a Hermitian structure on X .



The explicit correspondences are listed as follows:

- 1 $2\omega(-, -) = -\text{Im}h(-, -)$
- 2 $2h(-, -) = \omega(-, J-) - \sqrt{-1}\omega(-, -)$
- 3 $2g_{\mathbb{C}}(-, -) = \text{Re}h(-, -)$
- 4 $2h(-, -) = g_{\mathbb{C}}(-, -) - \sqrt{-1}g_{\mathbb{C}}(J-, -)$
- 5 $g_{\mathbb{C}}(-, -) = \omega(-, J-)$
- 6 $\omega(-, -) = g_{\mathbb{C}}(J-, -)$

Later, we may say a Hermitian manifold (X, h) or (X, ω) when we're emphasizing its Hermitian metric or fundamental form.

10.2 Curvatures of Hermitian manifold

Remark 10.11. For convenience, we always use the following notations.

- (1) $\text{Ric}^{(1)}(h) = \sqrt{-1}\text{tr}_h \Theta_h$ and $\text{Ric}^{(2)}(h) = \sqrt{-1}\text{tr}_g \Theta_h$, where Θ_h is the Chern curvature of (E, h) .
- (2) For a real $(1, 1)$ -form φ locally written as $\sqrt{-1}\varphi_{i\bar{j}}dz^i \wedge d\bar{z}^j$, $\text{tr}_g \varphi$ denotes the function $g^{i\bar{j}}\varphi_{i\bar{j}}$. In particular, the Chern scalar curvature is denoted by $\text{tr}_g \text{tr}_h \Theta_h$, where Θ_h is the Chern curvature of (E, h) .

Definition 10.12. Let (X, h) be a Hermitian manifold. Then

(1) the *first Chern-Ricci curvature* is locally given by

$$\text{Ric}^{(1)}(h) = \sqrt{-1} h^{k\bar{\ell}} \Theta_{i\bar{j}k\bar{\ell}} dz^i \wedge d\bar{z}^j.$$

(2) the *second Chern-Ricci curvature* is locally given by

$$\text{Ric}^{(2)}(h) = \sqrt{-1} h^{i\bar{j}} \Theta_{i\bar{j}k\bar{\ell}} dz^k \wedge d\bar{z}^\ell.$$

(3) the *Chern scalar curvature* of (E, h) is locally given by

$$s = g^{i\bar{j}} h^{k\bar{\ell}} \Theta_{i\bar{j}k\bar{\ell}}.$$

Proposition 10.13. The first Chern-Ricci curvature of (X, h) gives the first Chern class of TX up to a scalar.

Proof. A direct computation shows

$$\frac{\sqrt{-1}}{2\pi} h^{k\bar{\ell}} \left(-\frac{\partial^2 h_{k\bar{\ell}}}{\partial z^i \partial \bar{z}^j} + h^{m\bar{n}} \frac{\partial h_{k\bar{n}}}{\partial z^i} \frac{\partial h_{m\bar{\ell}}}{\partial \bar{z}^j} \right) = -\frac{\sqrt{-1}}{2\pi} \frac{\partial^2 \log \det(h_{k\bar{\ell}})}{\partial z^i \partial \bar{z}^j}.$$

Thus the first Chern-Ricci curvature of (TX, h) gives the Chern curvature of $(\det TX, \det h)$ up a scalar $1/2\pi$, and by Definition 9.30 the first Chern class of (TX, h) is defined to be first Chern class of $\det E$. $\square$

Definition 10.14. Let (X, h) be a Hermitian manifold and $v = v^i \frac{\partial}{\partial z^i} \in T_p X$ be a unit vector. The *holomorphic sectional curvature* in the direction v is defined as

$$\text{HSC}_p(v) := \Theta_{i\bar{j}k\bar{\ell}} v^i \bar{v}^j v^k \bar{v}^\ell.$$

Definition 10.15. Let (X, h) be a Hermitian manifold and $v = v^i \frac{\partial}{\partial z^i}, w = w^i \frac{\partial}{\partial z^i} \in T_p X$ be unit vectors. The *holomorphic bisectional curvature* in the direction v, w is defined as

$$\text{HBSC}_p(v, w) := \Theta_{i\bar{j}k\bar{\ell}} v^i \bar{v}^j w^k \bar{w}^\ell.$$

10.3 Useful formulas of Hermitian geometry

In this section we collect some useful formulas in Hermitian geometry. Unless otherwise specified, we assume (X, h) is a Hermitian n -manifold with fundamental form ω , and g is the corresponding Riemannian metric on the underlying real manifold.

Proposition 10.16.

$$\omega^n = (\sqrt{-1})^n n! \det(h_{i\bar{j}}) dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n.$$

Proof. Direct computation shows

$$\begin{aligned}
\omega^n &= (\sqrt{-1})^n (h_{i\bar{j}} dz^i \wedge d\bar{z}^j)^n \\
&= (\sqrt{-1})^n \sum_{\substack{\sigma \in S_n \\ \tau \in S_n}} h_{i_{\sigma(1)} \bar{j}_{\tau(1)}} \dots h_{i_{\sigma(n)} \bar{j}_{\tau(n)}} dz^{i_{\sigma(1)}} \wedge d\bar{z}^{j_{\tau(1)}} \wedge \dots \wedge dz^{i_{\sigma(n)}} \wedge d\bar{z}^{j_{\tau(n)}} \\
&= (\sqrt{-1})^n \sum_{\substack{\sigma \in S_n \\ \tau \in S_n}} (-1)^{|\sigma|} (-1)^{|\tau|} h_{i_{\sigma(1)} \bar{j}_{\tau(1)}} \dots h_{i_{\sigma(n)} \bar{j}_{\tau(n)}} dz^1 \wedge d\bar{z}^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^n \\
&= (\sqrt{-1})^n \sum_{\sigma \in S_n} \sum_{\rho \in S_n} (-1)^{|\rho|} h_{i_{\rho(1)} \bar{j}_1} \dots h_{i_{\rho(n)} \bar{j}_n} dz^1 \wedge d\bar{z}^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^n \\
&= (\sqrt{-1})^n n! \det(h_{i\bar{j}}) dz^1 \wedge d\bar{z}^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^n.
\end{aligned}$$

□

Corollary 10.17.

$$\omega^{n-1} = (\sqrt{-1})^{n-1} (n-1)! \sum_{\substack{i_1 < \dots < i_{n-1}, i_j \neq p \\ j_1 < \dots < j_{n-1}, j_l \neq q}} h(p, q) dz^{i_1} \wedge d\bar{z}^{j_1} \wedge \dots \wedge dz^{i_{n-1}} \wedge d\bar{z}^{j_{n-1}},$$

where $h(p, q)$ the cofactor of h without row p and column q .

Proof. Direct computation shows

$$\begin{aligned}
\omega^{n-1} &= (\sqrt{-1})^{n-1} (h_{i\bar{j}} dz^i \wedge d\bar{z}^j)^{n-1} \\
&= (\sqrt{-1})^{n-1} \sum_{\substack{\sigma \in S_{n-1} \\ \tau \in S_{n-1}}} h_{i_{\sigma(1)} \bar{j}_{\tau(1)}} \dots h_{i_{\sigma(n-1)} \bar{j}_{\tau(n-1)}} dz^{i_{\sigma(1)}} \wedge d\bar{z}^{j_{\tau(1)}} \wedge \dots \wedge dz^{i_{\sigma(n-1)}} \wedge d\bar{z}^{j_{\tau(n-1)}} \\
&= (\sqrt{-1})^{n-1} \sum_{\substack{i_1 < \dots < i_{n-1}, i_l \neq p \\ j_1 < \dots < j_{n-1}, j_l \neq q}} \sum_{\substack{\sigma \in S_{n-1} \\ \rho \in S_{n-1}}} (-1)^{|\rho|} h_{i_{\rho(1)} \bar{j}_1} \dots h_{i_{\rho(n-1)} \bar{j}_{n-1}} dz^{i_1} \wedge d\bar{z}^{j_1} \wedge \dots \wedge dz^{i_{n-1}} \wedge d\bar{z}^{j_{n-1}} \\
&= (\sqrt{-1})^{n-1} (n-1)! \sum_{\substack{i_1 < \dots < i_{n-1}, i_l \neq p \\ j_1 < \dots < j_{n-1}, j_l \neq q}} h(p, q) dz^{i_1} \wedge d\bar{z}^{j_1} \wedge \dots \wedge dz^{i_{n-1}} \wedge d\bar{z}^{j_{n-1}}.
\end{aligned}$$

□

Corollary 10.18.

$$\omega^{n-2} = (\sqrt{-1})^{n-2} (n-2)! \sum_{\substack{i_1 < \dots < i_{n-2}, i_l \neq p, s \\ j_1 < \dots < j_{n-2}, j_l \neq q, t}} h \begin{pmatrix} p, q \\ s, t \end{pmatrix} dz^{i_1} \wedge d\bar{z}^{j_1} \wedge \dots \wedge dz^{i_{n-2}} \wedge d\bar{z}^{j_{n-2}},$$

where $h \begin{pmatrix} p, q \\ s, t \end{pmatrix}$ the cofactor of h without rows p, s and columns q, t .

Proposition 10.19. $\omega^n/n!$ is the volume form of the underlying real manifold with respect to g .

Proof. Direct computation shows

$$\begin{aligned}\omega^n &\stackrel{(1)}{=} (\sqrt{-1})^n n! \det(h_{i\bar{j}}) dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n \\ &\stackrel{(2)}{=} n! 2^n \det(h_{i\bar{j}}) dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n \\ &\stackrel{(3)}{=} n! \text{vol},\end{aligned}$$

where

1. (1) holds from Proposition 10.16.
2. (2) holds from Exercise 8.28.
3. (3) holds from Proposition 10.4.

□

Proposition 10.20.

(1) If φ is a real $(1,1)$ -form, then

$$\varphi \wedge \omega^{n-1} = \frac{1}{n} \text{tr}_\omega \varphi \cdot \omega^n.$$

(2) If φ is a $(1,0)$ -form, then

$$\sqrt{-1} \varphi \wedge \bar{\varphi} \wedge \omega^{n-1} = \frac{1}{n} |\varphi|^2 \cdot \omega^n.$$

(3) If φ is a $(0,1)$ -form, then

$$\sqrt{-1} \varphi \wedge \bar{\varphi} \wedge \omega^{n-1} = -\frac{1}{n} |\varphi|^2 \cdot \omega^n.$$

Proof. For (1). Suppose $\varphi = \sqrt{-1} \varphi_{i\bar{j}} dz^i \wedge d\bar{z}^j$. Then by Proposition 10.16 one has

$$\frac{1}{n} \text{tr}_\omega \varphi \cdot \omega^n = (\sqrt{-1})^n (n-1)! h^{i\bar{j}} \varphi_{i\bar{j}} \det h dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n$$

and by Corollary 10.17 one has

$$\begin{aligned}\varphi \wedge \omega^{n-1} &= (\sqrt{-1})^n (n-1)! \varphi_{i\bar{j}} dz^i \wedge d\bar{z}^j \wedge \sum_{\substack{i_1 < \cdots < i_{n-1}, i_j \neq p \\ j_1 < \cdots < j_{n-1}, j_l \neq q}} h(p, q) dz^{i_1} \wedge d\bar{z}^{j_1} \wedge \cdots \wedge dz^{i_{n-1}} \wedge d\bar{z}^{j_{n-1}} \\ &= (\sqrt{-1})^n (n-1)! \sum_{1 \leq p, q \leq n} \varphi_{p\bar{q}} (-1)^{p+q} h(p, q) dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n.\end{aligned}$$

Note that by expansion of determinant one has

$$h^{i\bar{j}} \varphi_{i\bar{j}} \det h = h^{i\bar{j}} \varphi_{i\bar{j}} \sum_{k=1}^n h_{k\bar{j}} (-1)^{k+j} h(k, j) = \sum_{1 \leq i, j \leq n} \varphi_{i\bar{j}} (-1)^{i+j} h(i, j).$$

For (2). Suppose $\varphi = \varphi_i dz^i$. Then

$$\sqrt{-1} \varphi \wedge \bar{\varphi} = \sqrt{-1} \varphi_i \bar{\varphi}_j dz^i \wedge d\bar{z}^j$$

is a real $(1, 1)$ -form, and it's clear

$$\mathrm{tr}_\omega \sqrt{-1} \varphi \wedge \bar{\varphi} = |\varphi|^2$$

then by (1) we obtain desired result, and (3) follows from (2) directly. $\square$

Proposition 10.21.

$$\langle \mathrm{d}z^i \wedge \alpha, \beta \rangle = \langle \alpha, h^{p\bar{i}} \iota_p \beta \rangle$$

holds for α, β with appropriate bidegrees.

Proof. It follows from local computation directly. $\square$

Proposition 10.22. Let (E, h) be a Hermitian holomorphic vector bundle over a Hermitian manifold (X, g) and φ be a real $(1, 1)$ -form. Then

$$\{\varphi, \varphi\} \frac{\omega^{n-2}}{(n-2)!} = (|\mathrm{tr}_\omega \varphi|^2 - |\varphi|^2) \frac{\omega^n}{n!}.$$

Proof. Suppose $\varphi = \sqrt{-1} \varphi_{i\bar{j}}^\alpha \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j \otimes e_\alpha$. Then by Corollary 10.18 one has

$$\mathrm{LHS} = (\sqrt{-1})^n \varphi_{i\bar{j}}^\alpha \overline{\varphi_{k\bar{\ell}}^\beta} h_{\alpha\bar{\beta}} \mathrm{d}z^i \wedge \bar{z}^j \wedge \mathrm{d}z^\ell \wedge \bar{z}^k \wedge \sum_{\substack{i_1 < \dots < i_{n-2}, i_l \neq p, s \\ j_1 < \dots < j_{n-2}, j_l \neq q, t}} g \begin{pmatrix} p, q \\ s, t \end{pmatrix} \mathrm{d}z^{i_1} \wedge \mathrm{d}\bar{z}^{j_1} \wedge \dots \wedge \mathrm{d}z^{i_{n-2}} \wedge \mathrm{d}\bar{z}^{j_{n-2}}.$$

If we want to insert $\mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j \wedge \mathrm{d}z^\ell \wedge \mathrm{d}\bar{z}^k$ into $\sum_{\substack{i_1 < \dots < i_{n-2}, i_l \neq p, s \\ j_1 < \dots < j_{n-2}, j_l \neq q, t}} \mathrm{d}z^{i_1} \wedge \mathrm{d}\bar{z}^{j_1} \wedge \dots \wedge \mathrm{d}z^{i_{n-2}} \wedge \mathrm{d}\bar{z}^{j_{n-2}}$, there are the following four cases

$$\begin{cases} i = p, l = s, j = q, k = t \\ i = p, l = s, j = t, k = q \\ i = s, l = p, j = q, k = t \\ i = s, l = p, j = t, k = q \end{cases}$$

So case by case one has

$$\begin{aligned} & \varphi_{i\bar{j}}^\alpha \overline{\varphi_{k\bar{\ell}}^\beta} \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^j \wedge \mathrm{d}z^\ell \wedge \mathrm{d}\bar{z}^k \wedge \sum_{\substack{i_1 < \dots < i_{n-2}, i_l \neq p, s \\ j_1 < \dots < j_{n-2}, j_l \neq q, t}} \mathrm{d}z^{i_1} \wedge \mathrm{d}\bar{z}^{j_1} \wedge \dots \wedge \mathrm{d}z^{i_{n-2}} \wedge \mathrm{d}\bar{z}^{j_{n-2}} \\ &= \sum_{\substack{1 \leq p, q \leq n \\ 1 \leq s, t \leq n}} (\varphi_{p\bar{q}}^\alpha \overline{\varphi_{t\bar{s}}^\beta} - \varphi_{p\bar{t}}^\alpha \overline{\varphi_{q\bar{s}}^\beta} - \varphi_{s\bar{q}}^\alpha \overline{\varphi_{t\bar{p}}^\beta} + \varphi_{s\bar{t}}^\alpha \overline{\varphi_{q\bar{p}}^\beta}) (-1)^{p+q+t+s} \mathrm{d}z^1 \wedge \mathrm{d}\bar{z}^1 \wedge \dots \wedge \mathrm{d}z^n \wedge \mathrm{d}\bar{z}^n. \end{aligned}$$

On the other hand, direct computation shows

$$\begin{aligned} |\mathrm{tr}_\omega \varphi|^2 &= \varphi_{i\bar{j}}^\alpha \overline{\varphi_{k\bar{\ell}}^\beta} g^{i\bar{j}} g^{\ell\bar{k}} h_{\alpha\bar{\beta}} \\ |\varphi|^2 &= \varphi_{i\bar{j}}^\alpha \overline{\varphi_{k\bar{\ell}}^\beta} g^{i\bar{k}} g^{\ell\bar{j}} h_{\alpha\bar{\beta}} \end{aligned}$$

and thus

$$\text{RHS} = (\sqrt{-1})^n \varphi_{i\bar{j}}^\alpha \overline{\varphi_{k\bar{\ell}}^\beta} h_{\alpha\bar{\beta}} (g^{i\bar{j}} g^{\ell\bar{k}} - g^{i\bar{k}} g^{l\bar{j}}) \det g.$$

Note that Laplacian's theorem implies

$$\begin{aligned} (g^{i\bar{j}} g^{l\bar{k}} - g^{i\bar{k}} g^{l\bar{j}}) \det g &= (g^{i\bar{j}} g^{l\bar{k}} - g^{i\bar{k}} g^{l\bar{j}}) \sum_{\substack{1 \leq p, s \leq n \\ 1 \leq j, k \leq n}} (g_{p\bar{j}} g_{s\bar{k}} - g_{s\bar{j}} g_{p\bar{k}}) (-1)^{p+j+s+k} g \binom{p, j}{s, k} \\ &= (\delta_p^i \delta_s^\ell - \delta_s^i \delta_p^\ell - \delta_s^i \delta_p^\ell + \delta_p^i \delta_s^\ell) (-1)^{p+j+s+k} g \binom{p, j}{s, k}. \end{aligned}$$

This shows the RHS equals to the LHS. □

10.4 Kähler manifold

Definition 10.23. A Hermitian manifold (X, h) is called a *Kähler manifold*, if its fundamental form ω is d-closed¹.

Remark 10.24. Note that $d\omega = 0$ is equivalent to $\partial\omega = 0$, and is also equivalent to $\bar{\partial}\omega = 0$ since ω is a real $(1, 1)$ -form.

Remark 10.25 (local form). By Proposition 10.8 one has

$$\omega = \sqrt{-1} h_{i\bar{j}} dz^i \wedge d\bar{z}^j,$$

so Kähler condition $d\omega = 0$ can be computed explicitly as follows

$$\begin{aligned} d\omega &= \sqrt{-1} d(h_{i\bar{j}} dz^i \wedge d\bar{z}^j) \\ &= \sqrt{-1} \left(\frac{\partial h_{i\bar{j}}}{\partial z^k} dz^k \wedge dz^i \wedge d\bar{z}^j - \frac{\partial h_{i\bar{j}}}{\partial \bar{z}^k} dz^i \wedge d\bar{z}^k \wedge d\bar{z}^j \right) \\ &= 0. \end{aligned}$$

In other words, locally Kähler condition can be written as follows:

$$\begin{aligned} \partial_k h_{i\bar{j}} &= \partial_i h_{k\bar{j}}, \\ \partial_{\bar{k}} h_{i\bar{j}} &= \partial_{\bar{j}} h_{i\bar{k}} \end{aligned}$$

holds for all i, j, k .

Proposition 10.26. *Let (X, h) be a Kähler manifold. Then the first Chern-Ricci curvature coincides with the second Chern-Ricci curvature.*

¹ ω is called Kähler form and h is called Kähler metric.

Proof. Note that

$$\Theta_{i\bar{j}k\bar{\ell}} = -\frac{\partial^2 h_{i\bar{j}}}{\partial z^k \partial \bar{z}^\ell} + h^{p\bar{q}} \frac{\partial h_{k\bar{q}}}{\partial z^i} \frac{\partial h_{p\bar{\ell}}}{\partial \bar{z}^j}.$$

Thus if (X, h) is Kähler, then

$$\Theta_{i\bar{j}k\bar{\ell}} = \Theta_{k\bar{j}i\bar{\ell}} = \Theta_{i\bar{\ell}k\bar{j}}.$$

As a consequence, one has

$$\text{Ric}^{(1)}(h) = \sqrt{-1} h^{k\bar{\ell}} \Theta_{i\bar{j}k\bar{\ell}} dz^i \wedge d\bar{z}^j = \sqrt{-1} h^{k\bar{\ell}} \Theta_{k\bar{\ell}i\bar{j}} dz^i \wedge d\bar{z}^j = \text{Ric}^{(2)}(h).$$

This completes the proof. $\square$

Definition 10.27. A Kähler metric ω is called a *Kähler-Einstein metric*, if there exists $\lambda \in \mathbb{R}$ such that

$$\text{Ric}(\omega) = \lambda \omega.$$

Example 10.28. Any complex curve² X is Kähler since $d\omega = 0$ automatically holds.

Proposition 10.29. A submanifold of a Kähler manifold is still Kähler.

Proof. If (X, ω) is a Kähler manifold and Y is a submanifold, the restriction of ω to Y gives Kähler form of Y . $\square$

Proposition 10.30. Let (X, ω) be a compact Kähler n -manifold. Then $H^{2k}(X, \mathbb{R}) \neq 0$ for $0 \leq k \leq n$.

Proof. Note that $d(\omega^k) = 0$ holds for $0 \leq k \leq n$ since $d\omega = 0$, that is $[\omega^k] \in H^{2k}(X, \mathbb{R})$. By Proposition 10.19 one has $\omega^n = n! \text{vol}$, so the integral pairing

$$\int_X \omega^k \wedge \omega^{n-k} = n! \int_X \text{vol} \neq 0$$

implies $[\omega^k] \neq 0$ for $0 \leq k \leq n$. $\square$

Theorem 10.31. Let (X, h) be a Kähler manifold. Then around each point there exists a holomorphic coordinate $(z^1, \dots, z^n)$ such that

$$h_{i\bar{j}}(z) = \delta_{i\bar{j}} - \Theta_{i\bar{j}k\bar{\ell}}(p) z^k \bar{z}^\ell + O(|z|^2).$$

²In other words, a Riemann surface.

10.5 Levi-Civita connection encounters Chern connection

Let X be a complex n -manifold and $\{z^i = x^i + \sqrt{-1}x^I\}$ be a local coordinate of X , where $1 \leq i \leq n$ and $n+1 \leq I = i+n \leq 2n$. Then $\{x^i, x^I\}$ gives a local coordinate of underlying real manifold of X . Let g be a Riemannian metric on $T_{\mathbb{R}}X$ which is compatible with natural almost complex structure J on $T_{\mathbb{R}}X$ with Levi-Civita connection ∇ . Now consider $\mathbb{C}$ -linear extension of ∇

$$\widetilde{\nabla}: C^\infty(X, T_{\mathbb{C}}X) \rightarrow C^\infty(X, \Omega_{X, \mathbb{C}}^1 \otimes T_{\mathbb{C}}X).$$

It's clear $\widetilde{\nabla}$ gives a connection on $T_{\mathbb{C}}X$.

Notation 10.32. For $A \in \{1, \dots, n, \bar{1}, \dots, \bar{n}\}$, we denote

$$z^A = \begin{cases} z^i & A = i \\ \bar{z}^i & A = \bar{i}. \end{cases}$$

Proposition 10.33. Let $\{\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^i}\}$ be a local frame of $T_{\mathbb{C}}X$. Then

$$\widetilde{\nabla}_{\frac{\partial}{\partial z^A}} \frac{\partial}{\partial z^B} = \Gamma_{AB}^C \frac{\partial}{\partial z^C} = \frac{1}{2} h^{CE} \left(\frac{\partial h_{EB}}{\partial z^A} + \frac{\partial h_{AE}}{\partial z^B} - \frac{\partial h_{AB}}{\partial z^E} \right) \frac{\partial}{\partial z^C},$$

where $A, B, C, E \in \{1, \dots, n, \bar{1}, \dots, \bar{n}\}$.

Proof. By the proof of Koszul formula, it suffices to check

- (1) $\widetilde{\nabla}$ is compatible with $g_{\mathbb{C}}$.
- (2) $\widetilde{\nabla}$ is torsion-free, that is for $X, Y \in C^\infty(X, T_{\mathbb{C}}X)$, one has

$$\widetilde{\nabla}_X Y - \widetilde{\nabla}_Y X = [X, Y].$$

Above two claims can be checked easily by using the fact ∇ is Levi-Civita connection and the $\mathbb{C}$ -linearity of $\widetilde{\nabla}$. □

Corollary 10.34.

$$\begin{aligned} \overline{\Gamma_{ij}^k} &= \Gamma_{ij}^k = \frac{1}{2} h^{k\bar{\ell}} \left(\frac{\partial h_{j\bar{\ell}}}{\partial z^i} + \frac{\partial h_{i\bar{\ell}}}{\partial z^j} \right) \\ \overline{\Gamma_{i\bar{j}}^k} &= \Gamma_{i\bar{j}}^k = \frac{1}{2} h^{k\bar{\ell}} \left(\frac{\partial h_{j\bar{\ell}}}{\partial \bar{z}^i} - \frac{\partial h_{j\bar{i}}}{\partial \bar{z}^{\ell}} \right) \\ \overline{\Gamma_{ij}^k} &= \Gamma_{ij}^k = 0. \end{aligned}$$

Proof. It follows from

$$h_{ij} = h_{i\bar{j}} = h^{ij} = h^{\bar{i}\bar{j}} = 0.$$

□

Theorem 10.35. Let (X, h) be a Kähler manifold with induced Riemannian metric g on underlying real manifold, and suppose ∇ is Levi-Civita connection with respect to g . Then Chern connection with respect to h can be obtained from the restriction of $\mathbb{C}$ -linear extension of ∇ to $T^{1,0}X$.

Proof. Let $\widetilde{\nabla}$ be the $\mathbb{C}$ -linear extension of ∇ and $\{\frac{\partial}{\partial z^i}\}$ be a local frame of $T^{1,0}X$. Then by definition one has

$$\widetilde{\nabla} \frac{\partial}{\partial z^j} = \Gamma_{ij}^k dz^i \otimes \frac{\partial}{\partial z^k} + \bar{\Gamma}_{ij}^{\bar{k}} dz^i \otimes \frac{\partial}{\partial \bar{z}^k} + \Gamma_{ij}^k d\bar{z}^i \otimes \frac{\partial}{\partial z^k} + \bar{\Gamma}_{ij}^{\bar{k}} d\bar{z}^i \otimes \frac{\partial}{\partial \bar{z}^k}$$

By Corollary 10.34 one has $\bar{\Gamma}_{ij}^{\bar{k}} = 0$. If Kähler condition holds, then

$$\begin{aligned} \Gamma_{ij}^k &= h^{k\bar{\ell}} \frac{\partial h_{j\bar{\ell}}}{\partial z^i} \\ \Gamma_{ij}^k &= \bar{\Gamma}_{ij}^{\bar{k}} = 0. \end{aligned}$$

Thus $\widetilde{\nabla}|_{T^{1,0}X}$ gives a connection on $T^{1,0}X$, and by formula of Chern connection, it's exactly the Chern connection with respect to h . $\square$

10.6 Curvatures of Kähler metric

In this section, let (X, h) be a Kähler manifold with induced Riemannian metric g on underlying real manifold, and suppose $\{z^i = x^i + \sqrt{-1}x^I\}$ is a local coordinate of X , where $1 \leq i \leq n$ and $n+1 \leq I = i+n \leq 2n$.

Notation 10.36.

(1)

$$\begin{aligned} R_{ijkl} &= R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell}\right) \\ R_{ijKL} &= R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^K}, \frac{\partial}{\partial x^L}\right), \end{aligned}$$

where R is curvature tensor of Levi-Civita connection ∇ with respect to g .

(2)

$$\Theta_{i\bar{j}k\bar{\ell}} = \Theta\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial z^k}, \frac{\partial}{\partial \bar{z}^\ell}\right),$$

where Θ is Chern curvature with respect to h .

10.6.1 Ricci curvature and scalar curvature

Lemma 10.37.

$$\begin{aligned} R_{ijk\ell} &= R_{ijKL} \\ R_{ijK\ell} &= -R_{ijkL} \end{aligned}$$

Proof. It follows from Kähler condition. $\square$

Corollary 10.38.

$$R_{ij} = R_{IJ}$$

$$R_{iJ} = R_{Ji} = -R_{jI} = -R_{Ij}$$

Proof. It follows from Proposition 10.2 and Lemma 10.37. □

Theorem 10.39.

$$\Theta_{i\bar{j}} = \frac{1}{2}(R_{ij} + \sqrt{-1}R_{iJ})$$

where $\Theta_{i\bar{j}}$ is given by $\text{Ric}(h) = \sqrt{-1}\Theta_{i\bar{j}}dz^i \wedge d\bar{z}^j$, while R_{ij} and R_{iJ} are Riemannian Ricci curvatures.

Proof. Suppose $\{\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^I}\}$ is an orthonormal frame of real tangent bundle with respect to g , and thus $\{u_i := \frac{1}{\sqrt{2}}(\frac{\partial}{\partial x^i} - \sqrt{-1}\frac{\partial}{\partial x^I})\}$ gives an orthonormal frame of holomorphic tangent bundle with respect to h . Then

$$\begin{aligned} \Theta_{i\bar{j}} &= \sum_k \left(R\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}, u_k, \bar{u}_k\right) \right) \\ &= \sum_k \frac{1}{2} \left(\underbrace{R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, u_k, \bar{u}_k\right)}_{\text{part I}} + \underbrace{\sqrt{-1}R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^I}, u_k, \bar{u}_k\right)}_{\text{part II}} \right) \end{aligned}$$

For part I, one has

$$\begin{aligned} \sum_k R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, u_k, \bar{u}_k\right) &= \sum_k \frac{1}{2} R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} - \sqrt{-1}\frac{\partial}{\partial x^K}, \frac{\partial}{\partial x^k} + \sqrt{-1}\frac{\partial}{\partial x^K}\right) \\ &= \sum_k \frac{\sqrt{-1}}{2} (R_{ijkK} - R_{ijKk}) \\ &= \sum_k \sqrt{-1} R_{ijkK} \\ &= \sum_k \sqrt{-1} (-R_{kijK} - R_{jkiK}) \\ &= \sum_k \sqrt{-1} (R_{kijK} + R_{KijK}) \\ &= \sqrt{-1} R_{ij} \end{aligned}$$

Here we used Lemma 10.37 and first Bianchi identity. Similarly for part II one has

$$\begin{aligned}
\sum_k \sqrt{-1} R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, u_k, \bar{u}_k\right) &= \sum_k \frac{\sqrt{-1}}{2} R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} - \sqrt{-1} \frac{\partial}{\partial x^{\bar{k}}}, \frac{\partial}{\partial x^k} + \sqrt{-1} \frac{\partial}{\partial x^{\bar{k}}}\right) \\
&= \sum_k -R_{i j k \bar{k}} \\
&= \sum_k (R_{k i j \bar{k}} + R_{j k i \bar{k}}) \\
&= \sum_k (R_{k i j k} + R_{k i j \bar{k}}) \\
&= R_{ij}
\end{aligned}$$

This shows the desired result. $\square$

Corollary 10.40. *Let (X, h) be a Kähler manifold with induced Riemannian metric g on underlying real manifold. Then*

- (1) *h has positive Chern-Ricci curvature if and only if g has positive Ricci curvature.*
- (2) *h is Kähler-Einstein with Einstein constant λ if and only if g is an Einstein metric with Einstein constant λ .*

Corollary 10.41. *Let (X, h) be a Kähler manifold with induced Riemannian metric g on underlying real manifold. Let s_R be the Riemannian scalar curvature and s is the Chern scalar curvature. Then $s_R = 2s$.*

Proof. On one hand, direct computation shows

$$\begin{aligned}
s &= h^{i\bar{j}} \Theta_{i\bar{j}} \\
&= 2(g^{ij} - \sqrt{-1}g^{i\bar{j}}) \cdot \frac{1}{2}(R_{ij} + \sqrt{-1}R_{i\bar{j}}) \\
&= g^{ij}R_{ij} + g^{i\bar{j}}R_{i\bar{j}}
\end{aligned}$$

On the other hand,

$$\begin{aligned}
s_R &= g^{ij}R_{ij} + g^{i\bar{j}}R_{i\bar{j}} + g^{I\bar{j}}R_{I\bar{j}} + g^{I\bar{j}}R_{ij} \\
&\stackrel{(1)}{=} 2g^{ij}R_{ij} + 2g^{i\bar{j}}R_{i\bar{j}} \\
&= 2s
\end{aligned}$$

where (1) holds from Proposition 10.2 and Corollary 10.38. $\square$

10.6.2 Holomorphic sectional curvature and holomorphic bisectional curvature

Recall that for a Hermitian manifold (X, h) and unit vectors $v = v^i \frac{\partial}{\partial z^i}, w = w^i \frac{\partial}{\partial z^i} \in T_p X$, the holomorphic sectional curvature is defined by

$$\text{HSC}_p(v) = \Theta_{i\bar{j}k\bar{\ell}} v^i \bar{v}^j v^k \bar{v}^{\ell}$$

and the holomorphic bisectional curvature is defined by

$$\text{HBSC}_p(v, w) = \Theta_{i\bar{j}k\bar{\ell}} v^i \bar{v}^j w^k \bar{w}^\ell$$

Now suppose (X, h) is a Kähler manifold, we will see the holomorphic sectional curvature and holomorphic bisectional curvature are closely related to the Riemannian sectional curvature.

Suppose unit vectors $v, w \in T_p X$ are given by

$$\begin{aligned} v &= \frac{1}{\sqrt{2}}(x - \sqrt{-1}Jx) \\ w &= \frac{1}{\sqrt{2}}(y - \sqrt{-1}Jy) \end{aligned}$$

where x, y are real vectors with $g(x, x) = g(y, y) = 1$. Then the holomorphic bisectional curvature is computed by

$$\begin{aligned} R(v, \bar{v}, w, \bar{w}) &= \frac{1}{2}R(x - \sqrt{-1}Jx, x + \sqrt{-1}Jx, w, \bar{w}) \\ &= \sqrt{-1}R(x, Jx, w, \bar{w}) \\ &= -R(x, Jx, y, Jy) \\ &= R(x, y, y, x) + R(x, Jy, Jy, x) \end{aligned}$$

In particular, the holomorphic sectional curvature of v is exactly the sectional curvature of the plane spanned by x and Jx .

Note that the holomorphic bisectional curvature is a sum of two sectional curvatures. Hence the holomorphic bisectional curvature carries less information than the sectional curvature. On the other hand, the holomorphic bisectional curvature carries more information than the Ricci curvature, since for real vector x , one has

$$\text{Ric}_p(x) = \sum_{i=1}^n \left(R\left(\frac{\partial}{\partial x^i}, x, x, \frac{\partial}{\partial x^i}\right) + R\left(J\left(\frac{\partial}{\partial x^i}\right), x, x, J\left(\frac{\partial}{\partial x^i}\right)\right) \right)$$

As a consequence, positive sectional curvature implies positive holomorphic bisectional curvature, and positive holomorphic bisectional curvature implies positive Ricci curvature.

Definition 10.42. Let (X, h) be a Kähler manifold. (X, h) has *constant holomorphic sectional curvature* c if

$$\text{HSC}_p(v) = c$$

for all unit vector $v \in T_p X$.

Remark 10.43. In other words, the sectional curvature of all J -invariant planes equal to c , that is, $R(x, Jx, Jx, x) = c$ for all unit real vector x .

The definition given above is the the most natural way to define constant holomorphic sectional curvature, and now let us try to give another descriptions which is easy to use.

Proposition 10.44. *Let (X, h) be a Kähler manifold with constant holomorphic sectional curvature c . Then $R = cR_0$, where*

$$R_0(X, Y, Z, W) = \frac{1}{4} \{g(X, W)g(Y, Z) - g(Y, W)g(X, Z) - g(X, JZ)g(Y, JW) \\ + g(X, JW)g(Y, JZ) - 2g(X, JY)g(Z, JW)\}$$

Proof. See Proposition 7.3 of Chapter IX in [KN69]. □

Proposition 10.45. *Let (X, h) be a Kähler manifold. It has constant holomorphic sectional curvature c if and only if*

$$\Theta_{i\bar{j}k\bar{\ell}} = \frac{c}{2}(h_{i\bar{j}}h_{k\bar{\ell}} + h_{i\bar{\ell}}h_{k\bar{j}})$$

Proof. A direct computation shows

$$\begin{aligned} \Theta_{i\bar{j}k\bar{\ell}} &= R\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial z^k}, \frac{\partial}{\partial \bar{z}^{\ell}}\right) \\ &= cR_0\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial z^k}, \frac{\partial}{\partial \bar{z}^{\ell}}\right) \\ &= \frac{c}{4} \left\{ g\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^{\ell}}\right)g\left(\frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial z^k}\right) - g\left(\frac{\partial}{\partial \bar{z}^j}, \frac{\partial}{\partial \bar{z}^{\ell}}\right)g\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial z^k}\right) - g\left(\frac{\partial}{\partial z^i}, J\frac{\partial}{\partial z^k}\right)g\left(\frac{\partial}{\partial \bar{z}^j}, J\frac{\partial}{\partial \bar{z}^{\ell}}\right) \right. \\ &\quad \left. + g\left(\frac{\partial}{\partial z^i}, J\frac{\partial}{\partial \bar{z}^{\ell}}\right)g\left(\frac{\partial}{\partial \bar{z}^j}, J\frac{\partial}{\partial z^k}\right) - 2g\left(\frac{\partial}{\partial z^i}, J\frac{\partial}{\partial \bar{z}^j}\right)g\left(\frac{\partial}{\partial z^k}, J\frac{\partial}{\partial \bar{z}^{\ell}}\right) \right\} \\ &= \frac{c}{4}(h_{i\bar{\ell}}h_{k\bar{j}} + h_{i\bar{j}}h_{k\bar{\ell}} + 2h_{i\bar{j}}h_{k\bar{\ell}}) \\ &= \frac{c}{2}(h_{i\bar{\ell}}h_{k\bar{j}} + h_{i\bar{j}}h_{k\bar{\ell}}) \end{aligned}$$

On the other hand, it's clear (X, h) has constant holomorphic sectional curvature c if $\Theta_{i\bar{j}k\bar{\ell}} = \frac{c}{2}(h_{i\bar{j}}h_{k\bar{\ell}} + h_{i\bar{\ell}}h_{k\bar{j}})$. □

Remark 10.46.

- (1) Above formula differs a sign from the one given in Proposition 7.6 of Chapter IX in [KN69], since the curvature notation we defined here differs a sign from the one defined by Kobayashi.
- (2) In [Tia00], he called this by constant holomorphic bisectional curvature, and it maybe a bit confusing for beginners. If (X, h) has constant holomorphic sectional curvature, in this case you have that the bisectional curvature of X, Y equals

$$R(X, JX, JY, Y) = \frac{c}{2}(1 + g(X, Y)^2 + g(X, JY))$$

As you can see, this expression is not constant (as you vary X, Y among unit vectors), but it varies between $c/2$ and c according to the relative position of the 2-planes spanned by (X, JX)

and (Y, JY) . In particular you see that there is no such thing as a non-flat “Kähler” manifold with constant holomorphic bisectional curvature”, because if there was such a thing, in particular the holomorphic sectional curvature would be constant, but then the curvature tensor would be given by the above formula, and the bisectional curvature would actually be NOT constant unless $c = 0$, that is, all the curvatures are constant because the metric is flat.

11 Hodge theory on complex manifold

11.1 Hodge star and adjoint operators

Let (X, ω) be a compact Hermitian manifold. A (p, q) -form α can be locally written as

$$\alpha = \frac{1}{p! \times q!} \alpha_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q} dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{\bar{j}_1} \wedge \dots \wedge d\bar{z}^{\bar{j}_q}.$$

Then for $\alpha, \beta \in C^\infty(X, \Omega_X^{p,q})$, the local inner product is defined as

$$\langle \alpha, \beta \rangle = \frac{1}{p! \times q!} h^{i_1 \bar{k}_1} \dots h^{i_p \bar{k}_p} h^{\ell_1 \bar{j}_1} \dots h^{\ell_q \bar{j}_q} \alpha_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q} \overline{\beta_{k_1 \dots k_p \bar{\ell}_1 \dots \bar{\ell}_q}},$$

which is a smooth function on X .

Definition 11.1. An *inner product* on the space of (p, q) -form is defined as

$$(\alpha, \beta) := \int_X \langle \alpha, \beta \rangle \frac{\omega^n}{n!},$$

where $\alpha, \beta \in C^\infty(X, \Omega_X^{p,q})$. This also gives an inner product on $\Omega_{X, \mathbb{C}}^k = \bigoplus_{p+q=k} \Omega_X^{p,q}$.

Holding the inner product $(-, -)$, the formal adjoint operator of d is defined as an operator

$$d^*: C^\infty(X, \Omega_{X, \mathbb{C}}^k) \rightarrow C^\infty(X, \Omega_{X, \mathbb{C}}^{k-1})$$

satisfying $(\alpha, d\beta) = (d^*\alpha, \beta)$ for α, β with appropriate degrees, similarly one can define ∂^* and $\bar{\partial}^*$. In order to construct these adjoint operators, we need to introduce the well-known Hodge star operator.

Definition 11.2. There exists an operator

$$\star: C^\infty(X, \Omega_X^{p,q}) \rightarrow C^\infty(X, \Omega_X^{n-q, n-p})$$

such that

$$(\alpha, \beta) = \int_X \alpha \wedge \star \bar{\beta}.$$

Lemma 11.3.

- (1) $\star 1 = \omega^n / n!$.
- (2) $\star \omega = \omega^{n-1} / (n-1)!$.
- (3) $\overline{\star \psi} = \star \bar{\psi}$.
- (4) $\star \star = (-1)^{p+q}$ on $C^\infty(X, \Omega_X^{p,q})$.
- (5) $(\star \varphi, \star \psi) = (\varphi, \psi)$.

Proposition 11.4. $d^* = -\star d \star$.

Proof. For arbitrary $\alpha \in C^\infty(X, \Omega_{X, \mathbb{C}}^{p+q})$ and $\beta \in C^\infty(X, \Omega_{X, \mathbb{C}}^{p+q+1})$, we have

$$\begin{aligned}
(d\alpha, \beta) &= \int_X d\alpha \wedge \star \beta \\
&= \int_X d(\alpha \wedge \star \beta) - (-1)^{p+q} \alpha \wedge d\star \beta \\
&= (-1)^{p+q+1} \int_X \alpha \wedge d\star \beta \\
&\stackrel{(1)}{=} (-1)^{p+q+1} (-1)^{2n-(p+q+1)+1} \int_X \alpha \wedge \star \star d\star \beta \\
&= -(\alpha, \star d\star \beta),
\end{aligned}$$

where (1) holds from (4) of Lemma 11.3. □

Proposition 11.5.

$$\begin{aligned}
\partial^* &= -\star \bar{\partial} \star, \\
\bar{\partial}^* &= -\star \partial \star.
\end{aligned}$$

Proof. It follows from direct computations. □

Definition 11.6. Let (X, ω) be a compact Kähler manifold. The *Lefschetz operator* is defined as

$$\begin{aligned}
L: C^\infty(X, \Omega_X^{p,q}) &\rightarrow C^\infty(X, \Omega_X^{p+1,q+1}) \\
\alpha &\mapsto \omega \wedge \alpha.
\end{aligned}$$

Lemma 11.7. $\Lambda := L^* = (-1)^{p+q} \star L \star$ on (p, q) -forms.

Proof. For $\alpha \in C^\infty(X, \Omega_X^{p,q})$, $\beta \in C^\infty(X, \Omega_X^{p+1,q+1})$, direct computation shows

$$\begin{aligned}
(L\alpha, \beta) &= \int_X L\alpha \wedge \star \beta \\
&= \int_X \omega \wedge \alpha \wedge \star \beta \\
&\stackrel{(1)}{=} \int_X \alpha \wedge \omega \wedge \star \beta \\
&\stackrel{(2)}{=} \int_X \alpha \wedge (-1)^{p+q} \star \star \omega \wedge \star \beta \\
&= (\alpha, (-1)^{p+q} \star L \star \beta),
\end{aligned}$$

where

1. (1) holds from ω is a 2-form.
2. (2) holds from (4) of Lemma 11.3.

□

11.1.1 Local computations of adjoint operators

Proposition 11.8. *Let (X, h) be a Kähler manifold. Then locally*

$$\begin{cases} \partial = dz^i \wedge \nabla_i \\ \partial^* = -h^{i\bar{j}} \iota_{\bar{i}} \circ \nabla_{\bar{j}} = -h^{i\bar{j}} \nabla_{\bar{j}} \circ \iota_i \end{cases} \quad \begin{cases} \bar{\partial} = d\bar{z}^i \wedge \nabla_{\bar{i}} \\ \bar{\partial}^* = -h^{i\bar{j}} \iota_{\bar{j}} \circ \nabla_i = -h^{i\bar{j}} \nabla_i \circ \iota_{\bar{j}} \end{cases}$$

Proof. Here we only give the proof of the case ∂ and ∂^* , the proof for the other two cases are same. It suffices to check pointwisely, and at each point we may also choose normal coordinate in Theorem 10.31. For (p, q) -form α , locally written as $\alpha = \alpha_{J\bar{K}} dz^J \wedge d\bar{z}^K$. Then

$$\partial\alpha = \frac{\partial\alpha_{J\bar{K}}}{\partial z^i} dz^i \wedge dz^J \wedge d\bar{z}^K$$

and

$$\begin{aligned} dz^i \wedge \nabla_i \alpha &= dz^i \wedge \nabla_i (\alpha_{J\bar{K}} dz^J \wedge d\bar{z}^K) \\ &= dz^i \wedge \frac{\partial\alpha_{J\bar{K}}}{\partial z^i} dz^J \wedge d\bar{z}^K + \alpha_{J\bar{K}} \nabla_i (dz^J \wedge d\bar{z}^K) \\ &\stackrel{(1)}{=} \frac{\partial\alpha_{J\bar{K}}}{\partial z^i} dz^i \wedge dz^J \wedge d\bar{z}^K, \end{aligned}$$

where (1) holds from our choice of normal coordinate. To see formula of ∂^* , take arbitrary forms α, β with appropriate bidegrees, then

$$\begin{aligned} (\partial\alpha, \beta) &= (dz^i \wedge \nabla_i \alpha, \beta) \\ &\stackrel{(2)}{=} (\nabla_i \alpha, h^{p\bar{i}} \iota_p \beta) \\ &\stackrel{(3)}{=} -(\alpha, h^{p\bar{i}} \nabla_{\bar{i}} \circ \iota_p \beta), \end{aligned}$$

where

1. (2) holds from Proposition 10.21.
2. (3) holds from Stokes' theorem and the fact Chern connection is compatible with metric.

This shows

$$\partial^* = -h^{i\bar{j}} \nabla_{\bar{j}} \circ \iota_i \stackrel{(4)}{=} -h^{i\bar{j}} \iota_i \circ \nabla_{\bar{j}},$$

where (4) holds from $\iota_i \circ \nabla_{\bar{j}} = \nabla_{\bar{j}} \circ \iota_i$. □

Proposition 11.9. *Let (X, ω) be a Kähler manifold. Then locally*

$$\Lambda = \sqrt{-1} h^{i\bar{j}} \iota_i \circ \iota_{\bar{j}} = -\sqrt{-1} h^{i\bar{j}} \iota_{\bar{j}} \circ \iota_i.$$

Proof. For arbitrary forms α, β with appropriate bidegrees, direct computation shows

$$\begin{aligned}
(\omega \wedge \alpha, \beta) &= (\sqrt{-1} h_{i\bar{j}} dz^i \wedge d\bar{z}^j \wedge \alpha, \beta) \\
&\stackrel{(1)}{=} (\sqrt{-1} h_{i\bar{j}} d\bar{z}^j \wedge \alpha, h^{p\bar{i}} \iota_p \beta) \\
&\stackrel{(2)}{=} (\sqrt{-1} h_{i\bar{j}} \alpha, h^{p\bar{i}} h^{j\bar{q}} \iota_{\bar{q}} \circ \iota_p \beta) \\
&\stackrel{(3)}{=} (\alpha, -\sqrt{-1} h_{j\bar{i}} h^{p\bar{i}} h^{j\bar{q}} \iota_{\bar{q}} \circ \iota_p \beta) \\
&= (\alpha, -\sqrt{-1} h^{p\bar{i}} \iota_{\bar{i}} \circ \iota_p \beta),
\end{aligned}$$

where

1. (1) and (2) hold from Proposition 10.21.
2. (3) holds from $h_{i\bar{j}}$ is Hermitian, that is $\overline{h_{i\bar{j}}} = h_{j\bar{i}}$.

This shows

$$\Lambda = -\sqrt{-1} h^{i\bar{j}} \iota_{\bar{j}} \circ \iota_i \stackrel{(4)}{=} \sqrt{-1} h^{i\bar{j}} \iota_i \circ \iota_{\bar{j}},$$

where (4) holds from $\iota_i \circ \iota_{\bar{j}} = -\iota_{\bar{j}} \circ \iota_i$. □

11.2 Hodge theorem

Definition 11.10. The *Laplacian* $\Delta_\bullet$ is an operator defined by $\Delta_\bullet := \bullet \bullet^* + \bullet^* \bullet$, where $\bullet$ can be d, ∂ and $\bar{\partial}$.

Definition 11.11. A form α is called $\Delta_\bullet$ -harmonic if $\Delta_\bullet \alpha = 0$, where $\bullet$ can be d, ∂ and $\bar{\partial}$.

Notation 11.12. $\mathcal{H}^k$ denotes the space of Δ_d -harmonic k -forms, and $\mathcal{H}^{p,q}$ denotes the space of $\Delta_{\bar{\partial}}$ -harmonic forms of type (p, q) .

Lemma 11.13. α is $\Delta_\bullet$ -harmonic if and only if $\bullet \alpha = 0, \bullet^* \alpha = 0$, where $\bullet$ can be d, ∂ and $\bar{\partial}$.

Proof. Direct computation shows

$$\begin{aligned}
(\alpha, \Delta_d \alpha) &= (\alpha, dd^* \alpha) + (\alpha, d^* d \alpha) \\
&= \|d^* \alpha\|^2 + \|d \alpha\|^2.
\end{aligned}$$

This shows α is Δ_d -harmonic if and only if $d \alpha = d^* \alpha = 0$, the other cases are same. □

Theorem 11.14 (Hodge theorem). *Let (X, h) be a compact Hermitian n -manifold. Then*

- (1) $\mathcal{H}^{p,q}$ is finite dimensional.
- (2) *There is a decomposition $C^\infty(X, \Omega_X^{p,q}) = \mathcal{H}^{p,q} \oplus \Delta_{\bar{\partial}}(C^\infty(X, \Omega_X^{p,q}))$, which is orthogonal with respect to inner products in Definition 11.1.*

Corollary 11.15. *There is the following orthonormal decomposition*

$$C^\infty(X, \Omega_X^{p,q}) = \mathcal{H}^{p,q} \oplus \bar{\partial}(C^\infty(X, \Omega_X^{p,q-1})) \oplus \bar{\partial}^*(C^\infty(X, \Omega_X^{p,q+1})).$$

Corollary 11.16.

$$\begin{aligned}\ker \bar{\partial} &= \mathcal{H}^{p,q} \oplus \bar{\partial}^*(C^\infty(X, \Omega_X^{p,q-1})), \\ \ker \bar{\partial}^* &= \mathcal{H}^{p,q} \oplus \bar{\partial}(C^\infty(X, \Omega_X^{p,q+1})).\end{aligned}$$

Corollary 11.17. *The natural map $\mathcal{H}^{p,q} \rightarrow H^{p,q}(X)$ is an isomorphism. In particular, $H^{p,q}(X)$ is finite dimensional.*

In Riemannian geometry setting, the Hodge star induces an isomorphism between harmonic k -forms and harmonic $n - k$ -forms, which gives a proof of Poincaré duality. In complex geometry setting, in order to obtain the following

$$\star: \mathcal{H}^{p,q} \rightarrow \mathcal{H}^{n-q,n-p},$$

it suffices to prove

$$\star \circ \Delta_{\bar{\partial}} = \Delta_{\bar{\partial}} \circ \star.$$

However, since we have $\bar{\partial}^* = -\star \partial \star$, a direct computation shows $\Delta_{\bar{\partial}} \circ \star = \star \circ \Delta_{\partial}$. Thus the isomorphism $\star: \mathcal{H}^{p,q} \rightarrow \mathcal{H}^{n-q,n-p}$ fails in general since $\Delta_{\bar{\partial}} \neq \Delta_{\partial}$.

There are two methods to deal with this gap. The first method is that if X is compact Kähler manifold, then $\Delta_{\partial} = \Delta_{\bar{\partial}}$ (Theorem 12.5). This leads to

Corollary 11.18. *If (X, ω) is a compact Kähler n -manifold, then $\star: \mathcal{H}^{p,q} \rightarrow \mathcal{H}^{n-q,n-p}$ is an isomorphism. As a consequence, we have $H^{p,q}(X) \cong H^{n-q,n-p}(X)$.*

Another method is to consider the conjugate of the Hodge star

$$\begin{aligned}\bar{\star}: C^\infty(X, \Omega_X^{p,q}) &\rightarrow C^\infty(X, \Omega_X^{n-p,n-q}) \\ \alpha &\mapsto \star \bar{\alpha}\end{aligned}$$

and a direct computation shows

$$\bar{\star} \circ \Delta_{\bar{\partial}} = \Delta_{\bar{\partial}} \circ \bar{\star}.$$

Corollary 11.19. *If (X, h) is a compact Hermitian manifold, then $\bar{\star}: \mathcal{H}^{p,q} \rightarrow \mathcal{H}^{n-p,n-q}$ is an isomorphism. As a consequence, we have $H^{p,q}(X) \cong H^{n-p,n-q}(X)$.*

Remark 11.20. This is a special case of Serre duality.

12 Hodge decomposition

12.1 Kähler identities

Definition 12.1. Let A, B be two differential operators. The *commutator* of A, B is defined as

$$[A, B] := AB - (-1)^{\deg A \deg B} BA.$$

Lemma 12.2 (Jacobi identity). *Let A, B, C be differential operators. Then*

$$(-1)^{\deg A \deg C} [A, [B, C]] + (-1)^{\deg B \deg A} [B, [C, A]] + (-1)^{\deg C \deg B} [C, [A, B]] = 0.$$

Remark 12.3. In our case, the degree of $d, d^*, \partial, \partial^*, \bar{\partial}, \bar{\partial}^*$ is one, and the degree of L and Λ is zero³.

Proposition 12.4 (Kähler identities). *If (X, ω) is a compact Kähler manifold, then*

$$\begin{aligned} [\bar{\partial}^*, L] &= \sqrt{-1} \partial, \\ [\partial^*, L] &= -\sqrt{-1} \cdot \bar{\partial}, \\ [\Lambda, \bar{\partial}] &= -\sqrt{-1} \partial^*, \\ [\Lambda, \partial] &= \sqrt{-1} \cdot \bar{\partial}^*. \end{aligned}$$

Proof. By taking conjugates and adjoints, it suffices to prove the first identity, which is a first order identity of differential equation. But by Theorem 10.31, locally we have $h_{i\bar{j}} = \delta_{ij} + O(|\xi|^2)$. Thus it suffices to check Kähler identity for the case $U \subseteq \mathbb{C}^n$ equipped with standard Hermitian metric.

Suppose (p, q) -form α is locally given by $\alpha = \alpha_{J\bar{K}} dz^J \wedge d\bar{z}^{\bar{K}}$, then by Proposition 11.8 one has $\bar{\partial}^* \alpha = -\sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial \alpha}{\partial \bar{z}^{\ell}}$. Thus

$$\begin{aligned} [\bar{\partial}^*, L] \alpha &= \bar{\partial}^* (\omega \wedge \alpha) - \omega \wedge \bar{\partial}^* \alpha \\ &= -\sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial}{\partial \bar{z}^{\ell}} (\omega \wedge \alpha) + \omega \wedge \sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \\ &\stackrel{(1)}{=} -\sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} (\omega \wedge \frac{\partial \alpha}{\partial \bar{z}^{\ell}}) + \omega \wedge \sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \\ &= -\left\{ \sum_{\ell} (\iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \omega) \wedge \frac{\partial \alpha}{\partial \bar{z}^{\ell}} + \omega \wedge \sum_{\ell} \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \right\} + \sum_{\ell} \omega \wedge \iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \\ &= -\sum_{\ell} (\iota_{\frac{\partial}{\partial \bar{z}^{\ell}}} \omega) \wedge \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \\ &\stackrel{(2)}{=} \sqrt{-1} \sum_{\ell} dz^{\ell} \wedge \frac{\partial \alpha}{\partial \bar{z}^{\ell}} \\ &= \sqrt{-1} \partial \alpha, \end{aligned}$$

³You can try to understand this thing in a following way: operators $d, d^*, \partial, \partial^*, \bar{\partial}, \bar{\partial}^*$ take derivatives, but L and Λ are linear operators.

where

1. (1) holds from ω is a closed $(1,1)$ -form.
2. (2) holds from Proposition 10.8, that is $\omega = \sqrt{-1} \sum_{i=1}^n dz^i \wedge d\bar{z}^i$.

□

Theorem 12.5. *Let (X, ω) be a compact Kähler manifold. Then*

$$\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}.$$

Proof. Since

$$\Delta_d = (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial}),$$

firstly by the fourth Kähler identity, one has

- (1) The first term can be computed as

$$\begin{aligned} (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) &= (\partial + \bar{\partial})(\partial^* - \sqrt{-1}\Lambda\partial + \sqrt{-1}\partial\Lambda) \\ &= \partial\partial^* + \bar{\partial}\bar{\partial}^* - \sqrt{-1}\partial\Lambda\partial - \sqrt{-1}\cdot\bar{\partial}\Lambda\partial + \sqrt{-1}\cdot\bar{\partial}\partial\Lambda. \end{aligned}$$

- (2) The second term can be computed as

$$\begin{aligned} (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial}) &= (\partial^* - \sqrt{-1}\Lambda\partial + \sqrt{-1}\partial\Lambda)(\partial + \bar{\partial}) \\ &= \partial^*\partial + \partial^*\bar{\partial} - \sqrt{-1}\Lambda\partial\bar{\partial} + \sqrt{-1}\partial\Lambda\partial + \sqrt{-1}\partial\Lambda\bar{\partial}. \end{aligned}$$

By the third Kähler identity, one has

$$\partial^* = \sqrt{-1}[\Lambda, \bar{\partial}] = \sqrt{-1}\Lambda\bar{\partial} - \sqrt{-1}\bar{\partial}\Lambda.$$

Thus

$$\begin{aligned} \bar{\partial}\partial^* &= \bar{\partial}(\sqrt{-1}\Lambda\bar{\partial} - \sqrt{-1}\cdot\bar{\partial}\Lambda) = \sqrt{-1}\cdot\bar{\partial}\Lambda\bar{\partial}, \\ \partial^*\bar{\partial} &= (\sqrt{-1}\Lambda\bar{\partial} - \sqrt{-1}\cdot\bar{\partial}\Lambda)\bar{\partial} = -\sqrt{-1}\cdot\bar{\partial}\Lambda\bar{\partial} = -\bar{\partial}\partial^*. \end{aligned}$$

Now we have

$$\begin{aligned} \Delta_d &= \Delta_\partial - \sqrt{-1}\cdot\bar{\partial}\Lambda\partial - \sqrt{-1}\Lambda\partial\bar{\partial} + \sqrt{-1}\bar{\partial}\partial\Lambda + \sqrt{-1}\partial\Lambda\bar{\partial} \\ &= \Delta_\partial + \sqrt{-1}(\Lambda\bar{\partial}\partial - \bar{\partial}\Lambda\partial) + \sqrt{-1}(\partial\Lambda\bar{\partial} - \bar{\partial}\bar{\partial}\Lambda) \\ &= \Delta_\partial + \sqrt{-1}[\Lambda, \bar{\partial}]\partial + \sqrt{-1}\partial[\Lambda, \bar{\partial}] \\ &= \Delta_\partial + \partial^*\partial + \partial\bar{\partial}^* \\ &= 2\Delta_\partial. \end{aligned}$$

□

Corollary 12.6. *On a compact Kähler manifold, Δ_d -harmonic is equivalent to Δ_∂ -harmonic, and is equivalent to $\Delta_{\bar{\partial}}$ -harmonic.*

Corollary 12.7. Let (X, ω) be a Kähler manifold and α be a (p, q) -form. Then $\Delta_d \alpha$ is still a (p, q) -form.

Proof. It's clear to see $\Delta_{\partial} \alpha$ is still a (p, q) -form. $\square$

Exercise 12.8. Show that for compact Kähler manifold we have

$$\begin{aligned} [\Delta_d, L] &= 0 \\ [L, \Lambda] &= (k - n) \text{id} \quad \text{on } C^\infty(X, \Omega_{X, \mathbb{C}}^k). \end{aligned}$$

12.2 Hodge decomposition

Theorem 12.9. Let (X, h) be a compact Kähler manifold, $\alpha = \sum_{p+q=k} \alpha^{p,q}$. Then α is harmonic if and only if $\alpha^{p,q}$ is harmonic, that is

$$\mathcal{H}^k \otimes_{\mathbb{R}} \mathbb{C} = \bigoplus_{p+q=k} \mathcal{H}^{p,q}$$

with $\overline{\mathcal{H}^{p,q}} = \mathcal{H}^{q,p}$.

Proof. It follows from Δ_d preserves bidegree. $\square$

Theorem 12.10 (Hodge decomposition). Let (X, h) be a compact Kähler manifold. Then

$$H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(X)$$

with $\overline{H^{p,q}(X)} = H^{q,p}(X)$.

Proof. It follows from there are natural isomorphisms $H^k(X, \mathbb{C}) \cong \mathcal{H}^k \otimes \mathbb{C}$ and $H^{p,q}(X) \cong \mathcal{H}^{p,q}$. $\square$

Corollary 12.11. Let (X, h) be a compact Kähler manifold. Then

$$b_k = \sum_{p+q=k} h^{p,q}$$

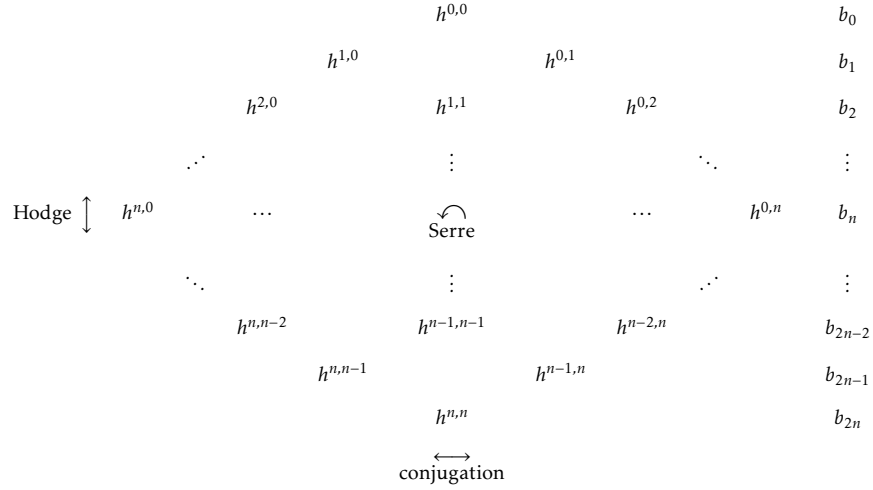
with $h^{p,q} = h^{q,p}$, where $b_k = \dim H^k(X, \mathbb{C})$ and $h^{p,q} = \dim H^{p,q}(X)$.

Corollary 12.12. b_k is even when k is odd.

Corollary 12.13. $b_k \neq 0$ when k is even.

Proof. $h^{k,k} \neq 0$ since $0 \neq \omega^k \in H^{k,k}(X)$. $\square$

There are many relations between $h^{p,q}$, and we can draw a picture as follows, called Hodge diamond since it has the same symmetry as a diamond.



Example 12.14.

$$H^{p,q}(\mathbb{CP}^n) = \begin{cases} \mathbb{C} & 0 \leq p = q \leq n \\ 0 & \text{otherwise} \end{cases}$$

Proof. It's known to all that the singular cohomology of $\mathbb{CP}^n$ with complex coefficient is

$$H^k(\mathbb{CP}^n, \mathbb{C}) = \begin{cases} \mathbb{C} & k \text{ is even} \\ 0 & k \text{ is odd} \end{cases}$$

Thus it's clear to compute Dolbeault cohomology of $\mathbb{CP}^n$ using the symmetry of Hodge diamond. □

12.3 Bott-Chern cohomology

In the proof Hodge decomposition, we used the Kähler metric. A natural question is to consider (in)dependence of the Kähler metric. In this section we will show our decomposition is independent of the choice of Kähler metric, by using Bott-Chern cohomology.

Definition 12.15. Let X be a complex manifold. The *Bott-Chern cohomology* is defined as

$$H_{\text{BC}}^{p,q}(X) := \frac{Z_{\text{BC}}^{p,q} := \{\alpha \in C^\infty(X, \Omega_X^{p,q}) \mid d\alpha = 0\}}{\partial\bar{\partial}C^\infty(X, \Omega_X^{p-1,q-1})}$$

Remark 12.16. There is a natural map

$$Z_{\text{BC}}^{p,q}(X) \rightarrow H^{p+q}(X, \mathbb{C})$$

which descends to

$$H_{\text{BC}}^{p,q}(X) \rightarrow H^{p+q}(X, \mathbb{C})$$

since $\partial\bar{\partial}\beta = d\bar{\partial}\beta$. On the other hand, there is also a natural map

$$Z_{BC}^{p,q}(X) \rightarrow H^{p,q}(X)$$

which descends to

$$H_{BC}^{p,q}(X) \rightarrow H^{p,q}(X)$$

since $\partial\bar{\partial}\beta = -\bar{\partial}\partial\beta$. So if we can prove there are isomorphisms between

$$\begin{aligned} H_{BC}^{p,q}(X) &\cong H^{p,q}(X) \\ \bigoplus_{p+q=k} H_{BC}^{p,q}(X) &\cong H^k(X, \mathbb{C}) \end{aligned}$$

then Hodge decomposition is canonical, that is independent of choice of Kähler metric since Bott-Chern cohomology is independent of the choice of Kähler metric.

Lemma 12.17 ($\partial\bar{\partial}$ -lemma). *Let (X, ω) be a compact Kähler manifold and α be a d -closed (p, q) -form. If α is $\bar{\partial}$ -exact or ∂ -exact, then there exists a $(p-1, q-1)$ -form such that*

$$\alpha = \partial\bar{\partial}\beta$$

Proof. Suppose α is $\bar{\partial}$ -exact. Then $\alpha = \bar{\partial}\gamma$ for some $(p, q-1)$ -form γ , and Hodge's theorem implies γ has decomposition

$$\gamma = a + \partial b + \partial^* c$$

where a is Δ_{∂} -harmonic, and b, c are forms with appropriate degrees. Direct computation shows

$$\begin{aligned} \alpha = \bar{\partial}\gamma &= \bar{\partial}a + \bar{\partial}\partial b + \bar{\partial}\partial^* c \\ &= -\partial\bar{\partial}b + \bar{\partial}\partial^* c \\ &= -\partial\bar{\partial}b - \partial^*\bar{\partial}c \end{aligned}$$

Now it suffices to show $-\partial^*\bar{\partial}c = 0$. A trick here is to note that

$$0 = \partial\alpha = -\partial\bar{\partial}\partial^*c \implies \partial^*\bar{\partial}c \in \ker \partial \cap \text{im } \partial^* = 0 \implies \partial^*\bar{\partial}c = 0$$

So we have

$$\alpha = \partial\bar{\partial}(-b)$$

as desired. □

Corollary 12.18. *Let (X, ω) be a compact Kähler manifold. Then*

- (1) $H_{BC}^{p,q}(X) \rightarrow H^{p,q}(X)$ is an isomorphism.
- (2) $\bigoplus_{p+q=k} H_{BC}^{p,q}(X) \rightarrow H^k(X, \mathbb{C})$ is an isomorphism.

Proof. Here we only prove the first isomorphism. From Remark 12.16, there is a canonical map $H_{\text{BC}}^{p,q}(X) \rightarrow H^{p,q}(X)$, and if we choose a Kähler metric, we have $H^{p,q}(X) \cong \mathcal{H}^{p,q}$, we will show our canonical map is both surjective and injective via this chosen metric.

- (1) To see surjectivity: For element in $H^{p,q}(X)$ we choose a $\Delta_{\bar{\partial}}$ -harmonic representative. Since $\Delta_{\bar{\partial}}$ -harmonic is equivalent to Δ_d -harmonic, so this representative is also d -closed.
- (2) To see injectivity: Suppose we have $[\alpha] \in H_{\text{BC}}^{p,q}(X)$ such that α is trivial in $H^{p,q}(X)$, that is $\bar{\partial}$ -exact. Then Lemma 12.17, that is $\partial\bar{\partial}$ -lemma implies it's trivial in Bott-Chern cohomology. $\square$

Corollary 12.19. *A Hermitian metric ω is Kähler if and only if it can be written locally as*

$$\omega = \sqrt{-1} \partial \bar{\partial} f$$

where f is a real-valued smooth function.

Proof. It's clear if ω is locally written as $\sqrt{-1} \partial \bar{\partial} f$, then it gives a Kähler metric. Conversely, a Kähler metric ω is an element in $H^{1,1}(X)$, and we have already shown that $H^{1,1}(X) = H_{\text{BC}}^{1,1}(X)$, and Dolbeault lemma implies Dolbeault cohomology vanishes on open subset which is sufficiently small, this completes the proof. $\square$

13 Positivity and vanishing theorems

13.1 Positivity of line bundle

Let (L, h) be a Hermitian holomorphic line bundle over a complex manifold X . Then the Chern curvature $\frac{\sqrt{-1}}{2\pi} \Theta_h$ is a real $(1, 1)$ -form, which corresponds to a Hermitian form on TX .

Definition 13.1. Let L be a holomorphic line bundle over X . L is called *positive* if it admits a Hermitian metric h such that the Hermitian form corresponding to $\frac{\sqrt{-1}}{2\pi} \Theta_h$ is positive definite.

Remark 13.2. The Kodaira embedding theorem implies positive line bundle is exactly ample divisor in algebraic geometry.

Remark 13.3 (local form). The curvature curvature locally is given by

$$\frac{\sqrt{-1}}{2\pi} \Theta_h = -\frac{\sqrt{-1}}{2\pi} \partial \bar{\partial} \log h = \frac{\sqrt{-1}}{2\pi} \frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j} dz^i \wedge d\bar{z}^j,$$

where $\varphi = -\log h$. Then line bundle L is positive if and only if the Hermitian matrix $(\frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j})$ is positive definite everywhere.

Proposition 13.4. *If X admits a positive holomorphic line bundle, then X is Kähler.*

Proof. The first Chern class of (L, h) is a Kähler form on X . $\square$

Proposition 13.5. L is positive if and only if $L^{\otimes m}$ is positive for some $m \in \mathbb{N}_{>0}$.

Proof. If h is a Hermitian metric locally given by $\varphi = -\log h$, then we have the Hermitian matrix corresponding to its curvature is of form

$$\left(\frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j} \right),$$

and h induces a Hermitian metric $h^{\otimes m}$ on $L^{\otimes m}$, and the Hermitian matrix corresponding to the Chern curvature of induced metric is of form

$$\left(m \cdot \frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j} \right).$$

This shows that if L is positive, then $L^{\otimes m}$ is positive. Conversely, if h_m is a Hermitian metric on $L^{\otimes m}$ locally given by $\varphi_m = -\log h_m$, then it is easy to check that $h = e^{-\varphi_m/m}$ defines a Hermitian metric on L , and thus if $L^{\otimes m}$ is positive, then L is positive. $\square$

Exercise 13.6. Let X be a compact complex manifold and L be a positive line bundle. For any holomorphic line bundle L' , there exists $N_0 \in \mathbb{N}$ such that $L' \otimes L^{\otimes N}$ positive for $N \geq N_0$.

Proof. The proof is quite similar to above Proposition, but compactness is necessary here. Over an open subset U_i , locally we have the Hermitian metric corresponding to $L' \otimes L^m$ looking like

$$\left(\frac{\partial^2 \varphi_{L'}}{\partial z^i \partial \bar{z}^j} + m \cdot \frac{\partial^2 \varphi_L}{\partial z^i \partial \bar{z}^j} \right)$$

So we can choose suffices large N_i such that $L' \otimes L^{\otimes N_i}$ is positive on U . Since X is compact, we can take a finite open covering $\{U_i\}$ of X and choose the largest N_i to be N we desired. $\square$

13.2 Positivity of vector bundles

Let (E, h) be a Hermitian holomorphic vector bundle of rank r over a complex n -manifold X with Chern connection ∇ . In local frame, its Chern curvature is given by

$$\Theta_h = \Theta_{i\bar{j}\alpha}^{\beta} dz^i \wedge d\bar{z}^j \otimes e^{\alpha} \otimes e_{\beta}$$

Definition 13.7.

- (1) (E, h) is said to be *Griffiths positive*, if for any non-zero $(u^i) \in \mathbb{C}^n$ and $(v^{\alpha}) \in \mathbb{C}^r$

$$\Theta_{i\bar{j}\alpha\bar{\beta}} u^i \bar{u}^j v^{\alpha} \bar{v}^{\beta} > 0.$$

- (2) (E, h) is said to be *Nakano positive*, if for any non-zero matrix $(u^{i\alpha})$

$$\Theta_{i\bar{j}\alpha\bar{\beta}} u^{i\alpha} \bar{u}^{j\beta} > 0.$$

(3) (E, h) is said to be *dual Nakano positive*, if for any non-zero matrix $(u^{i\alpha})$.

$$\Theta_{i\bar{j}\alpha\bar{\beta}} u^{i\beta} \bar{u}^{j\alpha} > 0.$$

Remark 13.8. The semi-positivity and negativity can be defined in the same way.

Proposition 13.9.

- (1) If (E, h) is Nakano positive or dual Nakano positive, then (E, h) is Griffiths positive.
- (2) (E, h) is Nakano positive if and only if (E^*, h^*) is dual Nakano negative.

Proof. For (1). If (E, h) is Nakano positive, then for non-zero $(u^i) \in \mathbb{C}^n$ and $(v^\alpha) \in \mathbb{C}^r$, consider matrix $(u^{i\alpha})$ defined by $u^{i\alpha} := u^i v^\alpha$, then

$$\Theta_{i\bar{j}\alpha\bar{\beta}} u^i \bar{u}^j v^\alpha \bar{v}^\beta = \Theta_{i\bar{j}\alpha\bar{\beta}} u^{i\alpha} \bar{u}^{j\beta} > 0$$

The same argument holds for the case (E, h) is dual Nakano positive. (2) follows from the relation between curvature form of (E, h) and (E^*, h^*) . $\square$

Proposition 13.10. Let

$$0 \rightarrow S \rightarrow E \rightarrow Q \rightarrow 0$$

be an exact sequence of holomorphic vector bundles.

- (1) If (E, h) is Griffiths or dual Nakano positive, then so is (Q, h^Q) .
- (2) If (E, h) is Griffiths or Nakano negative, then so is (S, h^S) .

Proposition 13.11. If Hermitian holomorphic vector bundle (E, h) is Griffiths positive, then $(E \otimes \det E, h \otimes \det h)$ is Nakano positive and dual Nakano positive.

13.3 Kodaira vanishing theorem

Theorem 13.12 (Bochner-Kodaira-Nakano identity). Let (X, ω) be a compact Kähler manifold and (E, h) a Hermitian holomorphic vector bundle. Then

$$\Delta_{\bar{\partial}_E} = [\sqrt{-1} \Theta_h, \Lambda] + \Delta_{\partial_E}.$$

Proof. A direct computation shows

$$\begin{aligned} \Delta_{\bar{\partial}_E} &= [\bar{\partial}_E, \bar{\partial}_E^*] \\ &= -\sqrt{-1} [\bar{\partial}_E, [\Lambda, \partial_E]] \\ &= -\sqrt{-1} [\Lambda, [\partial_E, \bar{\partial}_E]] - \sqrt{-1} [\partial_E, [\bar{\partial}_E, \Lambda]] \\ &= -\sqrt{-1} [\Lambda, \Theta_h] - \sqrt{-1} [\partial_E, \sqrt{-1} \partial_E^*] \\ &= [\sqrt{-1} \Theta_h, \Lambda] + \Delta_{\partial_E}. \end{aligned}$$

$\square$

Corollary 13.13 (Bochner-Kodaira-Nakano inequality). *Let (X, ω) be a compact Kähler manifold and (E, h) a Hermitian holomorphic vector bundle. Then for $\alpha \in C^\infty(X, \Omega_X^{p,q} \otimes E)$, one has*

$$([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) \leq (\Delta_{\bar{\partial}_E} \alpha, \alpha).$$

In particular, if α is $\Delta_{\bar{\partial}_E}$ -harmonic, then $([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) \leq 0$.

Proof. Direct computation shows

$$\begin{aligned} (\Delta_{\bar{\partial}_E} \alpha, \alpha) - ([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) &= (\Delta_{\partial_E} \alpha, \alpha) \\ &= \|\partial_E \alpha\|^2 + \|\partial_E^* \alpha\|^2 \geq 0. \end{aligned}$$

□

Corollary 13.14. *Let X be a complex manifold. If Hermitian holomorphic vector bundle (E, h) is Griffiths positive, then $(\det E, \det h)$ is a positive holomorphic line bundle. In particular, X is Kähler.*

Theorem 13.15 (Kodaira-Akizuki-Nakano vanishing). *Let X be a compact n -manifold, (L, h) a positive Hermitian holomorphic line bundle. Then*

$$H^{p,q}(X, L) = 0$$

for $p + q > n$.

Proof. Let X be endowed with the Kähler metric ω given by Chern curvature of L . Then there is an isomorphism $H^{p,q}(X, L) \cong \mathcal{H}^{p,q}(X, L)$. For $\alpha \in \mathcal{H}^{p,q}(X, L)$, by Corollary 13.13 one has

$$[\sqrt{-1}\Theta_h, \Lambda]\alpha \leq 0.$$

On the other hand,

$$([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) = 2\pi(p + q - n)\|\alpha\|^2 \geq 0.$$

Thus if $p + q > n$, one has $\alpha = 0$. This completes the proof. □

Corollary 13.16 (Kodaira vanishing). *Let X be a compact n -manifold and (L, h) be a positive holomorphic line bundle over X . Then*

$$H^q(X, K_X \otimes L) = 0$$

for $q > 0$.

Proof. It suffices to note that

$$H^q(X, K_X \otimes L) = H^{n,q}(X, L).$$

□

Definition 13.17 (Fano). A Fano manifold is a compact Kähler manifold with positive anti-canonical bundle $K_X^* = \det TX$.

Proposition 13.18. *Let X be a Fano manifold, then*

$$H^q(X, \mathcal{O}_X) = 0$$

for all $q > 0$.

Proof. Note that $\mathcal{O}_X = K_X \otimes K_X^*$. □

Theorem 13.19 (Serre vanishing). *Let X be a compact complex n -manifold and (L, h) be a positive holomorphic line bundle over X . For any holomorphic vector bundle E on X , there exists a constant m_0 such that for all $m \geq m_0$*

$$H^q(X, E \otimes L^{\otimes m}) = 0$$

for $q > 0$.

Proof. If X is endowed with Kähler metric ω given by Chern curvature of L and E is endowed with a Hermitian metric h , then $H^{p,q}(X, E \otimes L^{\otimes m}) \cong \mathcal{H}^{p,q}(X, E \otimes L^{\otimes m})$. For $\alpha \in \mathcal{H}^{p,q}(X, E \otimes L^{\otimes m})$, one has

$$\begin{aligned} ([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) + 2\pi m(p + q - n)\|\alpha\|^2 &\stackrel{(1)}{=} ([\sqrt{-1}\Theta_{E \otimes L^{\otimes m}}, \Lambda]\alpha, \alpha) \\ &\stackrel{(2)}{\leq} 0, \end{aligned}$$

where

1. (1) holds from $\Theta_{E \otimes L^{\otimes m}} = \Theta^E \otimes \text{id} + m(\text{id} \otimes \Theta^L)$.
2. (2) holds from Corollary 13.13, that is Bochner-Kodaira-Nakano inequality.

On the other hand, Cauchy inequality implies that

$$([\sqrt{-1}\Theta_h, \Lambda]\alpha, \alpha) \geq -C\|\alpha\|^2,$$

where constant C is the norm of $[\sqrt{-1}\Theta_h, \Lambda]$. So if we have $2\pi m(p + q - n) - C > 0$, the argument in proof of Kodaira vanishing theorem implies $\alpha = 0$. Consider $p = n, q > 0, m_0 \geq \frac{C}{2\pi}$, then for all $m \geq m_0$ and $q > 0$, one has

$$H^{n,q}(X, E \otimes L^{\otimes m}) = 0.$$

In other words, $H^q(X, K_X \otimes E \otimes L^{\otimes m}) = 0$. Thus in order to show $H^q(X, E \otimes L^{\otimes m}) = 0$, it suffices to consider $K_X^* \otimes E$ at beginning, and then we will obtain

$$H^q(X, K_X \otimes K_X^* \otimes E \otimes L^{\otimes m}) = H^q(X, E \otimes L^{\otimes m}) = 0.$$

This completes the proof. □

13.4 Nakano vanishing

Theorem 13.20 (Nakano vanishing). *Let X be a compact complex manifold and (E, h) a Hermitian holomorphic vector bundle over X .*

(1) *If (E, h) is Nakano positive, then*

$$H^{n,q}(X, E) = 0$$

for $q \geq 1$.

(2) *If (E, h) is dual Nakano positive, then*

$$H^{p,n}(X, E) = 0$$

for $p \geq 1$.

Proof. Here we only give the proof of the first one, the second can be derived from the same argument. Suppose $\{\frac{\partial}{\partial z^1}, \dots, \frac{\partial}{\partial z^n}\}$ is a holomorphic local frame which is orthonormal at point $p \in X$, and $\varphi \in H^{n,q}(X, E)$ which is locally written as $\varphi = \varphi_I^\alpha dz^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^I \otimes e_\alpha$ where $|I| = q$. Direct computation shows

$$\begin{aligned} \langle \sqrt{-1}[\Theta_h, \Lambda]\varphi, \varphi \rangle &= \langle \sqrt{-1}\Theta_h \Lambda \varphi, \varphi \rangle \\ &\stackrel{(1)}{=} -\langle \Theta_{k\bar{\ell}\alpha}^\gamma dz^k \wedge d\bar{z}^\ell \sum_m \iota_m \iota_{\bar{m}} \varphi_I^\alpha dz^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^I \otimes e_\gamma, \varphi_J^\beta dz^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^J \otimes e_\beta \rangle \\ &= -\Theta_{k\bar{\ell}\alpha\bar{\beta}} \varphi_I^\alpha \overline{\varphi_J^\beta} \langle dz^k \wedge d\bar{z}^\ell \wedge \sum_m \iota_m \iota_{\bar{m}} dz^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^I, dz^1 \wedge \dots \wedge dz^n \wedge d\bar{z}^J \rangle \\ &= \sum_k \Theta_{k\bar{\ell}\alpha\bar{\beta}} \varphi_I^\alpha \overline{\varphi_J^\beta} \langle d\bar{z}^\ell \wedge \iota_{\bar{k}} d\bar{z}^I, d\bar{z}^J \rangle \\ &= \sum_{I,J} \left\{ \Theta_{k\bar{\ell}\alpha\bar{\beta}} \left(\sum_{k \in I} \varphi_I^\alpha \right) \overline{\left(\sum_{\ell \in J} \varphi_J^\beta \right)} \right\} \\ &\stackrel{(2)}{\geq} 0, \end{aligned}$$

where

1. (1) holds from Proposition 11.9 since by Corollary 13.14 one has X is Kähler.
2. (2) holds from (E, h) is Nakano positive.

But by Corollary 13.13, that is Bochner-Kodaira-Nakano inequality, one has $\langle \sqrt{-1}[\Theta_h, \Lambda]\varphi, \varphi \rangle \leq 0$. This shows $\varphi = 0$ as before. $\square$

Corollary 13.21. *Let X be a compact complex n -manifold with $n \geq 2$. Then there doesn't exist metric h such that (TX, h) is Nakano positive.*

Proof. Suppose (TX, h) is Nakano positive, then

$$\begin{aligned} H^{1,1}(X) &= H^{0,1}(X, T^*X) \\ &\stackrel{(1)}{=} H^{n,n-1}(X, TX) \\ &\stackrel{(2)}{=} 0. \end{aligned}$$

where

1. (1) holds from Serre duality.
2. (2) holds from Nakano vanishing theorem.

However, by Proposition 13.9 and Corollary 13.14 one has X is a Kähler manifold, which leads to a contradiction since $H^{1,1}(X) \neq 0$ for a Kähler manifold. $\square$

Corollary 13.22. *Let X be a compact complex n -manifold with $n \geq 2$. Then there doesn't exist metric h such that (T^*X, h) is dual Nakano positive.*

Proof. Note that

$$H^{n-1,n}(X, T^*X) = H^{1,0}(X, TX) = H^0(X, \text{End}(TX)) \neq 0.$$

$\square$

Example 13.23. If $n \geq 2$, $(\mathbb{CP}^n, \omega_{\text{FS}})$ is dual Nakano positive and Nakano semi-positive, but not Nakano positive.

Proof. A direct computation shows the curvature of Fubini-Study metric is

$$\Theta_{i\bar{j}k\bar{\ell}} = h_{i\bar{j}}h_{k\bar{\ell}} + h_{i\bar{\ell}}h_{k\bar{j}}.$$

For $p \in X$, if we consider normal coordinate at p , then for any non-zero matrix (u^{ij}) , a direct computation shows

$$\begin{aligned} \Theta_{i\bar{j}k\bar{\ell}} u^{i\ell} \bar{u}^{jk} &= h_{i\bar{j}}h_{k\bar{\ell}} u^{i\ell} \bar{u}^{jk} + h_{i\bar{\ell}}h_{k\bar{j}} u^{i\ell} \bar{u}^{jk} \\ &= \sum_{i,j} |u^{ij}|^2 + u^{ii} \bar{u}^{jj} \\ &= \sum_{i \neq j} |u^{ij}|^2 + \frac{1}{2} \sum_{i,j} |u^{ii} + u^{jj}|^2, \\ &> 0, \end{aligned}$$

which implies $(\mathbb{CP}^n, \omega_{FS})$ is dual Nakano positive.

$$\begin{aligned}
\Theta_{i\bar{j}k\bar{\ell}} u^{ik} \bar{u}^{jl} &= h_{i\bar{j}} h_{k\bar{\ell}} u^{ik} \bar{u}^{jl} + h_{i\bar{\ell}} h_{k\bar{j}} u^{ik} \bar{u}^{jl} \\
&= \sum_{i,j} |u^{ij}|^2 + u^{ij} \bar{u}^{ji} \\
&= \frac{1}{2} \sum_{i,j} |u^{ij} + u^{ji}|^2 \\
&\geq 0,
\end{aligned}$$

which implies $(\mathbb{CP}^n, \omega_{FS})$ is Nakano semi-positive, and it's not Nakano positive by Corollary [13.21](#). □

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